

Sparse Autoencoders Can Capture Language-Specific Concepts Across Diverse Languages

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Abstract

Understanding the multilingual mechanisms of large language models (LLMs) provides insight into how they process different languages, yet this remains challenging. Existing studies often focus on individual neurons, but their polysemantic nature makes it difficult to isolate language-specific units from cross-lingual representations. To address this, we explore sparse autoencoders (SAEs) for their ability to learn monosemantic features that represent concrete and abstract concepts across languages in LLMs. While some of these features are language-independent, the presence of language-specific features remains underexplored. In this work, we introduce *SAE-LAPE*, a method based on feature activation probability, to identify language-specific features within the feed-forward network. We find that many such features predominantly appear in the middle to final layers of the model and are interpretable. These features influence the model’s multilingual performance and language output and can be used for language identification with performance comparable to fastText along with more interpretability. Our code and complete figures are available [here](#).

1 Introduction

Understanding the multilingual mechanisms of LLMs offers insight into how they process language across diverse linguistic contexts. However, such mechanisms remain challenging to interpret. Existing studies (Tang et al., 2024; Kojima et al., 2024) primarily focus on individual neurons, yet these neurons are often polysemantic, meaning they may activate in response to multiple unrelated concepts (Elhage et al., 2022). This characteristic limits both interpretability and reliability, as it becomes difficult to isolate language-specific units from cross-lingual representations.

A promising approach to address the issue of the polysemous nature of neurons is the use of sparse

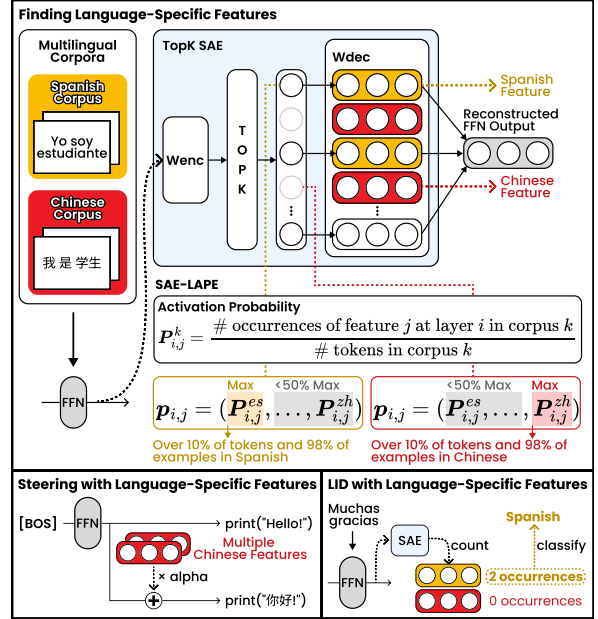


Figure 1: An illustration of language-specific features. **Top:** Finding language-specific features that are often activated in a target language using *SAE-LAPE*. **Bottom Left:** Steering multiple language-specific features simultaneously by multiplying them with a scaling factor α and adding the result to the FFN output. **Bottom Right:** Language identification (LID) by counting the occurrences of language-specific features in an SAE.

autoencoders (SAEs), an unsupervised method that can learn monosemantic features in the form of vectors that represent single concepts (Huben et al., 2024). SAEs have recently been shown to be capable of detecting particular concepts in LLMs, such as “Golden Gate Bridge”-related concepts, abstract grammatical concepts shared across languages (Templeton et al., 2024; Brinkmann et al., 2025), known entities (Ferrando et al., 2025), and “artificial text”-related concepts used for classifying such text (Kuznetsov et al., 2025). While some features appear to be language-agnostic, only few studies have explored the existence of language-specific features (see Section 6).

In this study, we investigate the presence of language-specific features by identifying them via SAEs and using them as units for analyzing the multilinguality of an LLM. By adapting *language activation probability entropy (LAPE)* method proposed by Tang et al. (2024), we introduce *SAE-LAPE*, a method to identify language-specific features in the SAE trained in the feed-forward network (FFN) of each layer. We further analyze the concepts represented by these features through automated interpretability techniques proposed by Paulo et al. (2025). Finally, we provide empirical evidence showing that the features are indeed language-specific. This results in several findings:

- **Language-specific features exist and are interpretable.** These features, predominantly active in a single language, are primarily located in the middle to final layers. Additionally, these features are interpretable, as their interpretation scores fall within the acceptable range reported by Paulo et al. (2025) across models.
- **Language-specific features influence both perplexity and language output of LLMs.** Suppressing these features increases perplexity in the corresponding language with relatively little impact on others, while activating them has minimal effect on that language but can influence others. Moreover, activating these features can shift the output towards the corresponding language.
- **Language-specific features can be used as a language classifier.** Because these features are activated for aspects of particular languages, we use them for a language identification (LID) task and this approach achieves results comparable to fastText (Joulin et al., 2017), yet more interpretable since each feature tends to be monosemantic with its own interpretation.

2 Background

2.1 TopK SAEs dan JumpReLU SAEs

SAEs are an unsupervised method used to discover interpretable feature directions in LLMs by reconstructing model activations as a sparse linear combination of learned overcomplete basis vectors representing these directions. The sparsity constraint encourages the model to represent each activation using only a small number of these feature directions, leading to more monosemous representations. To achieve this, SAEs use an encoder to map the activations $x \in \mathbb{R}^d$ into a sparse, higher-dimensional representation $z \in \mathbb{R}^n$, and a decoder to project

this representation back into the original activations. This process can be described as follows:

$$z = \text{ReLU}(W_{\text{enc}}(x - b_{\text{dec}}) + b_{\text{enc}}),$$

$$\hat{x} = W_{\text{dec}}z + b_{\text{dec}},$$

where $W_{\text{enc}} \in \mathbb{R}^{n \times d}$ is an encoder matrix; $W_{\text{dec}} \in \mathbb{R}^{d \times n}$ is a decoder matrix; $b_{\text{enc}} \in \mathbb{R}^n$ and $b_{\text{dec}} \in \mathbb{R}^d$ are the bias vectors for the encoder and decoder, respectively; $\hat{x}$ is the reconstruction of x ; and $\text{ReLU}(x) = \max(0, x)$ is an activation function (Glorot et al., 2011). We denote z_j as the activation of feature j and $d^j = W_{\text{dec}}[:, j] \in \mathbb{R}^d$ as the direction of feature j .

In this study, we use TopK SAEs (Gao et al., 2025), an improved variant that provides a better reconstruction-sparsity trade-off and increases feature monosemanticity. TopK SAEs enforce sparsity by retaining only the k largest values in SAE activations $z \in \mathbb{R}^n$ and zeroing the rest. In our setup, ReLU is applied prior to TopK to ensure non-negative activations. Furthermore, we use JumpReLU SAEs (Rajamanoharan et al., 2024), which improve reconstruction fidelity by replacing the standard ReLU with a discontinuous activation proposed by Erichson et al. (2020), defined as $\text{JumpReLU}(x) = xH(x - \theta)$, where H is the Heaviside step function and θ is a learned threshold.

2.2 High Frequency Latents

High-frequency latents (HFLs) are SAE features that appear in over 10% of tokens from the entire corpora, constituting a significant proportion (Sun et al., 2025). As noted by prior work, SAEs often still learn a few HFLs that may appear less interpretable, even under greater sparsity constraints (Cunningham and Conerly, 2024; Rajamanoharan et al., 2024; Sun et al., 2025). However, Sun et al. (2025) argued that these features may correspond to genuine dense language model features and can be somewhat interpretable, as suggested by their preliminary analyses.

3 Method

3.1 Finding Language-Specific Features

By adapting the method proposed by Tang et al. (2024), we introduce an approach called *SAE-LAPE* to identify language-specific features that are also HFLs in the FFN, where language-specific information resides (Tang et al., 2024; Kojima et al., 2024). Specifically, we provide an LLM with multilingual

corpora, and we compute the activation probability of feature d^j from an SAE at layer i for each language k as follows:

$$P_{i,j}^k = \mathbb{E}(\mathbb{I}(z_j^i > 0) \mid \text{language } k\text{'s corpus}) ,$$

where $\mathbb{E}$ is the expectation over the tokens in language k 's corpus; $\mathbb{I}$ is the indicator function, where a feature is considered active if its activation is nonzero; and z_j^i is the activation associated with feature d^j at layer i . Subsequently, we obtain the vector $p_{i,j} = (P_{i,j}^1, \dots, P_{i,j}^l)$ for l languages, and then apply L1-normalization to produce a valid probability distribution $p'_{i,j} = (P'_{i,j}^1, \dots, P'_{i,j}^l)$. We compute the entropy of this distribution, referred to as *language activation probability entropy (LAPE)*, to measure the degree of language specificity, where a lower value indicates greater specificity:

$$\text{LAPE}_{i,j} = - \sum_{k=1}^l P'_{i,j}^k \ln(P'_{i,j}^k) .$$

Thus far, our implementation follows LAPE (Tang et al., 2024). This method was originally used to identify language-specific neurons in the FFN activations (see Appendix A.2 for details). However, SAE activations are sparser compared to FFN activations. In addition, the features we aim to find are a type of HFLs. Therefore, we need to modify the original method to address these issues.

To find HFLs, we consider only features active in more than 10% of tokens and at least $N\%$ of examples in at least one language-specific corpus, ensuring that the concepts are predominantly associated with those languages. Next, we dynamically identify features as specific to one or more languages when their activation probability $P_{i,j}^k$ is at least $T\%$ of $\max(p_{i,j})$. Lastly, we select language-specific features as those specific to a single language.

3.2 Steering Language-Specific Features

Steering SAE features is a form of activation steering (Turner et al., 2024), in which model activations are modified during a forward pass using an SAE feature to influence downstream behavior. Generally, we can activate a concept in LLMs by adding a feature, or suppress it by subtracting one, at every token position (Templeton et al., 2024; Durmus et al., 2024):

$$x_{\text{new}} = x + \alpha z_j^{\max} d^j ,$$

where $\alpha \in \mathbb{R}$ is a scaling factor and $z_j^{\max}$ is the maximum value of the feature of interest d^j in the multilingual corpora.

In contrast to previous studies (e.g., Templeton et al., 2024; Brinkmann et al., 2025), where interventions were typically conducted on a single feature in the middle layer, we intervene on multiple features across multiple layers. We argue that this method works on language-specific features due to their tendency, as HFLs, to naturally and frequently co-occur within a specific language (see Section 5.1). We also use the same method with language-specific neurons as a baseline (see Appendix A.3).

3.3 LID with Language-Specific Features

Bricken et al. (2023) define the sensitivity of an SAE feature as the reliable activation of the feature when its associated concept is present in a given example. With language as the concept of interest, we use language-specific features for language classification to assess whether examples in a particular language predominantly activate the corresponding features over those of other languages.

Specifically, to classify languages using SAE features for a text input with FFN output $X = [x_1, \dots, x_t]$ consisting of t tokens, we compute the score for each language k as follows:

$$s_k = \sum_{i \in \mathcal{L}} \sum_{x \in X} \sum_{j \in \mathcal{F}_k^i} \mathbb{I}(z_{x,j}^i > 0) ,$$

where $\mathcal{L}$ is the set of all layers; $\mathcal{F}_k^i$ is the set of SAE feature indices associated with language k at layer i ; $z_{x,j}^i$ is the activation value of feature d^j for input x at layer i ; and $\mathbb{I}(z_{x,j}^i > 0)$ is the indicator function to ensure that only active features are considered. Then, we predict the language by selecting the index of the highest score in the vector $s = [s_1, \dots, s_l]$ over l languages. We also use the same method with language-specific neurons as a baseline (see Appendix A.4).

4 Experimental Setup

4.1 Models

We adopt Llama 3.2 1B (Meta AI, 2024b) as the primary model in this study due to its multilingual capabilities, explicitly supporting eight languages: English (en), German (de), French (fr), Italian (it), Portuguese (pt), Hindi (hi), Spanish (es), and Thai (th). It has also been trained on a broader collection of languages beyond these officially supported ones (Meta AI, 2024a). For this model, we use the publicly available TopK SAEs¹ released by EleutherAI

¹<https://huggingface.co/EleutherAI/sae-Llama-3.2-1B-131k>

(2024) for Llama 3.2 1B, with $k = 32$ and an expansion factor of 64, resulting in 131K features per layer. These SAEs are trained on the RedPajama-v2 (Weber et al., 2024) multilingual corpus using the model’s FFN outputs across all layers. See Appendix A.5 for the SAE configuration details.

We also use Gemma 2 (2B and 9B) (Team et al., 2024) in a subset of experiments to evaluate cross-model generalizability. Although primarily trained on English data, these models exhibit some multilingual capabilities. For both models, we use the JumpReLU SAEs from Gemma Scope² (Lieberum et al., 2024). We adopt the canonical variant, with 65K features per layer for Gemma 2 2B and 131K for Gemma 2 9B. These SAEs are trained mainly on English data using each model’s FFN outputs across all layers.

4.2 Datasets

To identify and analyze language-specific features, we focus on 15 languages across three categories: (1) 8 languages explicitly supported by Llama 3.2; (2) 4 languages where the primary model performs reasonably well³: Bulgarian (bg), Russian (ru), Turkish (tr), and Vietnamese (vi); and (3) 3 languages for broader typological coverage: Japanese (ja), Korean (ko), and Chinese (zh).

To create a language-specific corpus, we use training sets from three datasets: FLORES+ (NLLB Team et al., 2024), PAWS-X (Yang et al., 2019), and XNLI (Conneau et al., 2018). This yields multilingual corpora with 1,000 to 3,000 examples (approximately 40,000 to 120,000 tokens) per language.

4.3 Interpreting Language-Specific Features

To interpret language-specific features across layers, we use automated interpretability tools from Paulo et al. (2025). Specifically, we ask an LLM to generate an explanation of a given feature based on a set of its activating examples. Then, another LLM is tasked with classifying whether the feature is active in new examples based on that explanation. We then report the classification accuracy. Please refer to Appendix A.7 for more details on this evaluation.

We apply two scoring methods: detection scoring, which evaluates the entire sentence, and fuzzing scoring, which focuses only on randomly selected highlighted words within the sentence. Both methods have a random baseline accuracy of 50%

²<https://huggingface.co/google/gemma-scope>

³Over 40% accuracy on the XNLI task, comparable to the explicitly supported ones (see Table 7 in Appendix A).

and a median accuracy of 70–80% across models, as reported by Paulo et al. (2025).

4.4 Experiment Design

We first identify language-specific features, then use them for steering and detection to provide evidence that they are truly language-specific.

Finding Language-Specific Features. We use *SAE-LAPE* to identify language-specific features by setting the hyperparameters described in Section 3.1 to $N = 98$, so that language-specific features are active for the majority of examples, and $T = 50$, to ensure that no other language has an activation probability close to half of the maximum, thus reinforcing their language specificity. We then analyze both their properties and interpretations.

Steering Language-Specific Features. We conduct two experiments:

1. **Perplexity change:** We evaluate multilingual modeling capability by measuring perplexity (PPL) on the FLORES+ test set. We then compare the PPL of the model steered using the identified features with that of the language-specific neuron baseline to assess their influence.
2. **Unconditional text generation:** We steer the identified features without an input prompt (i.e., only a [BOS] token) 100 times, each with a random seed from 0 to 99, to observe their effect on the model’s output. In addition, we use Llama 3.2 default decoding strategy: random sampling with temperature = 0.6 and top_p = 0.9.

LID with Language-Specific Features. We evaluate whether the identified language-specific features can accurately detect the language of a given input and compare their performance against the language-specific neuron baseline and fastText (Joulin et al., 2017). For this purpose, we utilize the WiLI-2018 dataset (Thoma, 2018) to identify the analyzed 15 languages.

5 Results and Discussion

5.1 Finding Language-Specific Features

In total, using *SAE-LAPE*, we identified 540 language-specific features out of two million across layers in Llama 3.2 1B. Figure 2 shows that most features are concentrated in the middle to final layers⁴, with similar trends observed in Gemma 2 (see Appendix A.15.3).

⁴This distribution resembles that of the language-specific neurons shown in Figure 11 in Appendix A.

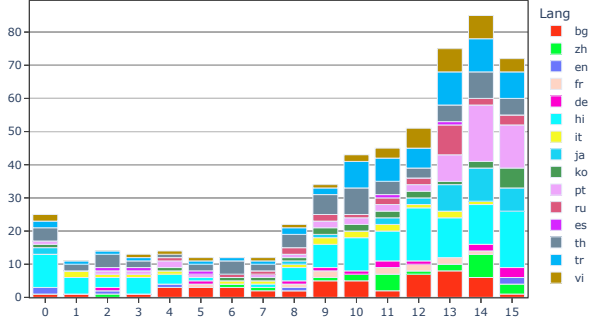


Figure 2: Distribution of the identified language-specific features with *SAE-LAPE* across layers of Llama 3.2 1B.

In addition, language-shared features across 2–8 languages (approximately half of the analyzed languages) exhibit a pattern consistent with that of language-specific features, as shown in Figure 7 in Appendix A. However, for features shared across 9–15 languages, as shown in Figure 8 in Appendix A, a contrasting pattern emerges, with these features are concentrated in the early to middle layers. This phenomenon supports Kojima et al.; Tang et al.’s (2024; 2024) view that the early layers of decoder-based LLMs handle cross-lingual transfer by converting lexical and syntactic inputs into language-independent semantic representations, the middle layers process these representations, and the final layers translate them back to the target language’s syntax and lexicon. See Appendix A.8 for details on language-shared features.

Further supporting this view, a logit lens-inspired analysis (nostalgabraist, 2020), which involved multiplying language-specific features with the token unembedding matrix, shows that features from shallower layers tend to map to more generic tokens (often in English or mixed-language) whereas features from deeper layers increasingly correspond to language-specific tokens (see Figures 34–42 in Appendix A). This progressive specialization suggests linguistic information becomes increasingly language-specific as model depth increases. This method is detailed in Appendix A.9.

To better understand the nature of these features, we analyze their properties. We observe that language-specific features tend to co-occur naturally and frequently within a specific language, and some exhibit moderate to high Pearson correlation in their activation values (see Figures 43–45 and Figures 46–48 in Appendix A). These properties enable the simultaneous steering of multiple language-specific features, as this approach aligns with their natural co-activation patterns.

5.1.1 Feature Interpretation Scores

Based on the automated evaluation for interpretable features proposed by Paulo et al. (2025), our SAE features are interpretable, as they have median accuracies of 0.75 and 0.76 for detection and fuzzing scoring, respectively, similar to the reported ones. Moreover, they achieve average accuracy, F1, precision, and recall scores of 0.74, 0.70, 0.85, and 0.64, respectively, across languages for both tasks (see Table 8 in Appendix A). This indicates that most of the generated explanations are faithful, as they can be used to determine whether the given sentences or tokens activate the corresponding features.

Moreover, we find that interpretation scores are moderately negatively correlated with entropy, indicating that more specific features are generally more interpretable (see Figure 49–50 in Appendix A). These scores also tend to have higher precision than recall because the explanations focus on top activating examples but are evaluated on a broader set, including weaker and non-activating ones (see Appendix A.7 for details). As a result, the explanations mainly reflect top examples from one or a few languages but miss weaker ones, which lowers recall.

5.1.2 Feature Interpretations

While many linguistic concepts from earlier layers remain in later layers, we focus on the distinctive aspects that emerge in deeper layers. A consistent, overall pattern observed across languages is a hierarchical processing of linguistic information, where the model builds understanding progressively. In earlier layers, the model’s features concentrate on foundational, low-level elements such as morphemes (e.g., suffixes, affixes, and word stems) as well as basic function words and punctuation. Progressing to middle layers, these features begin to integrate these basic components into larger, more cohesive units, such as multi-word phrases and clauses that convey specific actions, states, or relationships. Finally, in deeper layers, the model demonstrates a more abstract and sophisticated comprehension, with features that capture complex sentence-level structures, discourse patterns, pragmatic elements like conversational flow, and nuanced, context-specific meanings. See Appendix A.10 for language-specific interpretations.

Notably, we find highly specialized features that capture stylistic patterns in English. As shown in Figure 51–52 in Appendix A, English features in layer 15 can distinguish between text styles, with feature 67343 identifying conversational and narra-

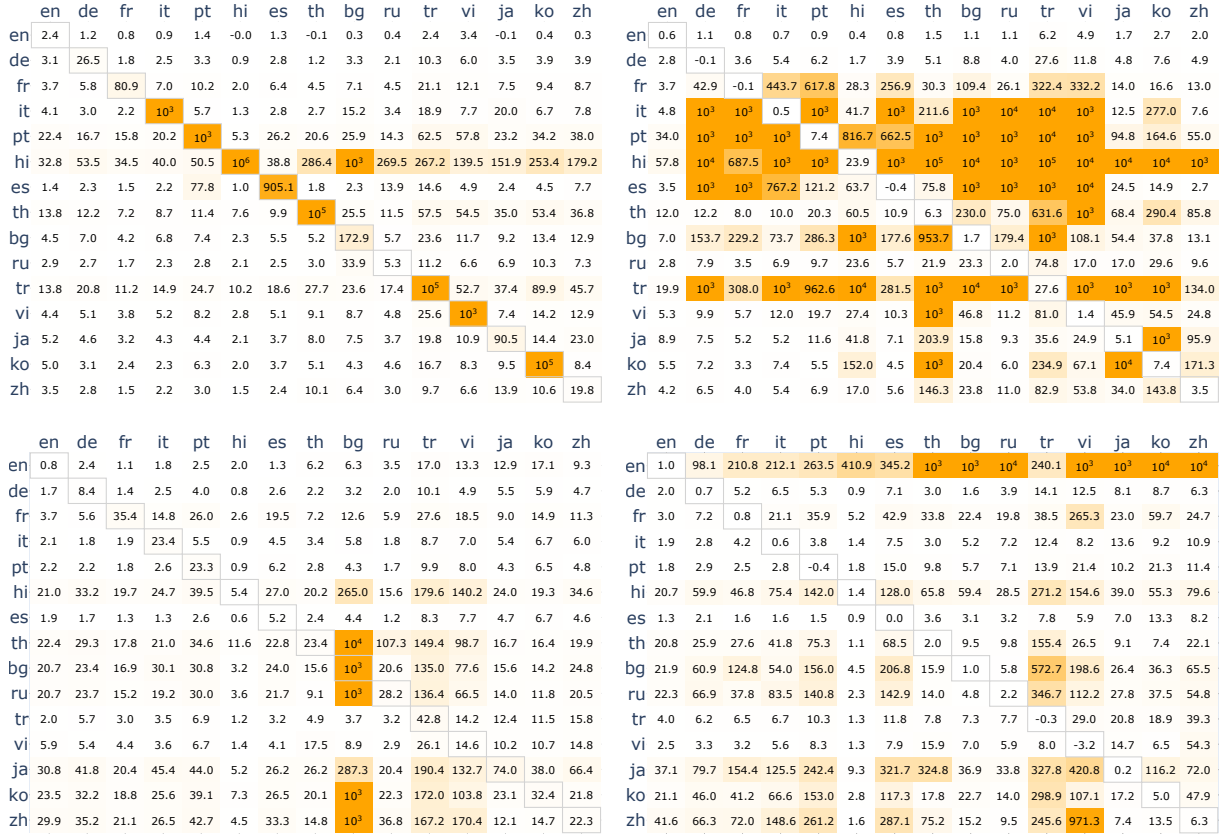


Figure 3: PPL changes in Llama 3.2 1B. **Top:** Steering using **language-specific features** with $\alpha = -0.2$ (left) and $\alpha = 0.2$ (right). **Bottom:** Steering using **language-specific neurons** with $\alpha = -0.2$ (left) and $\alpha = 0.2$ (right). Each cell (i, j) shows the PPL change for language j after intervening on language i. The heatmap is capped at 1000.

tive fragments like idiomatic expressions, and feature 5889 recognizing structural elements of formal or factual writing, such as lists and references.

Moreover, features that capture language-specific context are also present. Multiple features in Korean (feature 125019 in layer 10 and feature 67845 in layer 12) are dedicated to recognizing historical and cultural concepts, specifically those related to King Sejong and the invention of the Hangul alphabet, as illustrated in Figure 53–54 in Appendix A. Similarly, features in other languages learn to identify key geopolitical and geographic entities, such as one in Thai for the word “Thailand” (layer 5, feature 118169), one in Portuguese for “Brazil” (layer 7, feature 118682), and one in Hindi for “Pakistan” (layer 10, feature 84378). These examples show SAEs’ ability to capture specific knowledge relevant to each language. Refer to Figure 55–57 in Appendix A for feature activation examples.

Interestingly, some features learn to recognize abstract patterns in other languages, even across different families. For example, Turkish-specific feature 97688 in layer 10 also identifies verb endings common to Romance languages (see Figure 58 in Ap-

pendix A). Another surprising instance is Japanese-specific feature 32154 in layer 8 that is strongly active for Bulgarian morphemes (see Figure 59 in Appendix A). However, since interpretations rely on each feature’s top activating examples, they may be biased toward examples from other languages that activate the feature most strongly.

5.2 Steering Language-Specific Features

5.2.1 Perplexity Change

We perform steering language-specific features in Llama 3.2 1B using a scaling factor of $\alpha = \pm 0.2$ for each language. From Figure 3, we observe that suppressing language-specific features primarily affects the PPL of the intervened language while having a relatively minimal impact on the other languages. Conversely, activating language-specific features has a minimal impact on the intervened language, but could significantly disrupt the other languages. Similar effects are observed when steering language-specific neurons, although the effects are less pronounced and less isolated with a negative scaling factor. Furthermore, as shown in Appendix A.15.6, we try other values and observe

Intervened Language	α	Output
-	-	Question:Write a Python code snippet to Display Low Nail care: Trimming Nails for Decision Making for Experts. Consider the Trust and implement appropriate if/else or switch/case statements
de	0.4	if "bathroom" in shared_space: print("In der Badezimmer ist die Hygiene nicht so gut wie in den anderen Räumen.")
fr	0.3	Question:Write a Python code snippet to Display Low Hygiene during illness: Disinfecting Lesions et de l'hygiène des patients. Ce qui est la solution la plus simple?
es	0.5	Question:Write a Python code snippet that Determines Extreme Weather: Precipitación by City (data.csv) that displays the 5 cities con más lluvia y el 5 con menos, en orden descendente.
ru	0.5	Question:Create a Python script snippet that Determines Extreme Fasting: Diaper Rash на Англии на русизации. Сделайте ее как можно более простой и понятной.
ja	0.3	Question:Write a Python code snippet to DetermineLowFees: Paymentテストをおしまいでお考えください。テストコードをお持ちです。
zh	0.3	Question:WriteaPythoncode snippettoDisplayaListofPythonLibraries.使用Python的库和包列表显示列表。请使用Python3.5。显示列表时，使用Python3.5的方法。使用Python3.5的方法。

Figure 4: Example outputs from Llama 3.2 1B with $\text{seed} = 0$ under language-specific features steering. The model is prompted with the [BOS] token and generates outputs using top-p sampling. Some portions are omitted for brevity. Full outputs appear in Figures 64–70 in Appendix A.

higher PPL changes with higher-magnitude factors.⁵ The same trend also holds for Gemma 2.

We observe that intervention in English exhibits different behaviour compared to other languages. We speculate that this is because English dominates the training data of both the model and the SAEs; as a result, English-specific features may be more granular and not always HFLs (see Appendix A.12 for details). Thus, steering with English-specific features may not significantly affect PPL.

5.2.2 Unconditional Text Generation

Figure 4 shows that steering with language-specific features can guide the model to generate text in the target language. Interestingly, when generating code, the model correctly switches between using the target language for descriptive text (e.g., comments or string literals) and English for syntax, as shown in Figures 71–73 in Appendix A.

However, we acknowledge that the intervention is not always successful and sometimes leads to degraded outputs. To quantitatively evaluate the results, we prompt an LLM with a rubric assessing two criteria: language change occurrence (no change, change) and text coherence (incoherent, partially coherent, coherent). See Appendix A.13 for rubric details.

Figure 5 demonstrates that manipulating language-specific features enhances the likelihood of generating text in the target language during

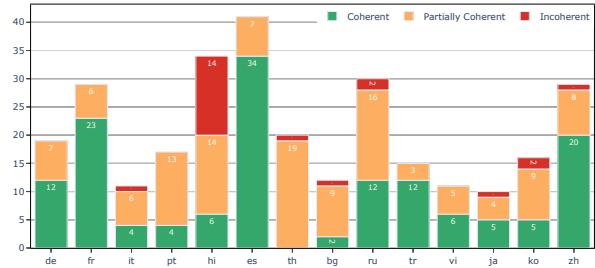


Figure 5: Changed text counts in Llama 3.2 1B per language with various scaling factors α . See Table 9 in Appendix A for details.

unconditional text generation. Unfortunately, we observe that further increasing the scaling factor, while increasing this likelihood, can lead to degenerate outputs, specifically a rise in incoherent and partially coherent generations, as shown in Table 9–10 and Figures 75–77 in Appendix A. This suggests that we may need to develop better steering strategies for language-specific features, such as diminishing or dynamic steering (Scalena et al., 2024) or filtering steering features (Arad et al., 2025). We leave this for future work.

5.3 LID with Language-Specific Features

The activation of language-specific features mostly for one language suggests their potential as a language classification feature. For Llama 3.2 1B, Figure 6 shows the performance of the SAE-based LID is slightly below fastText (Joulin et al., 2017), while the neuron-based LID performs poorly. For Gemma 2, the results are more mixed: SAE-based and neuron-based LID are broadly comparable to

⁵In contrast, activating and suppressing language-shared features across 15 languages have similar effects on PPL, as shown in Table 1 in Appendix A.

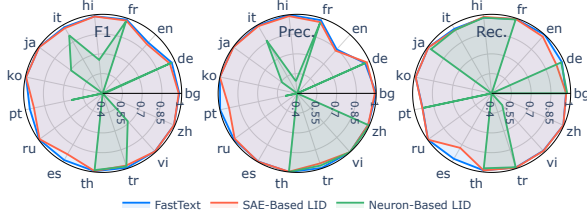


Figure 6: Results of SAE-based and neuron-based LID derived from Llama 3.2 1B, and fastText LID, applied to WiLI-2018 across 15 languages. Details are provided in Table 11 in Appendix A.

fastText overall, but exhibit weaker performance on certain languages (see Appendix A.15.7).

The SAE-based LID, however, offers more interpretability by showing which features activate on specific tokens in a given input, along with the interpretations of those features. To demonstrate this, we analyze the following English sentence from WiLI-2018: “Hamilton has been the primary songwriter, guitarist, and vocalist for Brothers Past, as well as co-producer for all of their recorded releases.” See Figure 78 in Appendix A for the full feature activation details. The lower layers (0–2) primarily recognize English function words, such as “has” and “been” (auxiliary verbs), “the” (article), “for” and “of” (prepositions), “and” and “as well as” (conjunction), and even punctuation marks like the comma (.). In middle layers (4–8), the activated features correspond to English phrase structures and common word combinations. Most notably, feature 34138 in layer 4 activates on multi-word segments forming complete semantic units. Higher-level features in layer 15 show activation across almost the entire sentence, demonstrating recognition of overall English structure, including punctuation patterns such as the use of commas in lists.

Overall, these features help us understand which aspects of particular languages are detected across layers for a given input and why the classifier makes such predictions, highlighting the potential of using SAEs as a concept classifier. However, the current approach may be biased toward languages with more features. This bias is particularly evident in mixed-language inputs, as shown in the misclassification case in Appendix A.14, and is also reflected in the results for Gemma 2 in Appendix A.15.7. We leave this for future work.

6 Related Work

Language-specific neurons. Multilingual LLM analyses often treat individual neurons as the unit

of analysis. Tang et al. (2024) proposed *language activation probability entropy* (LAPE) to detect language-specific neurons in the FFN activations, showing their impact on multilingual capabilities. Kojima et al. (2024) showed that identifying and manipulating such neurons in decoder LLMs can control the output language. Wang et al. (2024) found neuron activation patterns to vary across languages, shaped by task and example characteristics.

Language-shared and specific features. Few studies have explored the possible existence of such features in SAEs. Brinkmann et al. (2025) showed that abstract grammatical concepts are often captured by features shared among various languages. Jing et al. (2025) used counterfactual datasets to extract, evaluate, and manipulate linguistic features across four dimensions: morphology, syntax, semantics, and pragmatics. Deng et al. (2025) and Chou et al. (2025) proposed methods for identifying language-specific features in the residual stream using mean activation differences, following an approach similar to Soo et al. (2025). Both studies found that, at each layer, the top- k features ranked by mean activation differences between a target language and other languages are highly language-specific. Deng et al. (2025) further showed that ablating these features at selected layers substantially increases the cross-entropy loss for the target language while leaving other languages largely unaffected. In contrast, Chou et al. (2025) demonstrated that the causal effect of steering these features varies across layers and can control the output language. Unlike prior work, we target language-specific features that are also HFLs using *SAE-LAPE* in the FFN, analyze both their properties and interpretations, manipulate these features simultaneously across all layers, and compare the results with a language-specific neuron baseline using *LAPE*.

7 Conclusion

This study offers new insights into the internal representations and multilingual mechanisms of LLMs through language-specific features from SAEs. These features mainly appear in the middle to final layers and are interpretable. They also impact the model’s multilingual performance and output. Finally, the SAE-based classifier enables interpretable language identification, highlighting its potential for both interpretability and classification, with the latter possibly extending to tasks beyond language identification, such as safety.

Limitations

Language-Shared Features. This study primarily focuses on language-specific features. While our preliminary analysis revealed interesting properties of language-shared features, they warrant further investigation in future work.

Steering Methods. We used the most commonly adopted steering method for SAEs in this study. To enhance the efficacy of language-specific features in supporting the multilingual abilities of LLMs, future research could explore new steering strategies, such as diminishing steering, dynamic steering, or filtering steering features. This is important as steering multiple features simultaneously is not yet common practice and may be challenging.

SAE-Based Classifier. The current preliminary classifier is limited to language identification and may be biased toward languages with more features, which should be addressed to develop a more reliable classifier. Enhancing the use of SAEs as an interpretable classifier remains a promising direction for detecting specific concepts in LLMs, including tasks beyond language identification, such as safety monitoring.

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A Appendix

A.1 Feed-Forward Network

The feed-forward network (FFN) module of LLMs with a Transformer architecture (Vaswani et al., 2017) can be described as follows:

$$\mathbf{h} = \sigma(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1),$$

$$\text{FFN}(\mathbf{x}) = \mathbf{W}_2 \mathbf{h} + \mathbf{b}_2,$$

where $\mathbf{h}$ is the intermediate activations; $\mathbf{x} \in \mathbb{R}^d$ is the pre-FFN representation from the preceding attention sub-layer; σ is an activation function, such as $\text{ReLU}(x) = \max(0, x)$ by Glorot et al. (2011); $\mathbf{W}_1 \in \mathbb{R}^{4d \times d}$ and $\mathbf{W}_2 \in \mathbb{R}^{d \times 4d}$ are weight matrices; and $\mathbf{b}_1 \in \mathbb{R}^{4d}$ and $\mathbf{b}_2 \in \mathbb{R}^d$ are bias terms. More recently, Llama 3.2 uses Gated Linear Unit (GLU) by Shazeer (2020) as its activation function for improving performance, and the FFN module operation is defined as follows:

$$\text{FFN}(\mathbf{x}) = \mathbf{W}_2 (\mathbf{h} \otimes (\mathbf{V} \mathbf{x} + \mathbf{b}_v)) + \mathbf{b}_2,$$

where $\otimes$ denotes the element-wise product; $\mathbf{V} \in \mathbb{R}^{4d \times d}$ is a weight matrix; and $\mathbf{b}_v \in \mathbb{R}^{4d}$ is a bias term. In both cases, a neuron corresponds to a linear transformation applied to a single row of $\mathbf{W}_1$, followed by a non-linear activation that yields a component $\mathbf{h}_i$ of the intermediate representation. Hence, an FFN module at one layer has $4d$ neurons. Tang et al. (2024) consider a neuron to be activated when its value exceeds zero and to be deactivated when its value is set to zero.

A.2 Finding Language-Specific Neurons

To identify language-specific neurons in the intermediate activations of the FFN (see Appendix A.1), we adopt the *language activation probability entropy (LAPE)* method proposed by Tang et al. (2024). Specifically, we provide an LLM with multilingual corpora and compute the activation probability of neuron j at layer i of an LLM for each language k as follows:

$$P_{i,j}^k = \mathbb{E}(\mathbb{I}(\mathbf{h}_j^i > 0) \mid \text{language } k\text{'s corpus}),$$

where $\mathbb{E}$ is the expectation computed over the tokens in language k 's corpus; $\mathbb{I}$ is the indicator function; and $\mathbf{h}_j^i$ is the value associated with neuron j at layer i . The subsequent operations follow those described in Section 3.1. Moreover, Tang et al. (2024) imposed a predefined minimum threshold at the

95th percentile⁶ of all activation probabilities on $P_{i,j}^k$ to exclude negligible activation probabilities and used the remaining values to determine which languages the neurons are specific to. Specifically, they selected only the bottom 1% of neurons, sorted by LAPE scores, as language-specific neurons. This means that even if neurons are shared across multiple languages, they are still considered specific to those languages as long as their LAPE scores are low.

A.3 Steering Language-Specific Neurons

Inspired by Turner et al. (2024), and to align with the approach used for steering language-specific features, we scale the activation contribution of language-specific neurons as follows:

$$\mathbf{x}_{\text{new}} = \mathbf{x} + \alpha \mathbf{h}_j^{\max} \mathbf{d}^j,$$

where $\alpha \in \mathbb{R}$ is a scaling factor, $\mathbf{h}_j^{\max}$ denotes the maximum activation value of neuron j observed in the multilingual corpora, and $\mathbf{d}^j = \mathbf{W}_2[:, j] \in \mathbb{R}^d$ is the steering vector associated with neuron j .

A.4 LID with Language-Specific Neurons

To classify languages using neurons for a text input with FFN output $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_t]$ consisting of t tokens, we compute the score for each language k as follows:

$$s_k = \sum_{i \in \mathcal{L}} \sum_{\mathbf{x} \in \mathbf{X}} \sum_{j \in \mathcal{H}_k^i} \mathbb{I}(\mathbf{h}_{x,j}^i > 0),$$

where $\mathcal{L}$ is the set of all layers; $\mathcal{H}_k^i$ is the set of neuron indices associated with language k at layer i ; $\mathbf{h}_{x,j}^i$ is the activation value of neuron j at layer i ; and $\mathbb{I}(\mathbf{h}_{x,j}^i > 0)$ is the indicator function to ensure that only active neurons are considered. Then, we predict the language by selecting the index of the highest score in the vector $\mathbf{s} = [s_1, \dots, s_l]$ over l languages.

A.5 TopK SAEs Configuration

We use TopK SAEs by EleutherAI (2024) with $k = 32$ and an expansion factor of 64, such that $z \in \mathbb{R}^{64d}$. These SAEs are trained on the RedPajama-v2 (Weber et al., 2024) multilingual

⁶The 95th percentile of all activation probabilities reported by Tang et al. (2024) for Llama 2 70B is 0.505. However, for SAEs, we find it to be 0.0007 due to their sparsity. Thus, this threshold is not suitable for SAEs, as it excludes almost nothing. To overcome this, we further applied two filters to exclude features with low activation probabilities for particular languages.

corpus. RedPajama-v2 primarily consists of documents in English, German, French, Spanish, and Italian. Specifically, [EleutherAI \(2024\)](#) use the Sample-10B subset of this dataset (see Table 4 for the language distribution) and train their SAEs on the FFN outputs with $d = 2048$ across all layers. Furthermore, the specific configuration of the SAEs and their training process is described below.

A.5.1 SAE Architecture

The SAEs are configured with the following core parameters:

- **Expansion Factor:** The expansion factor is set to 64, determining the number of SAE features relative to the input dimension. Given the FFN output dimension of $d = 2048$, each SAE learns $64 \times 2048 = 131,072$ features.
- **Activation Function:** A TopK activation function is employed with $k = 32$, meaning that only the 32 highest-activated features are retained as non-zero for any given input.
- **Decoder Normalization:** Normalization is applied to the decoder weights, enforcing a unit norm.

A.5.2 Training Configuration

The training procedure is governed by the following settings:

- **Hookpoints:** SAEs are trained on the outputs of the FFN blocks from `layers.0.mlp` to `layers.15.mlp`, resulting in one SAE per layer (16 total).
- **Batching:** A batch size of 4 sequences is used, and gradients are accumulated across 2 steps, yielding an effective batch size of 8.
- **Learning Rate:** The learning rate is automatically determined based on the number of SAE features. A warm-up period of 1000 steps is applied.
- **Dead Feature Threshold:** Features that remain inactive (i.e., not firing) for 10,000,000 consecutive tokens are considered “dead”.
- **Context Length:** The SAEs are trained on sequences with a context length of 2048 tokens.
- **Initialization:** A fixed random seed of 0 is used to ensure reproducibility, initializing one SAE per hookpoint.

A.6 SAE Feature Interpretability

SAE features can be interpreted either manually by humans or automatically by LLMs ([Huben et al., 2024](#); [Bricken et al., 2023](#)). Manual interpreta-

tion involves analyzing a feature’s activating contexts (i.e., sentences) and generating an explanation, which is then evaluated using a predefined rubric. In contrast, automatic interpretation involves presenting these contexts to LLMs with predefined instructions, which generate an explanation that is also scored by LLMs. This automatic method was inspired by [Bills et al. \(2023\)](#), who used LLMs to interpret neurons, and was further improved by [Paulo et al. \(2025\)](#) to interpret SAE features.

A.7 Automated Interpretability Process

We use an automated interpretability pipeline by [Paulo et al. \(2025\)](#) to efficiently explain and evaluate language-specific features in Llama 3.2 1B’s SAEs. The process involves four key steps:

- **Activation Collection:** We gather activations from diverse multilingual corpora (XNLI, PAWS-X, and FLORES+) spanning 15 languages, caching contexts of 256 tokens for efficient processing.
- **Interpretation Generation:** For each language-specific feature, we prompt GPT-4.1 ([OpenAI, 2025](#)) with 40 top-activating examples, each centered on the activating token with 32 tokens of surrounding context.
- **Evaluation Methods:** We assess interpretation quality through detection scoring and fuzzing scoring, each using 100 examples (50 stratified by activation deciles and 50 non-activating). In detection scoring, the model determines whether any token in the full sentence activates the feature based on its explanation. In fuzzing scoring, the model performs the same evaluation but only over randomly selected highlighted words within the sentence.
- **Results:** This automated pipeline enabled us to interpret 540 language-specific features for approximately \$50: \$5 for generating explanations and \$45 for scoring them.

A.8 Language-Shared Features

Using *SAE-LAPE*, we identified 503 language-shared features from a total of 2 million across layers. Features shared across 2 to 8 languages (roughly half of those analyzed) display a pattern similar to that of language-specific features in Subsection 5.1, as illustrated in Figure 7. In contrast, features shared across 9 to 15 languages show a different trend, being concentrated in the early to the middle layers (see Figure 8).

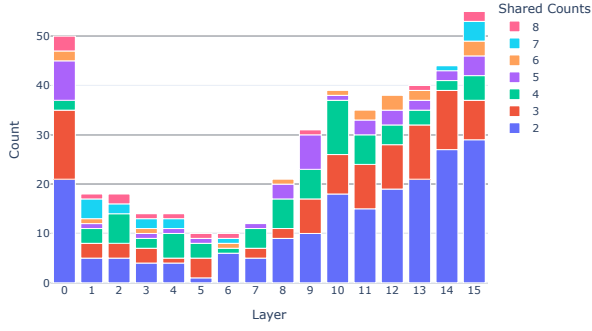


Figure 7: Distribution of the identified language-shared features in Llama 3.2 1B among 2 to 8 languages with SAE-LAPE across layers.

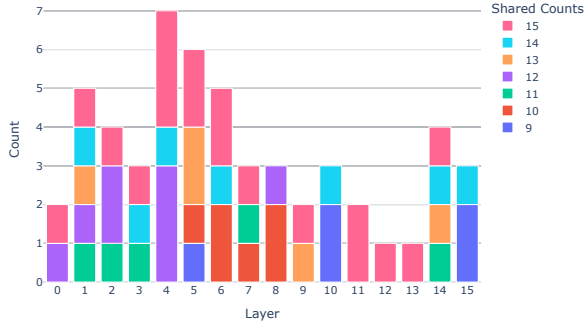


Figure 8: Distribution of the identified language-shared features in Llama 3.2 1B among 9 to 15 languages with SAE-LAPE across layers.

We also observe the intersection of language-shared features across languages. Figure 9 shows the feature intersection counts, while Figure 10 presents the Jaccard similarity (i.e., intersection over union) of the corresponding features. Remarkably, although the number of language-shared features varies across languages, the Jaccard similarity can indicate the relatedness between them to a certain extent, as languages with a similarity score close to one tend to be more closely related, as they share a greater number of concepts associated with languages.

Notably, English is most closely related to German, which aligns with their common origin in the Germanic language family. Interestingly, English also exhibits moderate similarity scores with Romance languages such as French, Italian, Portuguese, and Spanish, likely due to significant Latin influence on English vocabulary (Bryson, 2015). Within the Romance group, we observe particularly high similarity scores to each other, indicating a large number of shared concepts and a strong degree of mutual lexical and grammatical overlap. A similar trend is found in the Slavic language group, where Russian and Bulgarian show a high similar-

	en	de	fr	it	pt	hi	es	th	bg	ru	tr	vi	ja	ko	zh
en	44	39	41	37	39	24	41	24	26	33	33	35	31	33	31
de	39	71	66	66	65	35	67	40	51	56	59	55	47	47	38
fr	41	66	86	82	81	32	84	41	51	56	61	59	46	47	38
it	37	66	82	88	80	32	83	39	54	55	60	58	48	46	37
pt	39	65	81	80	90	31	88	39	50	55	60	58	45	45	37
hi	24	35	32	32	31	185	32	103	60	55	100	52	110	132	57
es	41	67	84	83	88	32	94	40	51	56	62	59	46	46	38
th	24	40	41	39	39	103	40	213	90	81	57	109	90	116	80
bg	26	51	51	54	50	60	51	90	178	144	62	62	52	61	42
ru	33	56	56	55	55	55	56	81	144	167	62	74	57	73	53
tr	33	59	61	60	60	100	62	57	62	62	159	70	111	122	55
vi	35	55	59	58	58	52	59	109	62	74	70	161	73	88	89
ja	31	47	46	48	45	110	46	90	52	57	111	73	224	195	119
ko	33	47	47	46	45	132	46	116	61	73	122	88	195	252	111
zh	31	38	38	37	37	57	38	80	42	53	55	89	119	111	155

Figure 9: Feature intersection counts of the identified language-shared features in Llama 3.2 1B across layers for each language.

	en	de	fr	it	pt	hi	es	th	bg	ru	tr	vi	ja	ko	zh
en	1	0.51	0.46	0.39	0.41	0.12	0.42	0.1	0.13	0.19	0.19	0.21	0.13	0.13	0.18
de	0.51	1	0.73	0.71	0.68	0.16	0.68	0.16	0.26	0.31	0.35	0.31	0.19	0.17	0.2
fr	0.46	0.73	1	0.89	0.85	0.13	0.88	0.16	0.24	0.28	0.33	0.31	0.17	0.16	0.19
it	0.39	0.71	0.89	1	0.82	0.13	0.84	0.15	0.25	0.28	0.32	0.3	0.18	0.16	0.18
pt	0.41	0.68	0.85	0.82	1	0.13	0.92	0.15	0.23	0.27	0.32	0.3	0.17	0.15	0.18
hi	0.12	0.16	0.13	0.13	0.13	1	0.13	0.35	0.2	0.19	0.41	0.18	0.37	0.43	0.2
es	0.42	0.68	0.88	0.84	0.92	0.13	1	0.15	0.23	0.27	0.32	0.3	0.17	0.15	0.18
th	0.1	0.16	0.16	0.15	0.15	0.35	0.15	1	0.3	0.27	0.18	0.41	0.26	0.33	0.28
bg	0.13	0.26	0.24	0.25	0.23	0.2	0.23	0.3	1	0.72	0.23	0.22	0.15	0.17	0.14
ru	0.19	0.31	0.28	0.28	0.27	0.19	0.27	0.27	0.72	1	0.23	0.29	0.17	0.21	0.2
tr	0.19	0.35	0.33	0.32	0.32	0.41	0.32	0.18	0.23	0.23	1	0.28	0.41	0.42	0.21
vi	0.21	0.31	0.31	0.3	0.3	0.18	0.3	0.41	0.22	0.29	0.28	1	0.23	0.27	0.39
ja	0.13	0.19	0.17	0.18	0.17	0.37	0.17	0.26	0.15	0.17	0.41	0.23	1	0.69	0.46
ko	0.13	0.17	0.16	0.16	0.15	0.43	0.15	0.33	0.17	0.21	0.42	0.27	0.69	1	0.38
zh	0.18	0.2	0.19	0.18	0.18	0.2	0.18	0.28	0.14	0.2	0.21	0.39	0.46	0.38	1

Figure 10: Jaccard similarity of identified language-shared features in Llama 3.2 1B across layers for each language.

ity score. Finally, East Asian languages such as Japanese, Korean, and Chinese exhibit moderate similarity scores with one another and may reflect historical and cultural influences, such as the borrowing of Chinese characters and vocabulary, contributing to a degree of conceptual overlap among these languages (Bryson, 2015).

Lastly, we conduct a PPL change experiment similar to the one described in Section 5.2.1, this time using language-shared features. Specifically, we perform steering using features shared among all 15 languages analyzed in this study, with $\alpha = \pm 0.2, \pm 0.3, \pm 0.4$. As shown in Table 1, these language-shared features consistently affect PPL evenly across languages when suppressed, leading to a greater increase in PPL. Interestingly, they also exhibit a similar effect when activated, in contrast to the behavior of language-specific features, which typically have minimal impact on their correspond-

ing language, as shown in Figure 3 in Section 5.

Language	Normal	Suppressed			Activated		
		$\alpha=-0.2$	$\alpha=-0.3$	$\alpha=-0.4$	$\alpha=0.2$	$\alpha=0.3$	$\alpha=0.4$
en	40.09	40.98	42.15	44.06	40.97	42.35	44.98
de	23.99	24.87	26.00	27.86	24.85	26.26	29.14
fr	18.37	18.71	19.26	20.21	19.04	19.93	21.55
it	20.55	21.18	22.01	23.43	21.21	22.24	24.24
pt	28.32	29.08	30.15	32.00	29.23	30.64	33.34
hi	9.92	10.19	10.57	11.23	10.31	10.94	12.31
es	25.24	25.74	26.53	27.91	26.16	27.50	30.09
th	16.14	16.78	17.71	19.32	16.99	18.42	21.56
bg	21.35	22.84	24.41	26.92	21.75	23.03	25.69
ru	20.80	21.79	22.86	24.60	21.24	22.25	24.42
tr	61.50	65.41	69.91	77.42	64.02	69.25	80.50
vi	44.93	46.14	48.00	51.74	46.72	49.40	54.76
ja	32.67	34.62	36.64	39.87	33.53	35.86	41.36
ko	40.27	42.60	45.05	48.93	41.88	45.51	54.11
zh	49.43	53.29	56.72	61.77	49.21	51.20	56.18

Table 1: PPL changes in Llama 3.2 1B across languages with features shared among all 15 languages in this study under various scaling factors α .

A.9 Top Tokens Promoted by Language-Specific Features

To identify which tokens are emphasized by language-specific features, we adopt a method inspired by the logit lens technique proposed by [nostalgebraist \(2020\)](#). Originally, the logit lens was introduced as a simple, although partial, tool for interpreting the intermediate residual stream. It works by multiplying the intermediate activations $h^i \in \mathbb{R}^d$ at layer i with the unembedding matrix $\mathbf{W}_U \in \mathbb{R}^{|\mathcal{V}| \times d}$, where $|\mathcal{V}|$ is the vocabulary size. The operation is defined as follows:

$$L^i = \text{softmax}(\mathbf{W}_U h^i).$$

When this operation is applied, the resulting token probability distributions L^i tend to be intuitively meaningful. Notably, these distributions often resemble the model’s final output distribution and begin to align with it progressively over the course of the network’s layers, frequently approaching it well before the final layer. This behavior helps track how the model gradually refines its predictions throughout the forward pass.

However, this method requires a text input so that the distribution resulting from the intermediate residual stream is meaningful. Applying this method directly to language-specific features would require identifying the activating inputs for each of these features, which is time-consuming and makes it difficult to isolate the effect of language-specific features themselves. Therefore, as described by [Arad et al. \(2025\)](#); [Bloom and Lin \(2024\)](#), to simplify our analysis, we directly multiply the features

with the token unembedding matrix as follows:

$$L^i = \text{softmax}(\mathbf{W}_U d^{i,s}),$$

where $d^{i,s}$ is the direction of a language-specific feature at layer i . Essentially, this is equivalent to applying the logit lens to a single token, but without the need to determine the corresponding token.

This method is possible due to the way the intermediate residual stream is computed:

$$h^i = x^i + \text{FFN}^i(x^i),$$

$$h^i = x^i + \hat{\text{FFN}}^i(x^i) + e,$$

$$h^i = x^i + \sum_{j \in \mathcal{T}_K} \mathbf{z}_j^i d^{i,j} + \mathbf{b}_{\text{dec}}^i + e,$$

where $h^i \in \mathbb{R}^d$ is the intermediate activations at layer i ; $x^i \in \mathbb{R}^d$ is the pre-FFN representation at layer i ; $\text{FFN}^i(x^i)$ is the FFN output at layer i ; $\hat{\text{FFN}}^i(x^i)$ is the FFN output reconstruction by an SAE at layer i ; $e \in \mathbb{R}^d$ is the reconstruction error; $\mathcal{T}_K \subseteq \{0, \dots, n-1\}$ is the indices of the top- k values; $\mathbf{z}_j^i$ is the activation of feature j at layer i ; $d^{i,j} = \mathbf{W}_{\text{dec}}^i[:, j] \in \mathbb{R}^d$ is the direction of feature j at layer i ; and $\mathbf{b}_{\text{dec}}^i$ is the bias vector for an SAE at layer i . Then, for a language-specific feature $d^{i,s}$, we can decompose the token probability distribution L^i into:

$$L^i = \text{softmax}(\mathbf{W}_U h^i),$$

$$L^i = \text{softmax}(\mathbf{W}_U (x^i + \mathbf{z}_s^i d^{i,s} + \mathbf{b}_{\text{dec}}^i + e)),$$

$$L^i \propto \exp(\mathbf{W}_U \mathbf{z}_s^i d^{i,s}),$$

$$L^i \propto \text{softmax}(\mathbf{W}_U d^{i,s}) \text{ when } \mathbf{z}_s^i = 1.$$

Here, $\mathbf{W}_U x^i$ can be regarded as a constant value for a particular token, $\mathbf{b}_{\text{dec}}^i$ is the fixed bias vector, and e is assumed to be small. Thus, based on this proportionality, $\text{softmax}(\mathbf{W}_U d^{i,s})$ can be interpreted as the contribution of a language-specific feature $d^{i,s}$ to the token probability distribution L^i .

By using this method, we found that features from shallower layers are associated with generic or mixed-language tokens, while deeper-layer features increasingly promoted coherent, language-specific vocabulary and complex morphological patterns. This pattern implies that deeper layers of the model capture increasingly language-specific linguistic concepts. See Figures 34–42 for the top 10 tokens promoted by all language-specific features.

A.10 Language-Specific Feature Interpretations

This section provides a detailed summary of language-specific feature interpretations across fifteen languages and highlights emerging patterns. While many feature interpretations identified in earlier layers persist in later ones, we focus on aspects that stand out in deeper layers. It is also worth noting that Llama 3.2 1B does not include complete token coverage for all languages in its tokenizer, which particularly affects morphologically rich or non-Latin script languages such as Korean, Japanese, and Chinese.

A.10.1 Spanish-Specific Features

A discernible pattern emerges in the Spanish interpretations, suggesting a hierarchical processing of linguistic information within the model. Early layers (layers 2, 3, and 5) tend to focus on more granular, morphological aspects of the language, such as morphemes (e.g., suffixes, affixes, and endings), word stems, and basic function words, emphasizing the constituent parts of words and their immediate grammatical roles (e.g., gender, number, and verb conjugation). As we move to the deeper layers (layers 11 and 13), the model appears to integrate these low-level concepts into a broader contextual and structural understanding. The focus shifts toward how words and morphemes combine to form phrases and contribute to sentence-level meaning. There is increased sensitivity to a wider range of function words that govern grammatical relationships (e.g., conjunctions, prepositions, pronouns), verb forms, phrase structures, and higher-order elements like punctuation and proper nouns. These deeper layers seem to capture the “connective tissue” of the language, which is essential for coherence, fluency, and the overall structure of sentences.

A.10.2 English-Specific Features

A more interesting pattern shows up in English, also suggesting a hierarchical pattern of linguistic processing. Early layers (layers 0 and 2) predominantly focus on low-level, foundational elements such as punctuation, function words, morphemes, and common short phrases, which are crucial for basic syntactic structure and token delimitation. Progressing to mid-layers (layers 4 and 8), the interpretations indicate a shift towards processing larger, multi-word phrases and clauses as cohesive semantic units that convey specific actions, states, relationships, and even context-dependent idiomatic expressions,

thereby starting to capture more complex meaning. Finally, interpretations from the deeper layers (layer 15) suggest the model comprehends higher-level linguistic phenomena. This includes identifying structural elements for enumerating information in lists, recognizing concepts specific to different text styles (formal or conversational), understanding discourse structures, and identifying pragmatic elements such as spoken language patterns and dialogue flow. This overall progression demonstrates an increasing sophistication from detecting basic building blocks of language to understanding nuanced semantic content and complex discourse-level characteristics.

A.10.3 French-Specific Features

The French interpretations reveal a consistent pattern of linguistic processing, progressing from foundational morphological elements in earlier layers to more complex syntactic and discourse-level structures in deeper layers. Initially, features in these earlier and middle layers (layers 5, 8, and 9) focus on identifying core word components such as morphemes, inflectional and derivational affixes, verb endings, and basic function words, crucial for word formation and elementary grammatical marking like tense or gender. As information propagates to the deeper layers (layers 11–14), the interpretations indicate an increasing capacity to recognize more abstract linguistic phenomena. This includes the identification of a broader range of function words (prepositions, conjunctions, pronouns, discourse markers), an understanding of phrase boundaries, syntactic relationships, sentence structuring, inter-clausal connections, and even the flow of conversation, suggesting a hierarchical abstraction of linguistic knowledge within the model.

A.10.4 German-Specific Features

The German interpretations also reveal a hierarchical processing pattern. Initial layers (layers 2 and 5) primarily focus on decomposing German words into fundamental morphological units, such as morphemes, word stems, and common inflectional endings. As information progresses to intermediate layers (layers 8–9), the model begins to integrate these morphological components into broader grammatical contexts, identifying function words, common collocations, and basic syntactic structures like phrase boundaries. Finally, in the deeper layers (layers 10, 11, 12, 14, and 15), the interpretations demonstrate a more sophisticated understanding, capturing complex syntactic dependen-

cies, multi-word expressions, phrasal patterns, and even specific grammatical constructs like verb-final clauses and subordinate clause markers, indicating a progression towards comprehending sentence-level structure and meaning.

A.10.5 Italian-Specific Features

The Italian interpretations demonstrate a progression from earlier to deeper layers. Initially, the earlier layers (layer 1), the focus is on identifying broad morphological components such as basic morphemes, affixes (prefixes and suffixes), roots, stems, and some function words, often related to word boundaries and fundamental grammatical constructs. As the analysis moves to early mid-layers (layers 2–4), there’s a refinement, with features capturing more specific linguistic elements like named entities, comparatives, diminutives, and sometimes broadening the context to common patterns in Romance or other European languages. Progressing further, late-mid layers (layers 6–12) exhibit increased sophistication, recognizing the agglutinative and inflectional nature of Italian morphology and specializing in highly frequent, specific affixes, occasionally providing concrete examples of common Italian suffixes. Finally, in the deeper layers (layers 13–14), interpretations often solidify the link between these morphological concepts and their syntactic roles, detailed grammatical categories, and the overall structure of morphologically rich languages, indicating a hierarchical build-up of linguistic understanding from foundational units to more complex grammatical functions.

A.10.6 Korean-Specific Features

The Korean interpretations show a progression from general structural elements to nuanced, context-specific knowledge across model layers. The earlier layers (layers 0 and 4) predominantly focus on foundational Korean linguistic units, such as morphemes, basic grammatical particles (e.g., for subject or object marking), and core semantic building blocks essential for elementary sentence structure and meaning. As the model progresses to deeper layers, the features demonstrate increasing specialization and abstraction. This includes the identification of more complex grammatical nuances like specific verb endings related to tense or politeness (emerging around layer 9) and honorifics (noted in layer 10), an understanding of abstract thematic content such as historical events like those involving King Sejong (in layer 10 and recurring in layer

12) or specific domains such as travel and communication (appearing by layer 13 and expanding in layer 15), and a refined recognition of proper nouns and specialized terminology (with initial indications around layer 7 and more developed instances in layer 10, becoming broadly evident by layer 15). Thus, there is a trajectory from a general, structural understanding of the language towards more nuanced, context-aware, and specialized knowledge representation in the model’s later layers.

A.10.7 Chinese-Specific Features

The Chinese interpretations demonstrate a progression from foundational to complex linguistic understanding. Early active layers (e.g., layer 2) primarily identify basic semantic tokens and morphemes. As the model deepens, mid-layers (layers 6–10) begin to distinguish larger phrasal units like noun and verb phrases, idioms, and also show an emerging specialization between features capturing semantic content and those focused on grammatical structure. In the deeper layers (layers 11–15), this specialization intensifies, with features becoming more granular, identifying specific entities like proper nouns, handling context-specific language (e.g., scientific or news domains), and robustly processing diverse elements such as punctuation, complex idioms, and structural connectors in parallel, indicating a sophisticated, multi-faceted comprehension of the language.

A.10.8 Russian-Specific Features

The Russian interpretations reveal a hierarchical pattern of feature abstraction, progressing from the identification of basic morphological units and general grammatical functions in earlier layers (e.g., layers 4, 6 focusing on morphemes, prefixes, and connectors) towards more granular and specific roles in the mid-layers (e.g., layers 7–12 detailing stems, suffixes, case, tense, and aspect). The deeper layers (layers 13–15) continue this morphological emphasis but increasingly integrate these features to interpret core semantic content, relationships, and their function in constructing meaning within complex sentences and specific contexts (e.g., in academic or formal text). However, we observed a notable deviation in feature 56594 in layer 10 (see Figure 60) which focuses on recognizing higher-level semantic categories like named entities before later layers synthesize morphological understanding for nuanced comprehension.

A.10.9 Japanese-Specific Features

The Japanese interpretations reveal a progression in linguistic abstraction. Earlier layers (e.g., layer 0) primarily identify foundational elements such as specific grammatical particles and basic semantic categories like nouns or key morphemes. As information flows to intermediate layers (layers 8–11), the interpretations demonstrate an increasing ability to recognize more complex Japanese structures, including noun and verb phrases, alongside a stronger focus on function words and initial indicators of contextual understanding. The deeper layers (layers 12–15) showcase highly refined and specialized feature interpretations. These capture nuanced grammatical functions, including specific particles and politeness markers, complex semantic relationships, domain-specific terminology (e.g., legal, technical), and a sophisticated grasp of Japanese morphology and its agglutinative nature. This evolution signifies a hierarchical processing of linguistic information, advancing from discrete components towards integrated, context-aware representations of language. However, we observed a deviation for feature 32154 in layer 8 (see Figure 59). Though predominantly responsive to Japanese linguistic patterns and examples, it also shows higher activation for specific Bulgarian morphemes, hinting that some features, while most relevant to the primary language of analysis, can also capture distinct patterns in other languages.

A.10.10 Vietnamese-Specific Features

The Vietnamese interpretations across layers reveal support for a hierarchical processing pattern for Vietnamese. Earlier layers (e.g., layers 0, 3, 4) predominantly focus on decomposing text into fundamental linguistic units, such as morphemes, syllables, function words, and elementary phrases, thereby establishing a foundational understanding. As information propagates to subsequent, mid-level layers (e.g., layers 7, 10, 12), features begin to assemble these components into more complex structures, identifying compound nouns, nuanced semantic roles, and key content elements. Consequently, deeper layers (e.g., layers 13–15) demonstrate the most abstract comprehension, interpreting larger semantic units like full phrases and collocations, recognizing tokens within specific discourse or pragmatic contexts (e.g., formal, conversational), and integrating diverse linguistic cues for a holistic understanding of the input. However, feature 49501 in layer 5 (see Figure 61), while mostly pro-

cessing Vietnamese, also strongly detects elements from Romanian and Eastern European place names, suggesting it captures some cross-lingual patterns. Similarly, feature 61383 in layer 9 (see Figure 62) for Vietnamese also shows high activation for Cambodian place names, indicating it can specialize in recognizing frequently mentioned foreign entities.

A.10.11 Bulgarian-Specific Features

The Bulgarian interpretations across its layers exhibit a hierarchical progression in complexity and abstraction. Initially, in earlier layers (e.g., layer 0–1), the focus is on foundational elements such as single Cyrillic letters, basic positional cues, and common suffixes tied to elementary word formation and fundamental grammatical features like gender and number. As processing moves to the middle layers (layers 3–9), the analysis deepens into more comprehensive morphological decomposition, identifying a wider array of morphemes like stems, prefixes, and diverse suffixes, alongside function words such as pronouns and conjunctions, and their roles in establishing core grammatical relationships, occasionally recognizing patterns shared with other Slavic languages. Finally, in the deeper layers (layers 10–15), the interpretations demonstrate an understanding of more complex syntactic roles and semantic nuances; features here often describe how these identified morphemes and function words contribute to constructing larger linguistic units, such as clauses, complex verb phrases, expressing modality, forming idiomatic expressions, and recognizing specific grammatical constructions like comparatives or superlatives, thereby indicating a grasp of higher-level sentence structure and meaning.

A.10.12 Portuguese-Specific Features

The Portuguese interpretations across the model layers exhibit a hierarchical pattern of linguistic abstraction. In the earlier layers (layers 0–5) primarily focus on identifying fundamental building blocks, such as specific high-frequency function words (e.g., the preposition “no” in layer 0) and basic morphological units like general function words, suffixes, and morphemes. Progressing to the middle layers (layers 7–12), the model demonstrates an enhanced capability to discern more specific morphological components (e.g., roots, stems, and affixes) and their roles in word formation (verb conjugations, noun/adjective forms), and begins to link these to semantic entities (e.g., “Brazil” in layer 7). Finally, the deeper layers (layers 13–15) showcase

a sophisticated synthesis, where interpretations reflect an understanding of how these accumulated morphological elements and function words, often in conjunction with punctuation, contribute to complex grammatical structures, sentence cohesion, nuanced meanings such as modality or conditional relationships, and the overall construction of coherent text. This trajectory suggests a process that starts with basic language elements and moves toward understanding how they work together to create complex meaning.

A.10.13 Turkish-Specific Features

The Turkish interpretations exhibit a hierarchical progression across its layers. In early layers (layers 0–7), the model begins with a foundational focus on deconstructing words into their core components, with features dedicated to identifying fundamental morphemes, suffixes, and roots essential for the language’s agglutinative structure. As processing moves to the middle layers (layers 8–12), a pattern of specialization emerges, with features becoming more attuned to specific grammatical categories like verb morphology and distinct semantic concepts such as named entities. Finally, in the deeper layers (layers 13–15), the model achieves a higher level of abstraction, developing features that interpret entire syntactic units, such as noun and verb phrases, and recognize language within specific semantic domains (e.g., administrative, legal, travel, and service-related contexts), all while continuously leveraging the foundational morphological features. An exception happens with feature 97688 in layer 10 (see Figure 58). Although this feature activates mainly on examples within the Turkish corpus, it shows higher activations in identifying verb endings in Romance languages.

A.10.14 Thai-Specific Features

The Thai interpretations reveal a hierarchical pattern of linguistic processing, progressing from foundational to abstract representations. In the earlier layers (layers 0–4), the model predominantly identifies fundamental morphological and syntactic units such as morphemes, syllables, and function words, focusing on the basic grammatical scaffolding of the Thai language. As we move to mid-layers (layers 5–9), these foundational elements are increasingly associated with concrete semantic categories and specific named entities, such as the literal concept of “Thailand.” In the deeper layers (layers 10–15), the interpretations demonstrate a more sophisticated

and abstract understanding, capturing complex semantic relationships like causality and temporality, as well as nuanced pragmatic functions. This progression shows a trajectory from deconstructing language into its structural components to synthesizing meaning and understanding its use in complex, often cross-lingual, contexts. Notably, feature 8775 in layer 12 (see Figure 63) describes phenomena that seem exclusive to other languages, such as politeness markers in Japanese and Korean, even though the feature itself is also highly activated within the Thai corpus.

A.10.15 Hindi-Specific Features

The Hindi interpretations exhibit a hierarchical progression in linguistic comprehension. Initially (layers 0–4), the model identifies fundamental, high-frequency morphemes such as postpositions, auxiliary verb components, and inflectional vowel diacritics, which form the basic syntactic building blocks of Hindi. As information propagates to the middle layers (layers 5–10), these components are assembled into more specific grammatical constructions, such as distinct verb conjugations and pronouns. Towards the end of this stage, particularly in layer 10, the model shows the first signs of semantic abstraction, moving beyond pure syntax with features that activate on specific geopolitical entities, including the country Pakistan, and related historical concepts. Finally, in the deepest layers (layers 11–15), this ability to interpret context becomes more sophisticated and widespread. The model builds upon its initial entity recognition to more consistently identify a broader range of named entities, locations, and organizations, completing a trajectory from recognizing grammatical form to comprehending contextual meaning.

A.11 Task Experiment

Although language-specific features influence PPL, our experiments on the XWinograd task (Muenighoff et al., 2023) suggest they do not reliably impact downstream task performance. We conducted two experiments to evaluate this. In the first (Table 2), suppressing language-specific features across all layers may significantly disrupt performance across languages. In contrast, the second experiment (Table 3) involved suppressing these features only in the middle layers, where their influence was presumed to be less critical, and resulted in negligible performance changes. These findings align with prior work showing that while

language-specific neurons may affect PPL, they do not necessarily degrade task performance (Mondal et al., 2025). This distinction may explain why researchers differentiate between language-specific and task-specific neurons (e.g., Tang et al., 2024; Kojima et al., 2024; Leng and Xiong, 2025), with the latter playing a more crucial role in enabling effective multilingual task execution. This also suggests that task-specific features are likely to exist in LLMs.

Task Language	Baseline	en	fr	jp	pt	ru	zh
en	0.8129	0.5196	0.8116	0.5166	0.5118	0.5768	0.5923
fr	0.6506	0.4940	0.6506	0.4699	0.4217	0.5181	0.5181
jp	0.5996	0.5328	0.6111	0.5172	0.5203	0.5255	0.5276
pt	0.6616	0.5171	0.6616	0.5019	0.5171	0.5133	0.5057
ru	0.5968	0.5397	0.5810	0.5587	0.5429	0.5556	0.5143
zh	0.6389	0.4861	0.6468	0.4861	0.4802	0.5655	0.5298

Table 2: Accuracy on XWinograd task under language-specific feature interventions applied across all layers of Llama 3.2 1B with a scaling factor of $\alpha = -0.2$. Each column corresponds to an intervention targeting a specific language.

Task Language	Baseline	en	fr	jp	pt	ru	zh
en	0.8129	0.8077	0.8095	0.8120	0.7931	0.8168	0.8090
fr	0.6506	0.6627	0.6024	0.6506	0.6747	0.6506	0.6386
jp	0.5996	0.6017	0.5923	0.6121	0.5912	0.5933	0.5850
pt	0.6616	0.6654	0.6692	0.6616	0.6046	0.6692	0.6654
ru	0.5968	0.6032	0.6063	0.6095	0.6317	0.5968	0.6413
zh	0.6389	0.6567	0.6290	0.6429	0.6429	0.6548	0.6369

Table 3: Accuracy on XWinograd task under language-specific feature interventions applied from layers 3–13 of Llama 3.2 1B with a scaling factor of $\alpha = -0.2$. Each column corresponds to an intervention targeting a specific language.

A.12 Relationship between Language-Specific Features and SAE training data

The existence of language-specific features could be related to the frequency of specific concepts (in this case, related to languages) in the SAE training data, which influences the dedicated learned features for those concepts, with the more frequent ones likely to have dedicated features, as described by Templeton et al. (2024). RedPajama-v2 is not uniformly distributed, with English being the most dominant one, as shown in Table 4. Interestingly, all major languages in RedPajama-v2 (en, de, fr, es, and it) have fewer language-specific features than the other languages (see Figure 19). This suggests that training SAEs on uniformly distributed multilingual corpora can help them learn language-specific concepts more evenly.

Language	Row Count	Percentage
en	4,730,801	74.02%
de	531,629	8.32%
fr	433,604	6.78%
es	459,208	7.18%
it	236,397	3.70%

Table 4: Language distribution and row counts in the RedPajama-Data-V2 Sample-10B subset

A.13 Text Generation Assessment Rubric

To assess steered generated text, we developed a rubric targeting two key criteria: (1) language occurrence and (2) language coherence. This rubric is designed to be applied manually by humans or automatically with an LLM, and aims to ensure consistency and transparency in evaluating whether and how language-specific features affect generated outputs. In this study, we use GPT-4.1 (OpenAI, 2025) to automatically evaluate the generated text according to this rubric.

A.13.1 Language Occurrence

This criterion captures whether a meaningful language change has occurred in the generated text. A change is considered to have occurred if the output contains at least one *morpheme* (the smallest meaningful unit of language) from a different language than English.

- **Score 0 (Unchange):** The text remains fully in English and contains no morphemes from another language. Minor orthographic changes that do not introduce new meaning or reflect actual cross-linguistic variation are *not counted* as a language change. For example:

- apple → äpple (still English, only orthographic variation)
- naive → naïve (still English usage)

- **Score 1 (Change):** At least one morpheme from a different language is introduced in the output, even if the change occurs only in part of the sentence. For example:

- apple → manzana (Spanish)
- He is a teacher → Er ist ein Lehrer (German)

A.13.2 Language Coherence

This criterion evaluates the overall intelligibility and fluency of the generated text when steered to-

ward multilingual expression. It is assessed along a three-point ordinal scale:

- **Incoherent:** The text is unintelligible and does not form meaningful sentences. It may consist of repeated characters, morphemes, or nonsensical word sequences. For example:

- blblblblbl èèèè \$\$\$\$\$
 - rerere banana moon qwerty

- **Partially Coherent:** The general meaning of the text can be inferred, but parts may require guessing or interpretation. This includes major grammatical or syntactic errors, repetitive yet intelligible structures, or repetitions at the end of the text. For example:

- He go to escuela because want aprender
 - The cat is sleep sleep the bed the bed

- **Coherent:** The text is readable and understandable, with only minor issues (e.g., minor grammatical or syntactic errors and mid-sentence code-switching that does not obscure meaning). Code-switching is acceptable as long as overall sentence-level coherence is maintained. For example:

- She goes to the bibliothèque every evening to study
 - I like to eat sushi y también tacos
 - Er ist ein Lehrer und he teaches German very well

This rubric allows researchers to both quantify the presence of cross-linguistic influence and evaluate the intelligibility of the generated text, supporting a robust evaluation of multilingual steering methods. Note that this rubric does not assess the correctness of the outputs. For example, the generated text may contain logical errors (as illustrated in Figures 71–73) or may include inaccurate information.

A.14 Misclassification Analysis of SAE-based LID

For the misclassified example, we analyze the following German sentence from WiLI-2018: “Olga Alexandrowna Girja (russisch Ольга Александровна Гиря; * 4. Juni 1991 in Langepas) ist eine russische Schachspielerin und seit 2009 Großmeister

der Frauen (WGM).” For complete details on feature activations, refer to Figures 79–83. The lower layers (0–3) do recognize some German elements, such as “isch”, “ist”, and “ß”. However, a critical shift occurs in the middle layers, where Russian features begin to dominate. By layer 6, Russian feature 121507 activates on Cyrillic script elements, such as “ль”, “га”, “Александ”, and “Г”, and intensifies in higher layers. The decisive factor appears to be feature 56594 in layer 10, which activates on proper nouns and named entities across both scripts, including the transliterated Russian name in Latin script (“ga”, “Alexand”, “row”, “na”, “Gir”, “ja”) and the Cyrillic characters (“О”, “ль”, “га”, “Александ”, “ров”, “на”, “Г”, “ир”, “я”). While German features persist in later layers, they are ultimately overshadowed by the Cyrillic script and Russian name, which trigger more consistent activations in the deeper layers. Notably, the SAE contains 13 German-specific features compared to 26 Russian-specific features, as shown in Figure 19, which may further amplify this imbalance. Taken together, this behaviour suggests that the current approach may be biased toward languages with a larger number of features, especially in mixed-language inputs.

A.15 Additional Results

For completeness, this section presents the results related to language-specific neurons and language-specific features for Llama 3.2 1B, Gemma 2 2B, and Gemma 2 9B.

A.15.1 Language-Specific Neuron Distributions

Figure 11–13 shows the distribution of language-specific neurons across layers for Llama 3.2 1B, Gemma 2 2B, and Gemma 2 9B.

A.15.2 Language-Specific Neuron Counts

Figure 14–16 shows the language-specific neuron counts across languages for Llama 3.2 1B, Gemma 2 2B, and Gemma 2 9B.

A.15.3 Language-Specific Feature Distributions

Figure 17–18 shows the distribution of language-specific features across layers for Gemma 2 2B and Gemma 2 9B.

A.15.4 Language-Specific Features Counts

Figure 19–21 shows the language-specific feature counts across languages for Llama 3.2 1B, Gemma 2 2B, and Gemma 2 9B.

A.15.5 Language-Specific Feature LAPE Scores

Figure 22–24 shows that the majority of the identified language-specific features in Llama 3.2 1B, Gemma 2 2B, and Gemma 2 9B have low LAPE scores.

A.15.6 Perplexity Change

Figures 25–27 illustrate the change in perplexity when steering with language-specific features and language-specific neurons in Llama 3.2 1B across different scaling factors α . The corresponding results for Gemma 2 2B are shown in Figures 28–31. Overall, the patterns are consistent across models, as explained in Section 5.2.1.

A.15.7 LID with Language-Specific Features

Figure 32 and Table 5 present the SAE-based, neuron-based, and fastText LID performance on WiLI-2018 across 15 languages for Gemma 2 2B. Figure 33 and Table 6 present the corresponding results for Gemma 2 9B. Overall, the results exhibit patterns largely consistent with those observed for Llama 3.2 1B in Section 5.3.

However, performance degrades for several languages. We hypothesise that this is due to a bias towards languages with a larger number of language-specific features, as discussed in Section 5.3. This bias is more pronounced in JumpReLU SAEs than in TopK SAEs, since JumpReLU does not constrain the number of activations per token. As a result, languages with more language-specific features may activate a disproportionately large number of features, thereby amplifying the bias, even when the activations are small. In contrast, TopK SAEs explicitly limit the number of activations per token to the top- k values, which may partially mitigate this effect. This behaviour is particularly evident for English, which has the fewest language-specific features among the 15 languages, as shown in Figure 20 for Gemma 2 2B and Figure 21 for Gemma 2 9B.

Moreover, these results may be influenced by the multilingual capabilities of Gemma 2 2B, Gemma 2 9B, and their corresponding SAEs, which are predominantly trained on English data (Team et al., 2024; Lieberum et al., 2024). For Turkish, both Gemma 2 2B and Gemma 2 9B exhibit poor performance, suggesting that the models and their SAEs may have limited multilingual capability for this language. In contrast, Gemma 2 2B exhibits poor performance on Italian, whereas Gemma 2 9B per-

forms well on this language, suggesting that Gemma 2 9B may have stronger multilingual capabilities for Italian than Gemma 2 2B due to its larger parameter count.

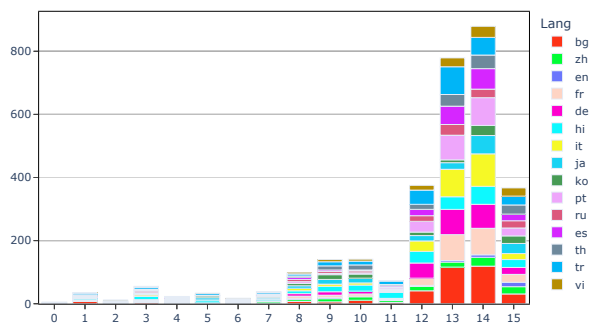


Figure 11: Distribution of the identified language-specific neurons with LAPE across layers of Llama 3.2 1B.

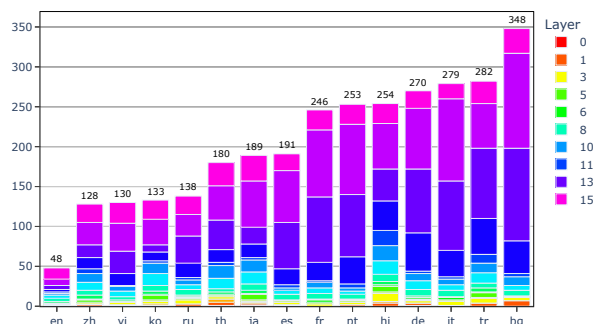


Figure 14: Language-specific neuron counts in Llama 3.2 1B across languages.

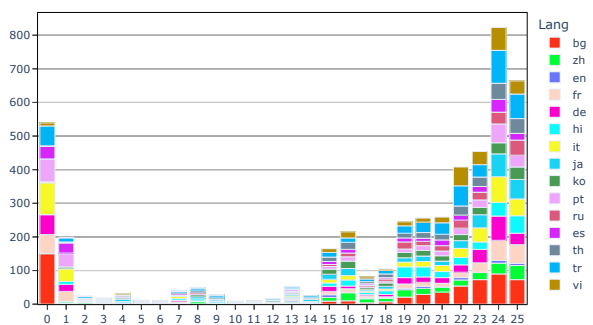


Figure 12: Distribution of the identified language-specific neurons with LAPE across layers of Gemma 2 2B.

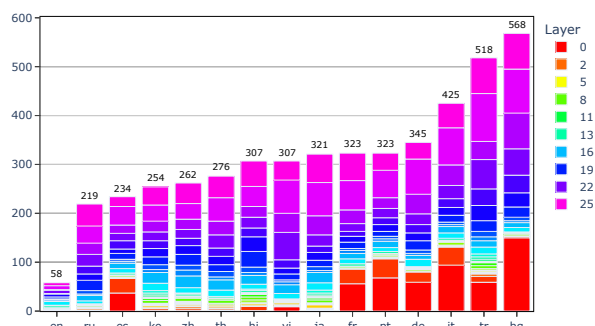


Figure 15: Language-specific neuron counts in Gemma 2 2B across languages.

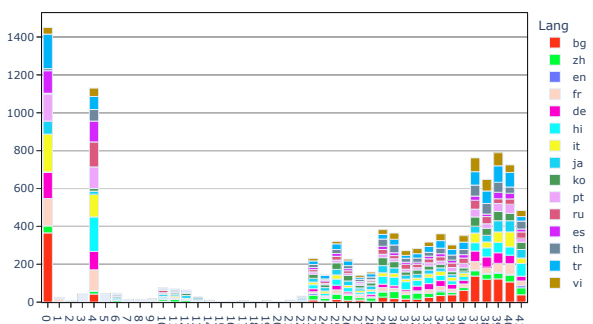


Figure 13: Distribution of the identified language-specific neurons with LAPE across layers of Gemma 2 9B.

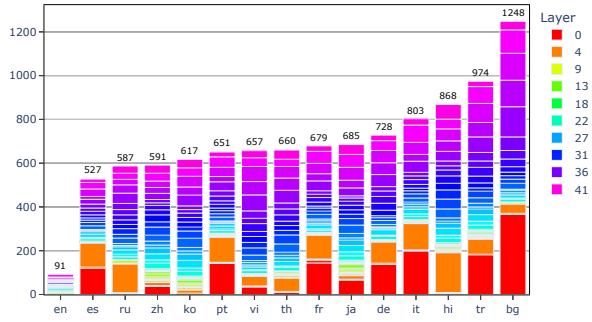


Figure 16: Language-specific neuron counts in Gemma 2 9B across languages.

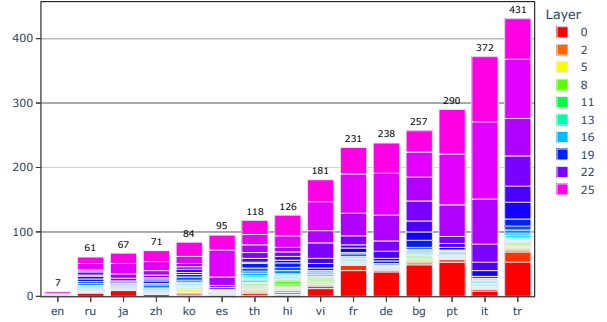


Figure 20: Language-specific feature counts in Gemma 2 2B across languages.

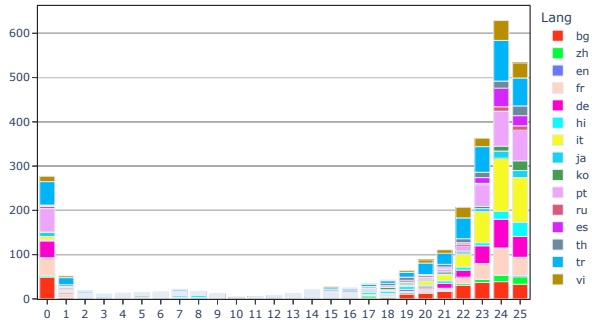


Figure 17: Distribution of the identified language-specific features with *SAE-LAPE* across layers of Gemma 2 2B.

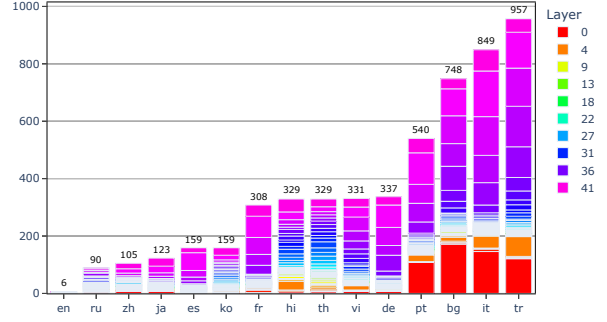


Figure 21: Language-specific feature counts in Gemma 2 9B across languages.

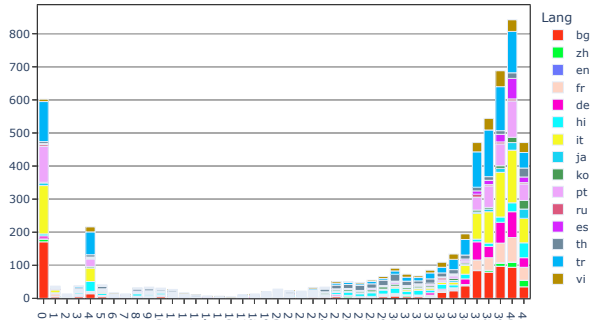


Figure 18: Distribution of the identified language-specific features with *SAE-LAPE* across layers of Gemma 2 9B.

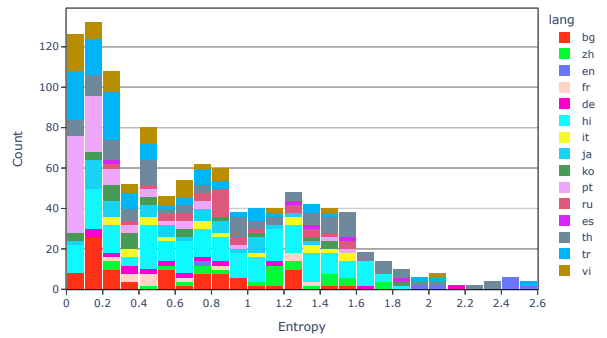


Figure 22: Distribution of the identified language-specific features in Llama 3.2 1B with *SAE-LAPE* across LAPE scores.

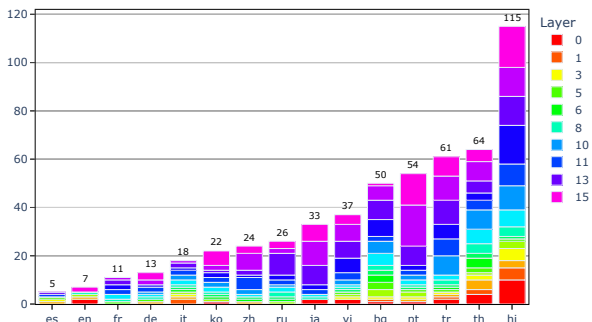


Figure 19: Language-specific feature counts in Llama 3.2 1B across languages.

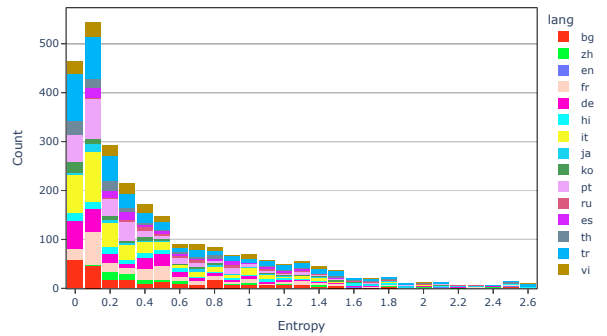


Figure 23: Distribution of the identified language-specific features in Gemma 2 2B with *SAE-LAPE* across LAPE scores.

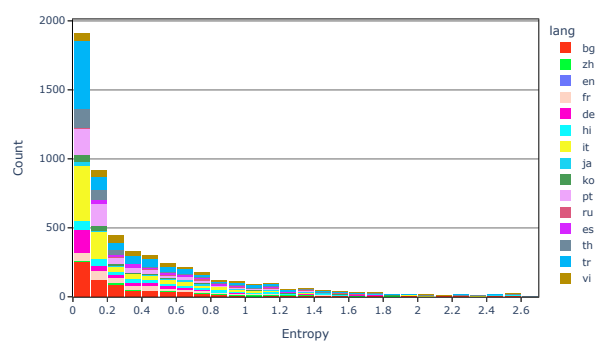


Figure 24: Distribution of the identified language-specific features in Gemma 2 9B with SAE-LAPE across LAPE scores.

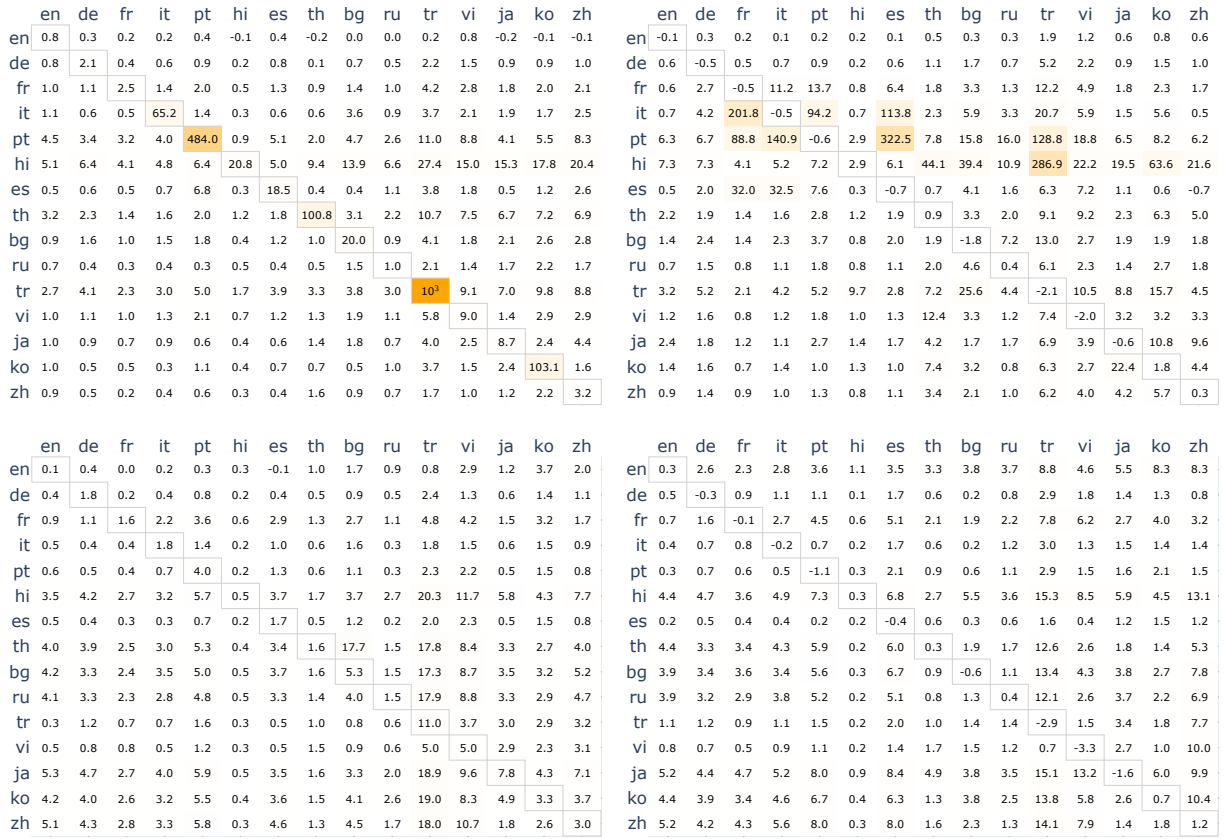


Figure 25: PPL changes in Llama 3.2 1B. **Top:** Steering using language-specific features with $\alpha = -0.1$ (left) and $\alpha = 0.1$ (right). **Bottom:** Steering using language-specific neurons with $\alpha = -0.1$ (left) and $\alpha = 0.1$ (right).

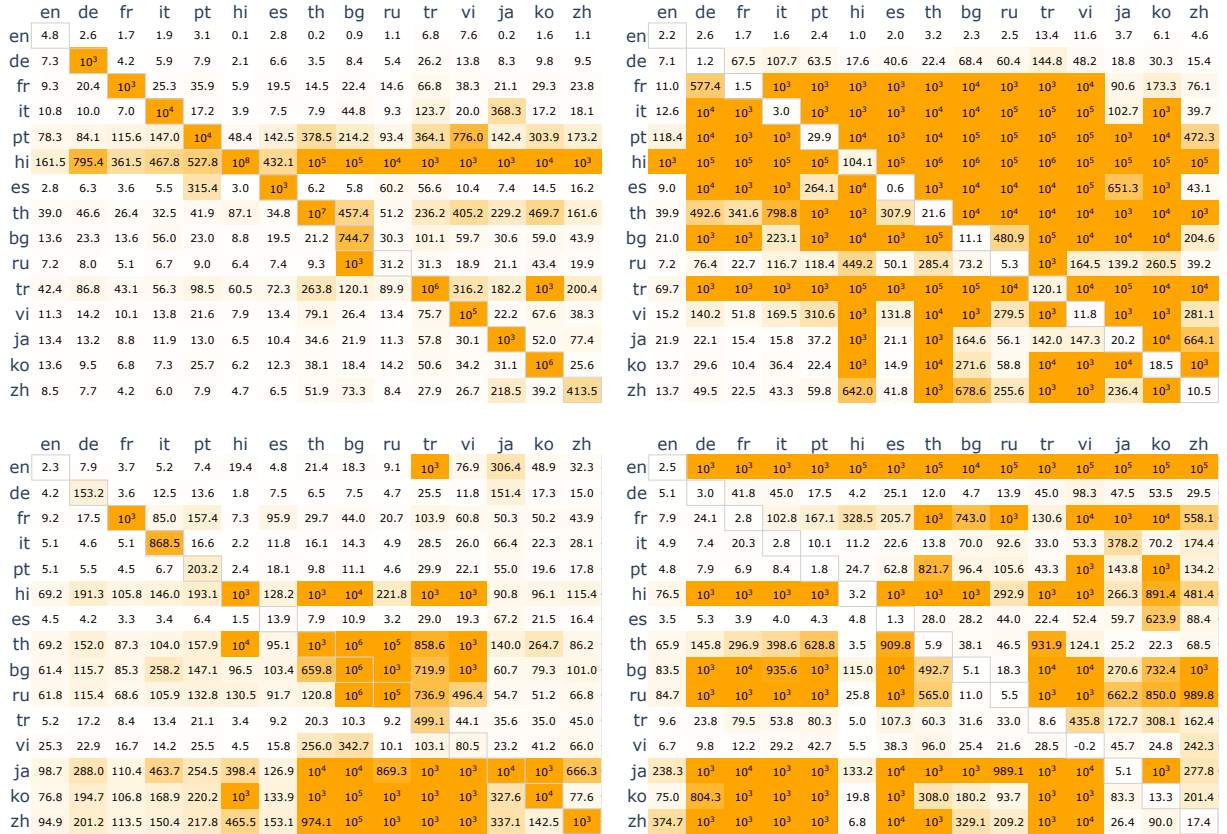


Figure 26: PPL changes in Llama 3.2 1B. **Top:** Steering using language-specific features with $\alpha = -0.3$ (left) and $\alpha = 0.3$ (right). **Bottom:** Steering using language-specific neurons with $\alpha = -0.3$ (left) and $\alpha = 0.3$ (right).

	en	de	fr	it	pt	hi	es	th	bg	ru	tr	vi	ja	ko	zh
en	7.8	4.7	3.2	3.3	5.4	0.3	5.0	0.8	1.7	2.2	13.3	13.4	0.9	3.5	2.4
de	14.1	10 ⁴	8.8	12.2	16.2	4.4	13.2	8.1	18.0	12.6	55.8	26.9	16.9	20.7	19.4
fr	19.0	86.9	10 ⁴	95.9	120.2	15.2	52.9	46.7	63.4	48.6	214.7	134.3	55.2	86.3	56.9
it	28.7	67.4	36.6	10 ⁴	66.1	86.5	21.2	29.6	134.3	25.0	10 ³	58.9	10 ³	45.2	43.9
pt	368.3	10 ³	10 ³	10 ³	10 ⁵	10 ³	10 ³	10 ⁴	10 ³	10 ³	10 ³	10 ⁴	10 ³	10 ³	10 ³
hi	10 ³	10 ⁴	10 ³	10 ⁴	10 ⁴	10 ⁵	10 ⁴	10 ³	10 ⁵	10 ⁵	10 ⁵	10 ⁵	10 ⁵	10 ⁵	10 ⁴
es	5.5	30.1	27.0	16.3	525.9	8.3	10 ³	69.9	19.5	55.8	698.7	29.1	27.3	52.5	27.4
th	96.8	182.5	90.3	114.3	139.9	10 ⁴	115.9	10 ⁴	10 ³	353.1	10 ³	10 ³	10 ³	10 ³	831.5
bg	40.8	95.4	61.1	10 ³	104.1	68.7	107.5	122.9	10 ³	244.9	978.2	480.7	115.4	291.6	199.4
ru	14.9	19.6	13.0	16.6	23.7	23.4	18.0	26.2	10 ⁴	10 ³	72.5	49.5	72.2	234.3	47.5
tr	119.1	457.0	199.2	239.3	435.3	746.4	318.7	10 ³	762.9	605.2	10 ⁴	10 ³	10 ³	10 ³	10 ³
vi	23.3	34.6	24.8	32.4	50.9	21.6	31.1	10 ³	76.8	35.9	221.4	10 ⁴	57.9	512.3	104.5
ja	27.0	32.0	20.5	29.2	31.4	18.1	24.2	181.2	65.0	31.6	149.0	72.5	10 ⁵	169.2	234.3
ko	30.2	24.3	17.9	19.2	92.4	19.2	36.9	635.7	87.1	45.8	150.2	227.3	123.2	10 ⁶	65.5
zh	16.9	17.6	9.6	13.6	17.2	25.1	14.1	419.3	695.7	26.5	66.7	135.8	10 ³	239.5	10 ⁴

	en	de	fr	it	pt	hi	es	th	bg	ru	tr	vi	ja	ko	zh
en	5.7	85.9	9.1	12.8	18.1	403.9	12.6	76.1	54.3	24.6	10 ⁴	674.4	10 ³	140.2	116.7
de	8.3	10 ³	7.7	66.3	48.1	3.5	17.2	19.2	14.9	9.4	55.3	24.8	10 ³	50.4	44.5
fr	19.1	53.3	10 ⁴	527.1	870.8	18.9	451.9	139.4	171.4	74.1	411.6	204.0	539.7	171.7	146.7
it	10.0	10.1	11.4	10 ³	51.6	5.6	25.6	118.7	32.8	11.1	121.6	158.0	445.4	78.1	116.3
pt	10.1	11.8	9.8	15.3	10 ³	5.3	48.7	39.1	25.2	10.1	128.5	64.2	586.8	58.0	59.6
hi	172.5	879.9	499.9	736.5	800.3	10 ⁶	510.9	10 ⁵	10 ⁵	10 ³	10 ³	10 ⁴	10 ³	10 ³	619.0
es	9.0	8.9	6.9	7.3	13.8	3.4	41.6	28.9	21.8	6.6	149.8	48.1	609.9	68.2	49.9
th	162.9	677.7	368.7	452.1	609.5	10 ⁶	336.5	10 ⁴	10 ⁷	10 ⁶	10 ³	10 ⁵	10 ³	10 ⁴	812.9
bg	143.5	526.8	346.4	10 ³	639.5	10 ⁴	380.4	10 ⁴	10 ⁷	10 ⁵	10 ³	10 ⁴	380.8	10 ³	929.5
ru	145.6	497.9	273.1	548.3	542.6	10 ⁴	346.5	10 ³	10 ⁷	10 ⁷	10 ³	10 ³	328.4	466.8	319.1
tr	10.4	42.6	23.5	57.1	65.0	8.6	23.1	99.8	26.1	22.1	10 ⁴	135.3	124.6	110.8	112.4
vi	78.4	93.9	60.4	49.6	88.2	19.2	51.0	10 ³	10 ⁴	61.2	374.4	10 ³	70.3	199.4	512.8
ja	236.2	10 ³	536.2	10 ³	10 ³	10 ³	521.4	10 ³	10 ³	10 ⁵	10 ³	10 ³	10 ⁵	10 ⁴	10 ⁴
ko	195.1	981.6	555.2	863.7	10 ³	10 ³	569.5	10 ⁵	10 ⁷	10 ⁴	10 ³	10 ⁴	10 ⁶	10 ⁶	10 ³
zh	226.4	976.4	542.9	711.0	921.7	10 ⁵	578.6	10 ⁵	10 ⁵	10 ⁵	10 ³	10 ⁴	10 ⁵	10 ⁴	10 ⁶

	en	de	fr	it	pt	hi	es	th	bg	ru	tr	vi	ja	ko	zh
en	5.3	10 ³	10 ³	10 ³	10 ³	10 ³	10 ³	10 ⁵	10 ⁵	10 ⁵	10 ⁴	10 ⁵	10 ⁵	10 ⁵	10 ⁵
de	10.6	7.2	451.9	415.6	56.1	34.6	95.8	61.2	11.6	52.0	139.6	10 ³	775.2	921.9	209.7
fr	16.6	105.0	6.4	322.7	476.3	10 ⁵	569.0	10 ⁵	10 ⁴	10 ⁵	423.2	10 ⁵	10 ⁴	10 ⁵	10 ⁴
it	10.4	17.6	117.4	7.2	22.1	10 ³	58.9	326.7	10 ³	10 ³	79.3	984.1	10 ⁴	10 ³	10 ⁴
pt	10.6	19.3	18.9	22.5	6.2	10 ⁴	153.9	10 ⁵	10 ³	10 ⁴	139.1	10 ⁴	10 ⁴	10 ⁵	10 ⁴
hi	209.9	10 ⁴	10 ⁴	10 ⁴	10 ⁴	6.5	10 ⁴	10 ⁴	10 ⁴	10 ³	10 ⁴	10 ⁴	10 ³	10 ³	10 ³
es	7.5	11.6	8.6	8.5	10.1	198.8	3.8	10 ³	601.1	10 ³	61.8	10 ³	10 ⁴	10 ⁴	10 ⁴
th	162.4	486.0	10 ³	10 ³	10 ³	10.6	10 ³	14.3	188.5	226.3	10 ³	497.9	61.2	61.3	226.6
bg	211.3	10 ⁴	10 ⁵	10 ³	10 ⁴	10 ³	10 ³	10 ³	12.5	48.2	10 ⁴	10 ⁴	10 ³	10 ³	10 ⁴
ru	211.2	10 ³	10 ⁴	10 ⁴	10 ⁴	386.0	10 ⁴	10 ⁴	20.7	10.8	10 ⁴	10 ⁴	10 ⁴	10 ³	10 ⁴
tr	21.3	153.3	10 ³	973.5	10 ³	23.2	10 ³	479.3	237.2	173.8	27.0	10 ³	10 ³	10 ³	10 ³
vi	13.9	26.7	49.3	202.7	224.0	20.4	198.8	391.0	99.8	65.7	86.0	5.8	122.8	95.4	10 ³
ja	10 ³	10 ⁴	10 ⁵	10 ³	10 ³	10 ³	10 ³	10 ³	10 ⁴	10 ⁴	10 ⁴	10 ⁴	10 ⁴	10 ⁴	676.3
ko	216.0	10 ⁴	10 ⁴	10 ⁴	10 ⁴	211.7	10 ⁴	10 ³	10 ³	10 ³	10 ⁴	10 ⁴	454.4	26.5	874.6
zh	10 ³	10 ⁴	10 ⁵	10 ⁵	10 ⁵	37.0	10 ⁵	10 ⁴	10 ⁴	10 ³	10 ⁴	10 ⁵	97.5	620.3	36.8

Figure 27: PPL changes in Llama 3.2 1B. **Top:** Steering using language-specific features with $\alpha = -0.4$ (left) and $\alpha = 0.4$ (right). **Bottom:** Steering using language-specific neurons with $\alpha = -0.4$ (left) and $\alpha = 0.4$ (right).

	en	de	fr	it	pt	hi	es	th	bg	ru	tr	vi	ja	ko	zh
en	1.3	0.3	0.1	0.0	0.1	0.1	0.2	0.0	0.0	0.1	0.1	0.2	-0.1	-0.0	0.2
de	3.1	57.2	2.7	4.2	6.8	3.6	6.1	3.8	2.7	5.3	9.1	7.5	13.6	4.7	21.8
fr	5.0	6.4	10 ⁴	7.4	9.0	2.5	8.9	4.3	2.9	5.4	8.9	8.9	10.8	4.5	14.1
it	3.2	5.1	3.4	10 ³	9.8	2.7	6.7	3.1	8.4	4.9	9.0	6.6	7.6	3.8	12.3
pt	0.0	5.4	2.5	5.2	10 ³	2.9	8.6	3.3	4.1	5.7	7.8	5.5	5.3	6.1	11.4
hi	0.0	1.4	0.8	1.0	1.4	5.9	1.7	1.3	0.8	1.0	3.5	3.5	3.6	2.5	6.8
es	0.7	1.6	0.7	1.5	3.3	0.7	11.3	0.8	0.4	0.6	2.0	1.5	1.4	1.0	5.1
th	1.7	5.8	2.2	3.0	2.9	2.1	4.5	13.7	3.1	4.5	4.8	5.1	5.4	2.8	12.9
bg	5.7	9.6	2.8	11.3	9.1	5.8	6.5	601.4	14.6	11.7	11.2	14.2	10.7	28.4	
ru	0.3	2.0	0.4	0.6	1.1	1.0	1.1	0.7	10.4	3.2	0.9	1.2	1.1	1.1	4.0
tr	2.6	11.1	5.1	8.0	10.2	9.9	14.2	11.1	9.0	11.4	10 ³	15.9	20.8	12.2	27.6
vi	2.3	5.4	2.8	3.0	5.9	3.0	2.8	5.2	3.0	3.0	9.3	10 ³	6.9	5.4	19.8
ja	-0.6	1.4	0.5	1.0	-0.3	0.6	0.4	-0.6	1.0	1.8	0.3	14.5	1.4	0.9	
ko	1.4	2.7	0.6	1.2	2.8	1.1	2.1	1.2	1.3	1.8	6.5	2.6	2.6	13.9	5.4
zh	-0.5	0.0	-0.0	-0.3	0.6	0.2	-0.4	1.0	1.1	0.6	-0.0	1.1	1.0	1.1	18.9

	en	de	fr	it	pt	hi	es	th	bg	ru	tr	vi	ja	ko	zh
en	1.3	-0.1	0.8	1.0	-0.7	0.6	-0.5	0.6	2.6	-0.3	-0.4	1.1	0.2	0.9	2.6
de	0.7	5.4	1.0	1.2	1.5	1.1	0.7	0.8	1.4	1.6	2.5	2.4	1.4	1.2	3.6
fr	0.8	1.1	3.2	1.5	1.9	1.0	2.7	0.6	2.0	1.8	1.3	1.4	1.3	1.0	4.2
it	0.7	1.3	1.1	8.9	2.8	1.0	2.3	0.9	3.9	1.7	2.1	2.0	1.7	1.3	5.4
pt	1.0	1.3	0.9	2.2	8.4	1.3	4.0	0.9	2.1	2.9	2.2	1.9	0.5	1.3	4.6
hi	-1.8	3.1	1.1	1.5	2.0	1.5	2.1	1.0	2.6	3.7	6.4	5.8	2.6	2.2	2.9
es	1.0	0.8	0.4	1.6	2.9	1.3	5.8	0.8	2.5	1.7	1.5	1.8	0.3	1.0	5.1
th	-2.1	2.9	1.1	1.1	1.3	0.6	1.0	2.6	2.3	4.0	5.4	6.1	2.1	1.3	3.9
bg	-1.0	4.5	1.4	4.8	3.1	1.7	3.4	1.5	15.5	4.5	5.4	6.1	3.0	2.6	6.9
ru	-1.6	3.3	1.1	2.3	3.4	1.4	2.9	0.9	1.9	2.9	3.8	5.6	2.3	1.8	5.5
tr	1.0	2.5	1.3	2.9	1.1	1.5	4.0	1.0	3.4	1.8	27.5	3.8	2.9	1.7	1.8
vi	0.5	1.8	0.7	1.4	2.2	1.2	1.3	1.0	1.2	2.2	5.2	25.4	0.1	1.3	1.4
ja	-2.2	3.8	2.3	3.1	1.3	0.7	2.3	1.9	3.2	4.8	8.2	5.1	10.3	3.6	7.9
ko	-1.2	4.0	1.4	1.8	3.2	1.1	2.3	1.0	21.8	6.1	10.1	7.6	3.4	3.6	3.2

	en	de	fr	it	pt	hi	es	th	bg	ru	tr	vi	ja	ko	zh
en	2.8	0.9	0.5	0.3	0.7	0.2	0.8	0.1	0.1	0.3	0.5	0.7	0.1	0.1	0.8
de	14.8	10^5	12.5	21.7	29.3	27.6	27.8	22.3	19.2	64.2	48.3	34.5	76.5	28.7	119.6
fr	23.9	39.3	10^4	118.4	42.9	47.3	39.1	73.7	26.9	46.4	61.0	60.0	74.1	50.5	106.1
it	14.8	26.0	15.2	10^5	71.5	18.6	31.2	16.7	42.3	21.4	43.3	29.2	46.0	23.3	65.2
pt	7.3	24.9	11.2	26.2	10^4	16.0	38.7	18.2	23.4	27.5	37.7	29.7	30.5	385.9	58.9
hi	4.6	9.4	4.5	6.2	8.9	10^5	9.1	6.8	5.7	6.9	16.9	14.3	19.1	11.5	32.8
es	4.2	7.3	3.6	11.6	102.5	3.8	307.0	4.0	4.8	5.4	9.8	7.4	8.9	5.5	22.3
th	9.3	23.5	9.7	13.3	14.6	13.0	21.5	10^4	15.5	19.1	26.4	25.8	34.0	18.4	60.2
bg	25.4	55.8	16.1	352.5	48.2	66.8	45.1	85.6	10^4	113.3	96.8	84.8	88.7	10^3	188.2
ru	2.0	7.8	2.0	3.4	4.7	4.2	4.9	3.4	658.4	10^3	6.8	4.8	6.3	4.8	14.6
tr	22.9	89.5	39.9	72.7	94.5	221.2	137.6	231.7	135.9	124.0	10^7	131.4	270.8	10^3	324.1
vi	12.0	28.0	14.1	16.8	27.6	18.9	18.3	10^4	19.2	20.8	56.0	10^7	62.2	51.2	157.5
ja	1.1	8.1	2.8	6.7	3.0	4.3	4.4	4.5	2.8	6.0	10.7	5.2	10^3	10.9	17.8
ko	6.1	9.6	3.0	5.2	9.6	5.4	7.8	5.8	6.4	7.3	22.5	9.9	17.4	10^5	24.1
zh	0.9	3.0	1.5	1.3	4.5	2.7	2.5	5.3	6.5	4.9	4.1	5.2	13.5	8.3	536.4

	en	de	fr	it	pt	hi	es	th	bg	ru	tr	vi	ja	ko	zh
en	-2.1	0.3	0.5	0.7	0.8	0.1	0.9	0.2	0.2	0.2	1.0	0.2	0.8	0.4	0.6
de	52.6	17.9	10^5	10^5	10^5	10^4	10^5	10^5	10^6	10^5	10^5	10^5	10^4	10^5	10^3
fr	180.8	10^5	7.2	10^4	10^5	10^4	10^4	10^7	10^7	10^6	10^5	10^5	10^4	10^5	10^4
it	34.3	10^5	10^4	11.1	10^4	10^5	10^4	10^6	10^6	10^6	10^5	10^6	231.0	10^5	568.6
pt	37.7	10^4	10^4	10^4	14.8	10^4	10^4	10^5	10^5	10^4	10^4	10^5	10^3	443.8	857.9
hi	17.2	21.0	6.9	11.9	18.3	4.7	16.4	266.2	250.9	161.9	223.9	66.2	743.8	10^4	96.0
es	8.7	678.6	10^3	10^3	10^3	26.4	1.1	22.0	10^3	217.6	190.9	10^5	28.0	16.0	28.7
th	14.7	16.9	20.9	21.5	91.8	10^4	24.5	9.5	909.8	38.3	10^4	10^4	98.9	10^3	277.5
bg	28.4	10^3	10^4	693.6	10^4	10^5	10^3	10^4	34.6	10^4	10^4	10^4	599.7	10^4	436.5
ru	6.1	754.5	82.1	10^3	10^3	10^4	228.8	10^4	153.6	6.4	10^3	10^3	624.7	10^4	248.2
tr	127.1	10^5	10^4	10^4	10^4	10^5	10^6	10^4	10^6	10^6	89.9	10^5	10^6	10^6	10^5
vi	30.7	10^4	10^4	10^5	10^5	10^7	10^5	10^6	10^6	10^5	10^5	38.1	10^5	10^6	10^5
ja	10.8	15.6	7.2	8.9	22.2	10^3	15.7	10^3	56.9	15.9	474.9	116.3	17.3	10^5	10^3
ko	8.0	7.6	4.4	9.3	6.2	10^4	8.1	328.9	28.1	10.6	355.5	36.5	10^3	4.5	10^3
zh	8.2	14.5	6.5	13.7	10.1	10^3	12.8	10^3	13.1	17.6	31.6	49.9	335.5	10^3	6.7

	en	de	fr	it	pt	hi	es	th	bg	ru	tr	vi	ja	ko	zh
en	3.5	17.1	10.0	171.1	3.4	3.5	6.3	2.9	11.4	2.3	8.2	9.4	3.8	4.1	11.3
de	2.8	24.5	4.0	5.6	6.8	4.4	5.8	3.5	5.6	6.2	12.3	9.3	6.6	4.7	13.9
fr	2.8	6.5	62.4	9.8	8.0	3.6	11.2	2.5	8.4	5.5	8.4	6.0	6.1	4.1	15.3
it	3.4	8.2	5.1	10^3	15.6	4.2	14.4	3.3	16.7	6.9	11.8	8.3	7.7	5.4	19.8
pt	2.5	8.8	4.0	10.6	47.7	6.6	17.0	3.9	11.4	13.4	13.0	8.9	6.2	6.7	17.9
hi	-0.3	20.6	7.4	11.3	15.5	9.4	16.3	4.7	17.6	16.7	36.2	35.8	11.5	8.2	16.7
es	2.8	6.2	2.7	13.7	11.5	5.7	20.4	3.6	11.1	6.1	10.4	7.9	5.0	5.2	17.2
th	-1.8	16.5	7.2	9.2	11.6	6.1	12.7	10^4	74.1	18.5	35.8	317.7	13.9	6.8	22.3
bg	2.7	30.3	12.0	32.3	27.6	11.9	27.0	8.4	10^5	82.7	45.2	36.0	18.6	11.7	33.9
ru	0.9	26.8	10.1	18.9	28.9	9.3	25.0	5.8	45.3	35.7	36.5	38.1	16.1	7.8	26.8
tr	3.7	16.0	7.0	17.6	8.9	7.2	28.7	5.9	15.5	9.4	10^3	38.7	13.4	8.2	14.2
vi	3.0	11.0	4.0	8.1	11.9	5.9	9.2	6.5	6.4	9.3	27.4	10^4	5.8	8.8	13.6
ja	-0.4	29.5	17.3	25.3	16.3	14.9	28.2	36.2	10^3	37.2	64.8	347.0	66.6	47.6	49.5
ko	2.7	33.9	12.0	17.9	25.1	39.0	26.2	23.4	10^4	112.1	79.1	10^3	22.1	10^4	29.1
zh	-1.6	18.3	8.4	13.5	20.2	26.3	22.8	23.1	10^3	35.2	38.0	87.4	25.8	22.2	57.1

	en	de	fr	it	pt	hi	es	th	bg	ru	tr	vi	ja	ko	zh
en	-1.2	10^4	10^4	10^4	10^5	10^4	10^5	10^4	10^5	10^6	10^3	10^4	10^4	10^3	10^5
de	3.5	2.4	12.2	41.0	39.7	50.9	29.0	12.0	34.6	66.8	293.4	35.2	36.5	41.8	67.6
fr	2.7	63.2	0.6	12.9	20.6	420.0	9.2	17.0	17.9	108.9	968.9	128.7	37.4	104.1	45.6
it	3.1	58.5	10.9	-0.0	664.0	427.6	268.1	18.0	141.2	10^3	10^3	265.3	28.1	567.4	51.2
pt	2.4	48.0	9.6	23.0	-1.3	131.2	29.1	153.5	14.1	98.6	299.8	154.6	75.1	36.9	45.8
hi	16.9	79.5	25.4	46.1	52.9	3.1	61.3	12.9	11.2	9.9	81.5	30.7	19.0	9.7	33.3
es	1.7	46.2	5.5	10.8	9.2	237.7	-1.1	160.2	11.1	117.0	263.5	472.8	66.3	49.3	56.0
th	15.7	58.4	29.6	102.3	157.3	14.3	84.3	6.2	27.0	11.1	165.7	698.9	22.4	17.1	36.2
bg	20.5	387.7	281.8	855.1	10^3	10^3	10^3	52.9	0.1	85.5	10^3	69.8	71.3	495.6	77.7
ru	17.6	108.1	52.9	132.4	105.3	192.9	101.7	18.5	4.1	2.2	929.6	40.3	38.1	84.2	46.3
tr	4.4	74.1	23.0	74.5	399.8	563.0	57.9	36.4	10^3	10^3	-2.3	48.1	33.9	199.8	43.8
vi	5.4	90.0	11.6	34.5	80.6	640.0	40.6	984.4	44.0	142.6	32.4	7.1	41.6	165.1	59.8
ja	36.4	446.4	109.6	348.7	10^3	242.4	428.8	75.6	79.3	151.1	733.2	805.8	4.2	10^4	92.0
ko	23.1	305.4	76.3	177.7	255.7	543.5	161.9	199.2	55.7	162.4	407.5	300.1	46.9	5.7	90.1
zh	32.1	544.1	71.6	148.4	220.8	40.6	202.7	24.4	63.7	117.2	289.5	187.4	21.8	19.2	10.8

Figure 29: PPL changes in Gemma 2 2B. **Top:** Steering using language-specific features with $\alpha = -0.2$ (left) and $\alpha = 0.2$ (right). **Bottom:** Steering using language-specific neurons with $\alpha = -0.2$ (left) and $\alpha = 0.2$ (right).

	en	de	fr	it	pt	hi	es	th	bg	ru	tr	vi	ja	ko	zh
en	4.4	1.8	1.1	1.0	1.6	0.5	1.8	0.3	0.3	0.6	1.4	1.4	0.5	0.2	1.8
de	42.0	10 ⁶	55.1	117.7	110.9	415.4	150.6	189.8	105.8	858.5	228.8	147.2	486.8	182.3	603.9
fr	110.2	656.8	10 ⁵	10 ³	339.4	10 ³	313.4	10 ⁴	312.4	10 ³	717.3	10 ³	10 ³	10 ³	10 ³
it	43.9	295.0	208.7	10 ⁶	10 ³	207.8	444.8	95.0	449.3	116.5	174.8	169.3	265.0	252.2	289.9
pt	27.4	84.3	37.4	139.5	10 ⁵	93.4	159.6	83.3	102.2	119.6	157.1	126.6	117.2	10 ⁴	212.0
hi	15.9	30.7	13.5	19.4	27.1	10 ⁷	28.5	26.5	20.7	25.6	52.5	39.9	64.7	40.4	102.2
es	11.4	21.4	10.7	716.4	290.9	12.8	10 ³	11.7	20.9	21.5	31.9	19.7	27.7	20.6	66.8
th	26.2	76.1	30.0	41.4	45.2	120.8	84.7	10 ⁶	103.3	77.3	94.6	181.0	268.4	10 ³	311.2
bg	77.8	321.1	71.5	10 ⁴	320.4	10 ³	207.0	10 ⁴	10 ⁶	10 ³	747.1	590.7	10 ⁵	10 ³	
ru	5.5	31.6	6.9	11.7	14.1	14.9	15.0	10.5	355.1	10 ⁴	30.0	14.9	20.2	16.0	39.2
tr	97.6	10 ³	649.1	10 ³	10 ³	10 ⁴	10 ³	10 ⁴	10 ⁴	10 ³	10 ³	10 ⁴	10 ⁵	10 ⁶	10 ³
vi	36.0	137.2	66.6	76.5	128.2	177.0	80.6	10 ⁶	152.6	176.8	408.5	10 ⁸	458.9	10 ⁴	10 ³
ja	5.3	25.8	8.3	38.7	11.1	21.1	12.7	15.8	11.7	22.7	38.5	17.2	10 ⁵	142.1	67.1
ko	15.8	24.0	8.3	14.0	23.6	18.7	19.4	25.4	19.2	19.8	90.8	28.2	131.3	10 ⁷	83.8
zh	4.6	9.8	4.7	5.3	15.3	17.0	12.0	18.1	25.4	22.9	13.7	14.0	63.7	122.0	10 ⁵

	en	de	fr	it	pt	hi	es	th	bg	ru	tr	vi	ja	ko	zh
en	-3.0	0.9	1.1	1.5	1.8	0.2	2.0	0.5	0.5	0.4	2.1	0.6	1.6	0.8	1.3
de	10 ³	75.3	10 ⁶	10 ⁵	10 ⁶	10 ⁵	10 ⁶	10 ⁷	10 ⁶	10 ⁷	10 ⁶	10 ⁶	10 ⁶	10 ⁷	10 ⁶
fr	10 ³	10 ⁶	28.5	10 ⁵	10 ⁵	10 ⁵	10 ⁷	10 ⁶	10 ⁸	10 ⁶	10 ⁷	10 ⁶	10 ⁶	10 ⁷	10 ⁶
it	180.1	10 ⁶	10 ⁵	40.6	10 ⁵	10 ⁷	10 ⁵	10 ⁸	10 ⁶	10 ⁴	10 ⁷	10 ⁶	10 ⁷	10 ⁵	10 ⁶
pt	595.3	10 ⁶	10 ⁵	10 ⁶	56.1	10 ⁷	10 ⁴	10 ⁶	10 ⁶	10 ⁶	10 ⁵	10 ⁷	10 ⁶	10 ⁶	10 ⁶
hi	47.2	298.9	31.4	60.2	113.4	12.0	58.4	10 ⁵	10 ⁴	10 ³	10 ⁵	10 ⁴	10 ⁶	10 ⁶	10 ⁴
es	22.4	10 ⁴	10 ⁴	10 ³	10 ³	10 ⁴	7.5	10 ⁴	10 ⁵	10 ⁴	10 ⁴	10 ⁴	274.2	238.1	115.0
th	45.4	480.0	10 ³	10 ³	10 ⁴	10 ⁷	10 ³	30.6	10 ⁵	10 ⁴	10 ⁶	10 ⁶	10 ⁴	10 ⁶	10 ⁵
bg	148.4	10 ⁶	10 ⁷	10 ⁶	10 ⁷	10 ⁷	10 ⁶	10 ⁶	167.3	10 ⁴	10 ⁷	10 ⁶	10 ⁶	10 ⁷	10 ⁵
ru	17.9	10 ⁴	10 ⁴	10 ³	10 ⁵	10 ⁵	10 ⁴	10 ⁵	845.7	14.1	10 ⁵	10 ⁵	10 ⁵	10 ⁵	10 ⁵
tr	10 ³	10 ⁷	10 ⁶	10 ⁵	10 ⁶	10 ⁷	10 ⁵	10 ⁷	10 ⁷	10 ⁷	648.2	10 ⁶	10 ⁷	10 ⁷	10 ⁷
vi	10 ³	10 ⁶	10 ⁶	10 ⁷	10 ⁷	10 ⁸	10 ⁷	10 ⁷	10 ⁷	10 ⁷	206.9	10 ⁹	10 ⁹	10 ⁷	10 ⁷
ja	26.8	58.2	23.9	30.9	106.7	10 ⁵	61.1	10 ⁵	10 ³	497.8	10 ⁴	10 ⁵	43.9	10 ⁶	10 ⁴
ko	23.7	79.9	22.9	147.3	36.2	10 ⁶	30.4	10 ⁵	10 ⁴	10 ³	10 ⁶	10 ³	10 ⁵	13.6	10 ⁴
zh	20.3	375.3	32.2	102.8	34.6	10 ⁵	48.5	10 ⁵	329.3	503.6	10 ³	10 ³	10 ⁴	10 ⁵	21.8

	en	de	fr	it	pt	hi	es	th	bg	ru	tr	vi	ja	ko	zh
en	7.2	10 ⁴	319.6	10 ³	94.0	18.5	148.8	8.5	53.2	9.0	45.4	45.8	12.3	10.0	30.2
de	6.4	445.0	10.4	15.1	17.8	12.7	17.5	9.9	14.8	17.8	34.0	25.1	17.6	12.4	34.4
fr	6.3	20.0	10 ³	40.2	25.5	9.5	28.0	6.8	26.4	13.7	23.8	15.7	15.3	10.2	36.2
it	7.4	24.5	24.4	10 ³	149.2	11.2	109.8	9.0	59.3	18.1	34.6	20.9	20.4	13.6	47.6
pt	5.7	27.6	14.8	57.2	10 ³	27.9	67.9	11.8	52.9	40.6	41.0	25.6	18.3	33.4	51.8
hi	5.6	98.6	30.0	47.1	63.3	190.3	65.8	22.8	10 ³	107.0	155.0	424.2	33.2	22.3	43.5
es	5.6	19.3	19.0	124.9	45.4	19.0	126.1	10.4	38.8	14.6	33.0	21.8	14.4	19.5	40.3
th	2.1	64.8	28.9	38.7	44.4	183.2	52.7	10 ⁶	10 ⁴	504.3	173.8	10.3	73.8	33.9	69.9
bg	15.7	142.9	64.8	198.9	140.4	279.1	124.7	68.0	10 ⁷	10 ³	254.0	203.7	136.0	63.1	172.4
ru	11.7	139.6	48.2	94.6	162.1	312.9	129.8	72.6	10 ⁴	10 ³	187.1	266.6	185.9	58.6	138.5
tr	8.7	63.6	31.8	166.2	29.3	25.9	169.3	31.5	57.5	30.0	10 ⁵	10 ³	37.0	26.2	43.8
vi	8.6	35.3	13.1	27.9	39.8	19.9	31.2	221.0	20.3	26.2	141.3	10 ⁷	19.1	142.0	46.4
ja	8.1	237.0	111.1	176.1	70.6	10 ³	148.5	10 ⁴	10 ⁶	10 ³	777.8	10 ⁵	10 ⁶	10 ³	931.7
ko	17.5	337.1	87.9	132.5	164.8	10 ⁵	198.1	10 ⁶	10 ⁷	10 ³	10 ³	10 ⁶	732.2	10 ⁶	10 ⁴
zh	4.5	75.5	36.3	60.6	93.3	10 ³	121.9	10 ³	10 ⁵	10 ³	200.2	10 ⁴	218.4	978.6	270.0

	en	de	fr	it	pt	hi	es	th	bg	ru	tr	vi	ja	ko	zh
en	-1.4	10 ⁵	10 ⁵	10 ³	10 ⁶	10 ⁵	10 ⁶	10 ⁶	10 ⁶	10 ⁶	10 ⁵	10 ⁵	10 ⁶	10 ⁵	10 ⁶
de	9.2	8.4	411.4	10 ³	10 ³	10 ⁴	942.2	10 ³	10 ⁴	10 ⁴	10 ⁴	10 ³	10 ⁴	10 ⁴	10 ⁵
fr	6.9	10 ³	2.7	54.1	93.1	10 ⁵	32.7	10 ⁴	10 ⁴	10 ⁴	10 ⁴	10 ⁴	10 ⁴	10 ⁵	10 ⁴
it	8.1	10 ³	125.8	2.8	10 ⁴	10 ⁵	10 ³	10 ⁵	10 ⁴	10 ⁵	10 ⁴	10 ⁵	10 ⁴	10 ⁴	10 ⁴
pt	6.3	10 ³	51.2	208.8	2.4	10 ⁵	192.6	10 ⁵	10 ³	10 ⁴	10 ⁴	10 ⁴	10 ⁴	10 ⁴	10 ⁴
hi	65.9	10 ³	671.7	10 ³	10 ³	7.2	10 ³	591.5	191.4	183.9	10 ³	483.6	126.6	60.1	118.4
es	4.9	10 ³	19.1	54.5	30.1	10 ⁵	1.0	10 ⁵	10 ³	10 ⁴	10 ³	10 ⁴	10 ⁴	10 ⁵	10 ⁴
th	66.9	10 ³	10 ⁵	10 ⁵	10 ⁵	10 ³	10 ³	20.3	10 ⁴	10 ³	10 ⁴	10 ⁴	104.2	10 ³	804.0
bg	71.1	10 ⁴	10 ⁴	10 ⁴	10 ⁵	10 ⁵	10 ⁵	10 ³	5.8	588.5	10 ⁵	10 ⁴	10 ³	10 ³	10 ³
ru	56.2	10 ³	722.1	10 ³	10 ³	10 ⁴	10 ³	248.1	9.7	5.8	10 ⁴	538.7	903.9	10 ³	672.6
tr	14.6	10 ³	10 ³	10 ³	10 ⁴	10 ⁴	10 ³	10 ³	10 ⁵	10 ⁵	8.4	10 ⁴	10 ³	10 ⁴	236.1
vi	13.6	10 ³	220.8	10 ³	10 ³	10 ⁴	895.2	10 ⁴	10 ⁴	10 ⁴	595.2	17.4	10 ³	10 ⁴	10 ³
ja	372.7	10 ⁴	10 ⁴	10 ⁴	10 ⁶	10 ⁴	10 ⁴	10 ⁵	10 ³	10 ⁴	10 ⁴	10 ⁴	10 ⁵	13.8	10 ⁵
ko	100.0	10 ⁴	10 ⁴	10 ⁵	10 ⁵	10 ⁴	10 ⁴	10 ⁴	10 ⁴	10 ⁴	10 ⁴	285.1	14.9	617.1	
zh	316.3	10 ⁴	10 ⁴	10 ⁴	10 ⁵	10 ³	10 ⁴	10 ³	10 ³	10 ⁴	10 ³	10 ³	116.8	302.9	35.6

	en	de	fr	it	pt	hi	es	th	bg	ru	tr	vi	ja	ko	zh
en	6.3	3.0	2.1	1.9	2.9	0.8	3.4	0.6	0.6	1.0	2.7	2.3	1.1	0.6	3.2
de	113.8	10 ⁶	642.8	10 ³	922.0	10 ⁴	10 ³	10 ⁴	10 ³	10 ³	10 ³	10 ³	10 ³	10 ³	10 ³
fr	903.2	10 ⁴	10 ⁷	10 ⁵	10 ⁴	10 ⁶	10 ⁴	10 ⁷	10 ³	10 ⁴	10 ⁴	10 ⁶	10 ⁵	10 ⁵	10 ⁵
it	128.0	10 ³	10 ³	10 ⁷	10 ⁴	10 ³	10 ³	10 ³	10 ³	10 ³	10 ³	10 ³	10 ⁴	10 ³	10 ³
pt	95.9	346.0	159.7	10 ³	10 ⁶	10 ³	10 ³	464.6	760.9	908.1	10 ³	584.1	579.4	10 ⁵	911.1
hi	39.1	87.8	36.5	53.8	72.1	10 ⁶	80.9	165.7	79.6	104.9	157.7	109.0	217.1	214.0	301.5
es	25.0	81.3	35.2	10 ³	392.6	82.7	10 ⁴	37.7	92.9	77.8	120.1	49.1	89.9	88.8	215.6
th	61.0	303.6	97.5	138.1	139.5	10 ³	402.9	10 ¹⁰	10 ³	829.2	373.6	10 ³	10 ⁴	10 ⁴	10 ³
bg	255.8	10 ³	633.8	10 ⁵	10 ⁴	10 ⁶	10 ³	10 ⁶	10 ⁸	10 ⁵	10 ⁵	10 ⁴	10 ³	10 ⁷	10 ⁴
ru	12.1	242.4	26.6	35.0	44.1	88.2	39.1	31.0	751.0	10 ⁴	10 ⁴	105.1	45.5	82.0	136.4
tr	415.3	10 ⁴	10 ⁴	10 ⁵	10 ⁵	10 ⁶	10 ⁵	10 ⁷	10 ⁶	10 ⁵	10 ¹⁰	10 ⁵	10 ⁶	10 ⁸	10 ⁵
vi	105.9	10 ³	850.3	737.3	10 ³	10 ⁴	636.7	10 ⁶	10 ³	10 ³	10 ⁴	10 ¹⁰	10 ³	10 ⁷	10 ⁴
ja	12.7	86.7	24.7	298.7	27.7	269.3	28.3	70.4	33.7	112.0	194.4	44.8	10 ⁶	10 ³	300.1
ko	33.8	57.0	19.7	35.5	58.5	101.9	43.7	752.7	65.7	54.8	10 ³	175.3	382.4	10 ⁸	10 ³
zh	11.1	23.1	11.1	13.5	95.4	358.4	48.2	94.4	127.8	110.8	35.0	36.8	752.3	10 ³	10 ⁶

	en	de	fr	it	pt	hi	es	th	bg	ru	tr	vi	ja	ko	zh
en	-3.7	1.8	1.9	2.5	3.1	0.4	3.6	0.8	0.9	0.8	3.7	1.2	2.6	1.3	2.5
de	10 ⁵	278.5	10 ⁶	10 ⁶	10 ⁶	10 ⁷	10 ⁶	10 ⁷	10 ⁷	10 ⁷	10 ⁶	10 ⁷	10 ⁷	10 ⁸	10 ⁷
fr	10 ⁶	10 ⁶	105.7	10 ⁵	10 ⁵	10 ⁷	10 ⁵	10 ⁸	10 ⁸	10 ⁷	10 ⁶	10 ⁷	10 ⁶	10 ⁷	10 ⁷
it	10 ⁴	10 ⁶	10 ⁵	125.8	10 ⁵	10 ⁷	10 ⁵	10 ⁸	10 ⁷	10 ⁷	10 ⁶	10 ⁶	10 ⁸	10 ⁶	10 ⁷
pt	10 ⁵	10 ⁶	10 ⁵	10 ⁵	184.2	10 ⁷	10 ⁵	10 ⁸	10 ⁷	10 ⁶	10 ⁶	10 ⁷	10 ⁶	10 ⁷	10 ⁷
hi	123.1	10 ⁴	283.8	10 ³	10 ³	25.1	469.5	10 ⁶	10 ⁶	10 ⁵	10 ⁶	10 ⁵	10 ⁷	10 ⁶	10 ⁷
es	49.8	10 ⁵	10 ⁵	10 ⁴	10 ⁴	10 ⁶	18.8	10 ⁶	10 ⁶	10 ⁵	10 ⁵	10 ⁷	10 ⁴	10 ⁴	10 ³
th	171.0	10 ⁵	10 ⁶	10 ⁶	10 ⁷	10 ⁸	10 ⁶	90.0	10 ⁶	10 ⁶	10 ⁷	10 ⁷	10 ⁶	10 ⁷	10 ⁶
bg	10 ³	10 ⁷	10 ⁸	10 ⁷	10 ⁷	10 ⁷	10 ⁷	10 ⁷	804.5	10 ⁵	10 ⁸	10 ⁷	10 ⁷	10 ⁷	10 ⁷
ru	52.3	10 ⁵	10 ⁵	10 ⁵	10 ⁵	10 ⁵	10 ⁵	10 ⁵	10 ⁵	10 ³	30.2	10 ⁶	10 ⁵	10 ⁶	10 ⁵
tr	10 ⁵	10 ⁷	10 ⁶	10 ⁶	10 ⁶	10 ⁷	10 ⁶	10 ⁷	10 ⁷	10 ⁷	10 ³	10 ⁷	10 ⁸	10 ⁷	10 ⁷
vi	10 ⁵	10 ⁷	10 ⁷	10 ⁷	10 ⁷	10 ⁸	10 ⁷	10 ⁸	10 ⁸	10 ⁷	10 ⁸	950.7	10 ¹⁰	10 ⁸	10 ⁸
ja	63.4	499.8	121.4	384.8	10 ³	10 ⁴	10 ³	10 ⁶	10 ⁵	10 ⁴	10 ⁵	10 ⁷	93.4	10 ⁴	10 ⁵
ko	57.9	10 ⁴	10 ³	10 ⁴	10 ³	10 ⁶	347.9	10 ⁶	10 ⁵	10 ⁵	10 ⁵	10 ⁵	10 ⁶	32.4	10 ⁵
zh	56.1	10 ⁴	10 ³	10 ³	865.3	10 ³	10 ³	10 ⁶	10 ⁴	10 ⁴	10 ⁵	10 ⁴	10 ⁵	10 ⁶	47.3

	en	de	fr	it	pt	hi	es	th	bg	ru	tr	vi	ja	ko	zh
en	18.5	10 ⁵	10 ⁵	10 ⁵	10 ⁴	121.2	10 ⁴	23.7	254.9	21.9	239.7	250.1	31.0	21.4	81.2
de	12.3	10 ³	25.1	37.2	40.6	42.6	42.4	29.7	34.9	47.8	89.0	68.7	40.0	34.8	76.3
fr	11.8	57.6	10 ⁴	271.1	208.1	26.0	104.7	18.6	79.3	35.6	56.9	39.1	31.9	21.4	75.2
it	13.5	60.8	325.5	10 ⁵	10 ³	29.8	10 ³	26.2	221.8	42.8	92.3	48.0	53.8	28.5	103.5
pt	11.8	75.8	136.6	674.8	10 ⁴	143.7	764.0	39.5	245.8	95.7	125.6	69.9	40.9	231.5	190.2
hi	18.7	725.5	116.8	190.0	249.4	10 ⁴	263.7	10 ³	10 ⁵	10 ⁴	845.9	10 ⁴	99.8	76.0	102.7
es	9.7	50.0	615.3	10 ³	10 ³	59.2	10 ⁴	30.3	132.9	31.2	95.8	56.5	30.9	67.1	91.6
th	11.8	270.4	110.9	151.6	149.2	10 ⁵	186.3	10 ⁷	10 ⁶	10 ⁴	10 ³	10 ⁶	10 ³	10 ³	225.5
bg	47.3	745.2	379.8	10 ³	751.6	10 ⁴	570.1	10 ³	10 ⁸	10 ⁶	10 ³	10 ³	10 ³	966.9	10 ³
ru	38.6	798.5	227.8	465.8	979.4	10 ⁴	752.9	10 ³	10 ⁶	10 ⁶	947.3	10 ³	10 ³	10 ³	10 ³
tr	17.3	274.1	236.8	10 ³	80.2	123.2	757.4	528.8	215.1	87.4	10 ⁶	10 ⁴	90.3	89.2	112.8
vi	19.1	104.0	38.9	96.8	134.1	66.3	98.2	10 ⁴	57.8	68.4	10 ³	10 ⁸	46.6	10 ³	135.3
ja	28.8	10 ³	830.3	10 ³	276.8	10 ⁶	614.0	10 ⁷	10 ⁷	10 ⁶	10 ⁴	10 ⁶	10 ⁷	10 ⁵	10 ⁶
ko	54.3	10 ⁴	844.7	10 ³	10 ³	10 ⁸	10 ³	10 ⁸	10 ⁹	10 ⁷	10 ⁴	10 ⁸	10 ⁷	10 ⁷	10 ⁸
zh	20.4	346.6	157.1	263.2	449.1	10 ⁵	836.7	10 ⁵	10 ⁷	10 ⁵	10 ³	10 ⁶	10 ³	10 ⁴	10 ⁴

	en	de	fr	it	pt	hi	es	th	bg	ru	tr	vi	ja	ko	zh
en	-1.3	10 ⁵	10 ⁵	10 ⁵	10 ⁶	10 ³	10 ⁵	10 ⁶	10 ⁶	10 ⁶	10 ⁵	10 ⁵	10 ⁶	10 ⁵	10 ⁶
de	19.8	19.0	10 ⁴	10 ⁴	10 ⁴	10 ⁵	10 ⁴	10 ⁵	10 ⁵	10 ⁵	10 ⁵	10 ⁵	10 ⁴	10 ⁶	10 ⁶
fr	13.5	10 ⁴	6.2	216.5	432.7	10 ⁶	100.6	10 ⁶	10 ⁶	10 ⁵	10 ⁵	10 ⁵	10 ⁵	10 ⁵	10 ⁵
it	16.0	10 ⁴	936.5	7.3	10 ⁴	10 ⁶	10 ⁴	10 ⁶	10 ⁵	10 ⁶	10 ⁵	10 ⁵	10 ⁶	10 ⁶	10 ⁶
pt	12.9	10 ⁴	231.9	10 ³	9.1	10 ⁶	632.2	10 ⁶	10 ⁴	10 ⁵	10 ⁴	10 ⁵	10 ⁶	10 ⁶	10 ⁶
hi	257.9	10 ⁵	10 ⁴	10 ⁵	10 ⁵	13.6	10 ⁴	10 ⁴	10 ⁴	10 ³	10 ⁴	10 ³	10 ³	578.2	498.2
es	10.4	10 ⁴	72.1	333.6	79.0	10 ⁴	4.8	10 ⁶	10 ⁵	10 ⁵	10 ⁴	10 ⁵	10 ⁴	10 ⁶	10 ⁶
th	327.2	10 ⁵	10 ⁷	10 ⁷	10 ⁷	10 ³	10 ⁷	10 ³	10 ⁷	52.8	10 ⁵	10 ⁵	10 ⁶	10 ⁵	10 ⁴
bg	183.8	10 ⁵	10 ⁶	10 ⁶	10 ⁶	10 ⁶	10 ⁶	10 ⁶	10 ⁴	16.2	10 ³	10 ⁶	10 ⁵	10 ⁵	10 ⁴
ru	130.8	10 ⁴	10 ³	10 ⁴	10 ⁴	10 ⁴	10 ³	10 ³	19.3	11.6	10 ⁵	10 ³	10 ⁴	10 ⁴	10 ⁴
tr	38.3	10 ⁵	10 ⁴	10 ⁴	10 ⁵	10 ⁵	10 ⁴	10 ⁵	10 ⁶	10 ⁶	31.2	10 ⁵	10 ⁵	10 ⁵	10 ⁴
vi	26.5	10 ⁴	10 ⁴	10 ⁵	10 ⁵	10 ⁵	10 ⁵	10 ⁵	10 ⁵	10 ⁵	10 ⁴	34.4	10 ⁵	10 ⁵	10 ⁵
ja	10 ³	10 ⁵	10 ⁵	10 ⁶	10 ⁶	10 ⁵	10 ⁶	10 ⁵	10 ⁶	10 ⁵	10 ⁵	10 ⁵	29.1	10 ⁶	10 ⁴
ko	313.4	10 ⁶	10 ⁵	10 ⁶	10 ⁶	10 ⁵	10 ⁵	10 ⁵	10 ⁵	10 ⁵	10 ⁵	10 ⁵	10 ³	30.6	10 ³
zh	10 ³	10 ⁶	10 ⁵	10 ⁵	10 ⁶	10 ⁴	10 ⁶	10 ⁴	10 ⁴	10 ⁵	10 ⁴	10 ⁴	605.5	10 ³	79.7

Figure 31: PPL changes in Gemma 2 2B. **Top:** Steering using **language-specific features** with $\alpha = -0.4$ (left) and $\alpha = 0.4$ (right). **Bottom:** Steering using **language-specific neurons** with $\alpha = -0.4$ (left) and $\alpha = 0.4$ (right).

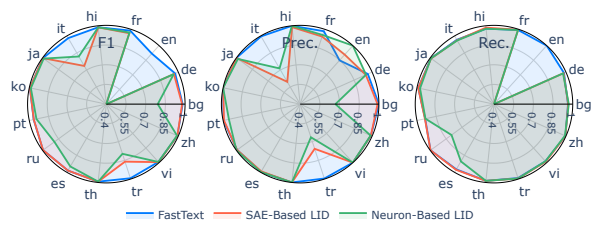


Figure 32: Results of SAE-based and neuron-based LID derived from Gemma 2 2B, and fastText LID, applied to WiLI-2018 across 15 languages.

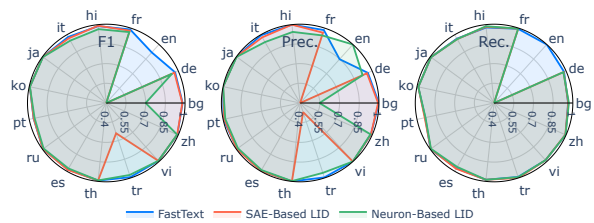


Figure 33: Results of SAE-based and neuron-based LID derived from Gemma 2 9B, and fastText LID, applied to WiLI-2018 across 15 languages.

Language	SAEs Classifier			FastText			Neurons Classifier		
	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1
bg	0.990	0.966	0.978	1.000	0.964	0.982	0.669	0.970	0.792
de	0.948	0.980	0.964	0.965	0.984	0.974	0.949	0.976	0.963
en	0.907	0.098	0.177	0.849	1.000	0.918	1.000	0.016	0.031
fr	0.941	0.996	0.968	0.984	0.994	0.989	0.954	0.994	0.974
hi	0.990	0.986	0.988	1.000	0.978	0.989	1.000	0.978	0.989
it	0.596	0.962	0.736	0.996	0.966	0.981	0.714	0.960	0.819
ja	1.000	0.992	0.996	1.000	0.994	0.997	0.986	0.994	0.990
ko	1.000	0.990	0.995	1.000	0.990	0.995	1.000	0.976	0.988
pt	0.994	0.938	0.965	0.998	0.936	0.966	0.955	0.934	0.944
ru	0.994	0.998	0.996	0.994	0.998	0.996	1.000	0.800	0.889
es	0.992	0.978	0.985	0.996	0.974	0.985	0.998	0.904	0.949
th	1.000	0.988	0.994	1.000	0.988	0.994	1.000	0.988	0.994
tr	0.758	0.996	0.861	0.996	0.992	0.994	0.666	0.996	0.798
vi	1.000	0.990	0.995	1.000	0.986	0.993	1.000	0.986	0.993
zh	0.990	0.996	0.993	0.990	0.996	0.993	0.988	0.992	0.990
Macro Avg	0.940	0.924	0.906	0.984	0.983	0.983	0.925	0.898	0.873
Accuracy	0.924			0.983			0.898		

Table 5: Results of SAE-based and neuron-based LID derived from Gemma 2 2B, and fastText LID, applied to WiLI-2018 across 15 languages. **Bold** text indicates the performance is equal or greater than the counterpart.

Language	SAEs Classifier			FastText			Neurons Classifier		
	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1
bg	0.994	0.966	0.980	1.000	0.964	0.982	0.547	0.970	0.700
de	0.957	0.980	0.968	0.965	0.984	0.974	0.921	0.978	0.949
en	0.000	0.000	0.000	0.849	1.000	0.918	1.000	0.096	0.175
fr	0.963	0.996	0.979	0.984	0.994	0.989	0.941	0.996	0.968
hi	1.000	0.984	0.992	1.000	0.978	0.989	0.945	0.988	0.966
it	0.978	0.962	0.970	0.996	0.966	0.981	0.936	0.962	0.949
ja	1.000	0.994	0.997	1.000	0.994	0.997	0.998	0.994	0.996
ko	1.000	0.990	0.995	1.000	0.990	0.995	1.000	0.990	0.995
pt	0.992	0.940	0.965	0.998	0.936	0.966	0.983	0.934	0.958
ru	0.996	0.996	0.996	0.994	0.998	0.996	0.986	0.994	0.990
es	0.998	0.978	0.988	0.996	0.974	0.985	0.990	0.962	0.976
th	1.000	0.988	0.994	1.000	0.988	0.994	0.998	0.988	0.993
tr	0.473	1.000	0.643	0.996	0.992	0.994	0.959	0.994	0.976
vi	1.000	0.986	0.993	1.000	0.986	0.993	1.000	0.986	0.993
zh	0.990	0.994	0.992	0.990	0.996	0.993	0.986	0.996	0.991
Macro Avg	0.889	0.917	0.897	0.984	0.983	0.983	0.946	0.922	0.905
Accuracy	0.917			0.983			0.922		

Table 6: Results of SAE-based and neuron-based LID derived from Gemma 2 9B, and fastText LID, applied to WiLI-2018 across 15 languages. **Bold** text indicates the performance is equal or greater than the counterpart.

Task	Accuracy	Std. Error
XNLI		
bg	0.3944	0.0098
de	0.4562	0.0100
en	0.5450	0.0100
es	0.4313	0.0099
fr	0.4618	0.0100
hi	0.3964	0.0098
ru	0.4659	0.0100
th	0.4092	0.0099
tr	0.4261	0.0099
vi	0.4402	0.0100
zh	0.3402	0.0095
PAWS-X		
de	0.5510	0.0111
en	0.6030	0.0109
es	0.5675	0.0111
fr	0.5700	0.0111
ja	0.5370	0.0112
ko	0.5385	0.0111
zh	0.4985	0.0112
XWinograd		
en	0.8129	0.0081
fr	0.6506	0.0527
ja	0.5996	0.0158
pt	0.6616	0.0292
ru	0.5968	0.0277
zh	0.6389	0.0214

Table 7: Llama 3.2 1B evaluation results for XNLI, PAWS-X, and XWinograd tasks in the zero-shot setting with lm-evaluation-harness (Gao et al., 2024). **Green text** indicates the explicitly supported languages by Llama 3.2 1B.

Layer	Lang	Feature ID	Top Tokens
layers.2.mlp	Chinese	18693	nam, da, plet, 堂, 公, umba, Bundes, da, uga, 文
layers.6.mlp	Chinese	126429	ряд, ibar, essed, cur, ar, tu, cur, quer, Ecc, ined
layers.7.mlp	Chinese	82671	imum, olith, Economics, marsh, curr, obel, .settings, Lord, esh, 滿
layers.9.mlp	Chinese	41091	rosso, acen, agn, agner, /all, ,No, 无, opens, -No, UGHT
layers.10.mlp	Chinese	86769	aling, aber, □, 指, □, Sud, aled, lej, □, 便
layers.10.mlp	Chinese	119196	COMPAT, jig, shalt, ulia, PLUS, urn, 有, 其, 而, èles
layers.11.mlp	Chinese	240	amet, Erf, tings, multiplic, atori, 影, mast, mul, barr, amon
layers.11.mlp	Chinese	38263	rani, employed, خانه, iola, iad, ueil, ROAD, instr, ox, ût
layers.11.mlp	Chinese	40026	们, udes, obbled, avers, umb, odes, 們, omain, AMES, chn
layers.11.mlp	Chinese	65660	therefore, draws, همج, drawing, Pract, Draws, vždy, есть, originally, real
layers.11.mlp	Chinese	73417	wore, lacked, ebi, simmer, 户, 家伙, 消息, tasted, atop, حصص
layers.12.mlp	Chinese	104224	quals, 中, 的, 繁, WikiLeaks, □, 之一, 的情, LLP, awner
layers.13.mlp	Chinese	113658	app, APP, ww, raw, WS, owe, tw, tw, the, App
layers.13.mlp	Chinese	118430	Under, Seats, ivé, Am, -Am, infos, round, works, ové, Under
layers.14.mlp	Chinese	12944	unca, ۱۱۱, 第一次, 納, 纳, पहल, 首, 訪, lan, ersten
layers.14.mlp	Chinese	12987	Water, enity, ill, seq, asy, illac, ugo, abis, Vol, 雅
layers.14.mlp	Chinese	49167	, 它,, 将,, 为,, 本,, 其,, 从,, 这,, 每,, 大,, 或
layers.14.mlp	Chinese	66584	间, 师, ullo, urbed, ativos, 周期, hoa, atures, iders, amentos
layers.14.mlp	Chinese	88535	□, reative, 司, □, 余, 任, 文, 但, 示, 属
layers.14.mlp	Chinese	103432	程度, 斗, 脸, chez, 眼睛, 身上, 令人, 赫, 事情, 反应
layers.14.mlp	Chinese	112457	□, 锦, 到, 则, 干, 场, 苏, -Fi, □, シヨ
layers.15.mlp	Chinese	11994	奪, an, S, We, Hi, 學校, Tr, In, 諾, Trav
layers.15.mlp	Chinese	38094	提交, 注册, 访问, 客户, 培训, 活动, 登录, 安装, 显示, 输出
layers.15.mlp	Chinese	114546	民主, 社, 家, 嘉, 世, 利, 邦, 耳, 彼, 美

Figure 34: Top 10 tokens promoted by multiplying Chinese-specific features with the token embedding matrix of Llama 3.2 1B. **Bold** text indicates tokens associated with Chinese.

Layer	Lang	Feature ID	Top Tokens
layers.5.mlp	French	63694	France , ois, Antoine, France, france, 프랑스, Quebec, 法国, agne, Québec
layers.8.mlp	French	86519	負, aff, malar, -controls, 'aff, gos, ilde, Bowling, 励, +")
layers.9.mlp	French	16701	ulaire, Marseille, pou, agne, travail, pour, Bordeaux, télé, istique, éo
layers.9.mlp	French	44983	iste, pou, istes, jeunes, site, ulaire, quelque, toujours, ét, ité
layers.11.mlp	French	25732	pou, ernes, encore, après, ainsi, aussi, existence, europé, aup, quan
layers.11.mlp	French	110940	même, seulement, contre, trait, ulant, deme, jamais, dont, trait, ache
layers.12.mlp	French	22084	ét, quelque, dé, trois, même, sous, pour, é, ré, thé
layers.12.mlp	French	41725	sobre, Wed, urb, ough, یش, tot, Bowen, ow, corp , pastor
layers.13.mlp	French	6455	l , le, les, la, (l, л, .l, {l, _l, =l
layers.13.mlp	French	77011	c, il, ils, comment, comment, Comment, il, commenting, pas, j
layers.14.mlp	French	45764	."] , ' , ' , ' , ' , \ , ' , , ... , ' , '

Figure 35: Top 10 tokens promoted by multiplying French-specific features with the token embedding matrix of Llama 3.2 1B. **Bold** text indicates tokens associated with French.

Layer	Lang	Feature ID	Top Tokens
layers.2.mlp	German	56064	baum, öt, feld, ö, mann, kre, platz, \xf, ости, auf
layers.5.mlp	German	31366	zburg, utsch, GmbH, immer, ammer, german, Germany, lage, we, nummer
layers.8.mlp	German	79729	grave, eros, здесь, Serv, exposures, ck, vis, requestOptions, serialVersionUID, args
layers.9.mlp	German	9083	rings, chers, bere, -ring, wel, シク, nation, □, ring, bere
layers.10.mlp	German	44371	lod, zf, ,:,: , nod, Lod, för, zial, öö, Kee, pal
layers.11.mlp	German	27401	unge, unters, über, fig, ge, verst, nach, über, groß, fig
layers.11.mlp	German	52681	durch, gesch, erw, unter, eins, auf, auss, gleich, ange, voll
layers.12.mlp	German	107855	lichkeit, 来说, gaben, 大会, behand, ây, Whilst, tas, py, .</
layers.14.mlp	German	312	und, Und, Und, und, och, og, oder, UND, e, bzw
layers.14.mlp	German	48455	“ ” , .&, ." , ,: , " , " " " , " " " " , " " "
layers.15.mlp	German	17109	Pokemon, Taco, Spider, Elvis, Tesla, Viking, Batman, Dodge, Apollo, Marvel
layers.15.mlp	German	75403	ch, f, p, st, m, ke, el, h, de, pen
layers.15.mlp	German	130833	* " MC. CD. ã. □. PC

Figure 36: Top 10 tokens promoted by multiplying German-specific features with the token embedding matrix of Llama 3.2 1B. **Bold** text indicates tokens associated with German.

Layer	Lang	Feature ID	Top Tokens
layers.1.mlp	Italian	28636	term, ertiary, mag , WAYS, 중애, absor, terms, ità, 的事情, -term
layers.1.mlp	Italian	47594	endorsements, storm, шиб, -di, -files, Damage, повре, broken, presence, endorsement
layers.2.mlp	Italian	33390	Italian , Italy , Italians , tratt, izione, Italian , zzo, Italy , Giul, italiana
layers.3.mlp	Italian	120200	Ris , Luigi , enne, vent, igli , agli , italian , del, onta, atto
layers.4.mlp	Italian	114161	Italians , italiana, italian, Italia, Italian, Luigi , Milano, Italian, ucch , italiani
layers.6.mlp	Italian	109638	romo, gian, gi, FLT, @",, emo , uctions, ollower, *,, тко
layers.7.mlp	Italian	52987	radix, ھ, Hall, flex, nce, Conce, Liked, hall, Empty, heim
layers.8.mlp	Italian	13547	alcon, aldi, уди, □, Sudan, atti, ate, gli , addir, @g
layers.9.mlp	Italian	5106	ò , aggi , più , ancora, migli , egg , storia, dei, ansa, ott
layers.9.mlp	Italian	58424	suppl, ott , att, butt, ilk, aggi , migli , pubb , impe, ott
layers.10.mlp	Italian	38479	oltre , nell, tutte, tratt, anche, nel, quanto, rag, tut, nella
layers.10.mlp	Italian	108570	già , oltre, semp, questa, pubb , anche, tratt, altro, risult, tutte
layers.11.mlp	Italian	14702	tuy, apid, 拾, نام, uellement, ément, urray, 匕, tiler, utin
layers.11.mlp	Italian	35463	milan , trou, timings, ater, Visual, Milan , Visual, -case, ميل, Pasta
layers.12.mlp	Italian	3234	sobre, complet , oron, selv, destin, gore, ill, completo , attend, semp
layers.13.mlp	Italian	30409	ec, vivo, es, pi , impe, var, collo, e, ecc, -ar
layers.13.mlp	Italian	44286	l , le , la , {l, .l, (l, :l, л, lo , les
layers.14.mlp	Italian	90522	di , del di , -di, della, dello, Di, Di, di, dell

Figure 37: Top 10 tokens promoted by multiplying Italian-specific features with the token embedding matrix of Llama 3.2 1B. **Bold** text indicates tokens associated with Italian.

Layer	Lang	Feature ID	Top Tokens
layers.0.mlp	Japanese	100720	dou, inu, inalg, Means, WCS, histor, iz, SPI, jid, Phon
layers.0.mlp	Japanese	112017	hci, 館, ogui, itsu, achi, eka, unilateral, oga, PEC, piler
layers.8.mlp	Japanese	32154	elda, innie, alt, roz, programma, contro, DRAM, ambi, arious, itar
layers.9.mlp	Japanese	34376	ihu, □, ativní, granularity, Morav, Suppress, upro, upert, _REFERER, grav
layers.11.mlp	Japanese	29972	azaar, 도어, ご, sidewalks, /org, 代, ôte, -match, Teams, Commit
layers.11.mlp	Japanese	126209	afén,) は, Generic, Leh, apos, フ, 略, owo, Generic, generic
layers.12.mlp	Japanese	25492	orm, していた, .\$. Own, om, angep, ramids, SEQ, lical, *,
layers.12.mlp	Japanese	121526	HEL, UFFER, Password, isol, uffers, udd, elda, brief, っ, hypers
layers.13.mlp	Japanese	44429	viz, Nicht, Wallace, kHz, vak, luqt, Neck, ault, 性的, Gel
layers.13.mlp	Japanese	72538	シ, 発, マ, では, リ, 広, ライ, プ, サ, メ
layers.13.mlp	Japanese	77268	oga, abinet, lint, シント, 舗, のだ, 目を, Nations, нив, 平
layers.13.mlp	Japanese	80366	ropol, Orn, inally, ment, privacy, Ist, ist, privacy, advertiser, tong
layers.13.mlp	Japanese	84606	loth, Sawyer, Skyl, □, rees, root, opc, 制, cee, oser
layers.13.mlp	Japanese	101925	があり, 措, orts, Vlad, をした, から, ulaire, には, utterstock, olor
layers.13.mlp	Japanese	121228	イス, フィ, ット, ラ, イ, ロ, O, ク, 素, ばかり
layers.13.mlp	Japanese	125863	pora, #undef, Alpine, drift, decorators, omit, pulse, RECEIVE, Actual, Zub
layers.14.mlp	Japanese	17056	кан, rum, umin, こ, □, なが, 耳, とも, rez, 乃
layers.14.mlp	Japanese	17850	かに, □, unal, なる, な, により, 故, とする, alar, が出
layers.14.mlp	Japanese	34692	たら, 間に, 頂, ざ, わ, ば, 級, 細, ね, 節
layers.14.mlp	Japanese	43240	And, And, OMEM, -validation, Demp, 、小, /angular, 和, 児, AndUpdate
layers.14.mlp	Japanese	44808	ctors, ypc, sez, omen, TW, inters, isode, Ens, ypr, -dist
layers.14.mlp	Japanese	77483	vol, 及び, 福, ve, ordered, 的手, abb, ikers, pios, té
layers.14.mlp	Japanese	85299	ле, usi, era, 998, uk, okoj, □, and, -life, 258
layers.14.mlp	Japanese	97905	asured, aff, her, .cl, bose, anson, icken, screened, dent, ash
layers.14.mlp	Japanese	98700	pay, issy, □, gre, patrick, iky, 丁, □, âk, igham
layers.14.mlp	Japanese	119697	hai, -blog, ht, pomoc, 안, hd, Blog, hest, valid, col
layers.15.mlp	Japanese	4131	ghi, edic, 国際, NI, edu, ERIC, LM, eds, TJ, 桑
layers.15.mlp	Japanese	12716	redi, es, каж, ve, 的, dom, Jeho, 至少, 把, stick
layers.15.mlp	Japanese	25135	, しかし, itr, それは, けれど, そんな, I, 俺は, iyi, ььь
layers.15.mlp	Japanese	35007	像, 意思, 事情, 等, 表示, について, ㄸ, olang, によ, 利用
layers.15.mlp	Japanese	95097	abase, しており, ㄸ, しました, arer, ください, Meh, usz, hindsight, ацц
layers.15.mlp	Japanese	95151	いた, つ, いて, っても, えない, たく, えば, いている, らしい, さんが
layers.15.mlp	Japanese	102746	ジャ, ドラ, スマ, ランド, ハイ, ブリ, イヤ, ヌ, ミュ, キュ

Figure 38: Top 10 tokens promoted by multiplying Japanese-specific features with the token embedding matrix of Llama 3.2 1B. **Bold** text indicates tokens associated with Japanese.

Layer	Lang	Feature ID	Top Tokens
layers.0.mlp	Korean	87423	kim, velt, Office, PMC, orney, ordinances, agre, %/, agony, office
layers.4.mlp	Korean	99278	Kim, Seoul, Jae, Kim, .kr, Samsung, Korean, Korea, Shin, Koreans
layers.6.mlp	Korean	109962	Highlands, ussels, iteur, tails, Ire, ativ, \$sql, plant, jsp, beginnings
layers.7.mlp	Korean	115377	vain, ane, ANE, lett, att, anes, roc, 区, igne, olina
layers.8.mlp	Korean	94922	ante, iente, innie, ensed, λλη, ssh, flips, us, Liter, itar
layers.9.mlp	Korean	31611	beck, acles, ataka, batt, -wrap, etat, pty, umn, Kear, chat
layers.9.mlp	Korean	53931	iosa, gli, chet, conl, Giov, 89, adero, ewolf, worm, edit
layers.10.mlp	Korean	96918	-Feb, partial, juan, pur, ipro, capitals, URATION, 膜, [[, experiences
layers.10.mlp	Korean	125019	当, Mori, Petty, Yug, Touch, che, bare, VIC, .jp, omi
layers.11.mlp	Korean	64252	viz, avour, aura, 今日, 琴, UPDATED, apl, intend, intention, ITT
layers.11.mlp	Korean	102511	r, 으로, FR, aar, l, war, 은, Protocol, айд, 이라
layers.12.mlp	Korean	27775	ptime, epochs, ascending, located, plays, inf, idual, Microsystems, 舖, layers
layers.12.mlp	Korean	67845	Kim, Korea, Korean, Seoul, Park, Kim, Hyundai, kim, Koreans, Je
layers.13.mlp	Korean	78512	plays, 바람, play, Maar, 편, Boost, maz, robots, antics, 帖
layers.14.mlp	Korean	87496	상, 령, 선, 이상, 나, 신, 증, 등, 한, 왕
layers.14.mlp	Korean	107903	asc, ey, anz, erc, acic, eh, ans, eyi, ekl, erv
layers.15.mlp	Korean	17864	이야기, 그것, 있었다, 없었다, 사람들이, 인정, 행동, 사람은, 이야, 그를
layers.15.mlp	Korean	41829	적, 찰, 리로, 리카, 림, □, 어나, 린이, 니아, 택
layers.15.mlp	Korean	57880	?, ? , "?", "?", "?" , "?", "(, "?", "?", "?:
layers.15.mlp	Korean	60936	통, 작, 소, 지, 그, 기, 판, 누, 빙, 식
layers.15.mlp	Korean	114116	LETE, Passive, 설명, flashlight, 게임, careful, REVIEW, ▼, 中文, 问
layers.15.mlp	Korean	131027	., reckon, örper, 림, ., 간, 모습, ., 손, лива

Figure 39: Top 10 tokens promoted by multiplying Korean-specific features with the token embedding matrix of Llama 3.2 1B. **Bold** text indicates tokens associated with Korean.

Language	Detection				Fuzzing			
	Acc.	F1	Prec.	Rec.	Acc.	F1	Prec.	Rec.
Bulgarian	0.79	0.77	0.89	0.73	0.77	0.76	0.83	0.77
Chinese	0.61	0.59	0.69	0.60	0.68	0.65	0.72	0.64
English	0.59	0.66	0.59	0.80	0.66	0.62	0.74	0.62
French	0.82	0.78	0.94	0.68	0.82	0.81	0.89	0.75
German	0.86	0.83	0.96	0.74	0.83	0.81	0.89	0.75
Hindi	0.78	0.70	0.96	0.58	0.78	0.72	0.92	0.63
Italian	0.69	0.58	0.84	0.53	0.68	0.62	0.75	0.61
Japanese	0.70	0.60	0.92	0.49	0.73	0.67	0.87	0.58
Korean	0.69	0.53	0.93	0.43	0.69	0.58	0.87	0.50
Portuguese	0.85	0.82	0.93	0.77	0.82	0.81	0.86	0.80
Russian	0.74	0.65	0.93	0.52	0.75	0.71	0.86	0.64
Spanish	0.74	0.70	0.85	0.60	0.70	0.71	0.73	0.72
Thai	0.70	0.67	0.83	0.63	0.69	0.68	0.78	0.70
Turkish	0.78	0.71	0.94	0.61	0.78	0.74	0.89	0.68
Vietnamese	0.76	0.68	0.92	0.59	0.77	0.70	0.90	0.63
Average	0.74	0.68	0.87	0.62	0.74	0.71	0.83	0.67

Table 8: Average interpretation scores of language-specific features in Llama 3.2 1B for detection and fuzzing across languages.

Layer	Lang	Feature ID	Top Tokens
layers.4.mlp	Russian	48343	dra, Ha, Pavel, standart , turned, Ha, Vladimir , amounted, Vlad, Ukr
layers.6.mlp	Russian	121507	acz, port, đrong, zik, swer, clist, eed, lish, Niet, activity
layers.7.mlp	Russian	110894	ört, gun, sectional, pix, ø, (typeof, Schools, bond, Havana, Vendor
layers.8.mlp	Russian	69357	Ell, Qing, 말 , ля, 리 , Lou, 리 , Antar, tail, sire
layers.8.mlp	Russian	109317	isse, Maid, MainActivity, allowed, .low, AIN, -м, -shell, 】 , weed
layers.9.mlp	Russian	30251	itious, ijo, sexual, @ , & , ri, anky, Tec, , , @n
layers.9.mlp	Russian	59767	Corona, Forward, Forward, Fits, _CORE, urette, Loader, Arg, øj, Demo
layers.10.mlp	Russian	56594	Sergey, Moscow, kh, Dmitry, Russian, Mikhail, .ru, Sergei, Dmit, enko
layers.11.mlp	Russian	8452	п, в, Р, дв, с, к, П, ок, с, Г
layers.11.mlp	Russian	74675	Major, Major, ав, Band, major, bands, -band, major, д , Bands
layers.12.mlp	Russian	31110	flat, iones, ereco, ushing, Flat, Ledger, бы , amat, 兽 , iona
layers.12.mlp	Russian	48228	undy, ickle, ела , Buen, acidad, apo, idos, вед , ISH, веч
layers.13.mlp	Russian	29244	группы, статьи, oki, войны, osoby, устройства, культуры, таки, ины, карти
layers.13.mlp	Russian	40052	hol, hol, hower, 等 , чи , aul, iked, Hol, FEMA, дня
layers.13.mlp	Russian	50197	eer, yii , MAD, mad, об , ыс, ц , коп , Americas, 叶
layers.13.mlp	Russian	68491	akh, ax, лях, ках, ших, kh, ных, овых, анных, щих
layers.13.mlp	Russian	91930	mark, me, yours, ithe, компании , :f, plast, test, him, autom
layers.13.mlp	Russian	97322	г, dda, essional, □ , ось , ittest, endif, прав , jure, osci
layers.13.mlp	Russian	103291	inside, within, v, inside, Within, within, Inside, Inside, Within, during
layers.13.mlp	Russian	115738	еп, евые, ев, еры , echa, eyn, ebi, imenti, ева , 矢
layers.13.mlp	Russian	120158	ango, Invisible, USD, uppies, ctest, ANGO, Ghost, s, Provider, cooper
layers.14.mlp	Russian	44860	возмож, сог, ое, выб, больш, ён, ож, внутрен, енные, соврем
layers.14.mlp	Russian	71946	име, возв, вов, wash, мужчин, уже, locked, вещ, когда, предпоч
layers.15.mlp	Russian	56964	ленин, енными, ственных, ственного, лением, нему, анием, ательных, ческих, енности
layers.15.mlp	Russian	65387	rh, fl, помощи , Santana, ral, north, mid, Mellon, действительно, ывает
layers.15.mlp	Russian	119968	atum, alling, affle, аться, ать , iat, айте , ats, af, аю

Figure 40: Top 10 tokens promoted by multiplying Russian-specific features with the token embedding matrix of Llama 3.2 1B for layers 14–15. **Bold** text indicates tokens associated with Russian.

Layer	Lang	Feature ID	Top Tokens
layers.2.mlp	Spanish	17631	Ñ , Herrera, ع , Españ , arde, íg , itas, Este , ina, Greatest
layers.3.mlp	Spanish	4518	normalize, normalization, iales, constructors, ices, nacional, cales, Nuevo, Puerto, Penal
layers.5.mlp	Spanish	32141	Gal , gal, едь, uced, Tx, ga , Tor, tor, ocol, fram
layers.11.mlp	Spanish	94207	ya, desarroll , mayor, cual, hoy, más, suger, menos, que , ci
layers.13.mlp	Spanish	53088	es , Es, Es, es , fue, era, puede, -es, ha, ES

Figure 41: Top 10 tokens promoted by multiplying Spanish-specific features with the token embedding matrix of Llama 3.2 1B. **Bold** text indicates tokens associated with Spanish.

Layer	Lang	Feature ID	Top Tokens
layers.0.mlp	Vietnamese	11601	utura , 立, idelberg, ora, ту, яб, ём, ighth, 雅, .heap
layers.0.mlp	Vietnamese	102922	benh , tat , tapi , untas , beck, ulate, '];, condition, Conditions, Covered
layers.3.mlp	Vietnamese	14622	ilot, анов, iste, elligence, ista, angl, ekk, istes, dit, esta
layers.4.mlp	Vietnamese	110587	nes, bt, bit, ô, simulation, optimizer, advanced, .logical, uga, cell
layers.5.mlp	Vietnamese	49501	Romanian, Romania, arium, Orioles, 楚, ovna, fortunes, escape, alat, ernes
layers.7.mlp	Vietnamese	28785	千, actor, iere, watch, regor, alian, heel, bă, μή, watch
layers.8.mlp	Vietnamese	79242	stro, autob, benign, IDGE, Mandatory, istance, stin, pon, Jaw, 諸
layers.9.mlp	Vietnamese	61383	enga, ella, angle, ěj, Zuk, obel, expert, apro, cassette, issor
layers.10.mlp	Vietnamese	42004	ăs, 族自治, outh, stan, 馥, atti, aste, Gujar, boiler, считается
layers.10.mlp	Vietnamese	42167	cuent, Ū, Sez, yn, strument, ène, ossal, ifferent, Contemporary, Cancelled
layers.11.mlp	Vietnamese	4065	ho , Ho , Ho , anness, anguage, osy, urd, acie, oppins, цез
layers.11.mlp	Vietnamese	58535	anh , ph , nh , tinh , Vietnamese , Nguyen , Thanh , Ho , Viet , sinh
layers.11.mlp	Vietnamese	66074	Mon , Ph , Y , yt, dle, antine, C, 论, loit, Cone
layers.12.mlp	Vietnamese	44940	μά, труда, Teh, conto, ulled, ccd, thresh, ໙໓, ialized, marketing
layers.12.mlp	Vietnamese	56617	geries, multiples, унк, čka, анка, 治, iership, nič, yanında, Cleveland
layers.12.mlp	Vietnamese	63803	Than , Van , VAN , Van , hung , -Tr , Tran , Thi , Nguyen , THAN
layers.12.mlp	Vietnamese	92851	emp, нова, umb, mega, seb, edy, ENCY, cé, ୧୧, ora
layers.12.mlp	Vietnamese	100653	œ, 鐘, eg, Gift, ovna, esthes, loop, 刻, olean, sex
layers.12.mlp	Vietnamese	124904	iali, promin, Dest, lier, Round, ROID, Dipl, mittel, ୨୩, 剩
layers.13.mlp	Vietnamese	7310	Paste, ivity, Paste, ities, adamente, last, hol, inh , umpt, 金
layers.13.mlp	Vietnamese	19743	ори, loi, ада, coli, athing, qed, aign, 法, icates, rored
layers.13.mlp	Vietnamese	65213	-i, adden, UFFER, кры, oles, Olymp, Mix, linked, RATION, RING
layers.13.mlp	Vietnamese	75072	ët, affer, ition, nov, iger, Bris, egra, trib, ril, ови
layers.13.mlp	Vietnamese	77971	qt, оба, formation, athlete, atorial, connect, RPC, ñ, dae , cong
layers.13.mlp	Vietnamese	98734	IV , 행, Giang , Ho , Hoa , Hut , Sq , VO , Vu , Iv
layers.13.mlp	Vietnamese	110071	uts, locator, weld, oard, Presenter, nen, locator, soever, OX, awan
layers.14.mlp	Vietnamese	54719	ầu , ôn , àm , ư , à , ăm , ộn , ờ , ướng , áp
layers.14.mlp	Vietnamese	59851	hoot, ala, col, MODULES, ALA, olas, cio, thiếu , plode, po
layers.14.mlp	Vietnamese	71834	Gi , Hi , Th , Tr , Hi , T , intervening, Bi , Hv , B
layers.14.mlp	Vietnamese	73696	cho , urs, nguyên , unique, vers, uchs, iaz, mixing, ica, tập
layers.14.mlp	Vietnamese	111142	than , -than , Than , than , THAN , Than , niž , ăn , than , než
layers.14.mlp	Vietnamese	121534	Cu , Vu , trans, cu , vn , ected, duc , Tu , Vo , □
layers.14.mlp	Vietnamese	130249	lein, je, chie, etable, ambre, erp, евич, emon, nie, je
layers.15.mlp	Vietnamese	489	arda, variation, appearance, Grad, enders, 359, adio, Scot, imize, knowledge
layers.15.mlp	Vietnamese	56098	tâm , lợi , biện , chứng , ám , thần , đỡ , xấu , chỗ , thảo
layers.15.mlp	Vietnamese	92076	Quảng , À , Hồ , Ninh , □, Urb, Đài , Nguyễn , Khu , Ó
layers.15.mlp	Vietnamese	114319	VN , Ngh , Thanh , Ninh , VN , Kinh , Trang , Hoa , Trung , Nha

Figure 42: Top 10 tokens promoted by multiplying Vietnamese-specific features with the token embedding matrix of Llama 3.2 1B. **Bold** text indicates tokens associated with Vietnamese.

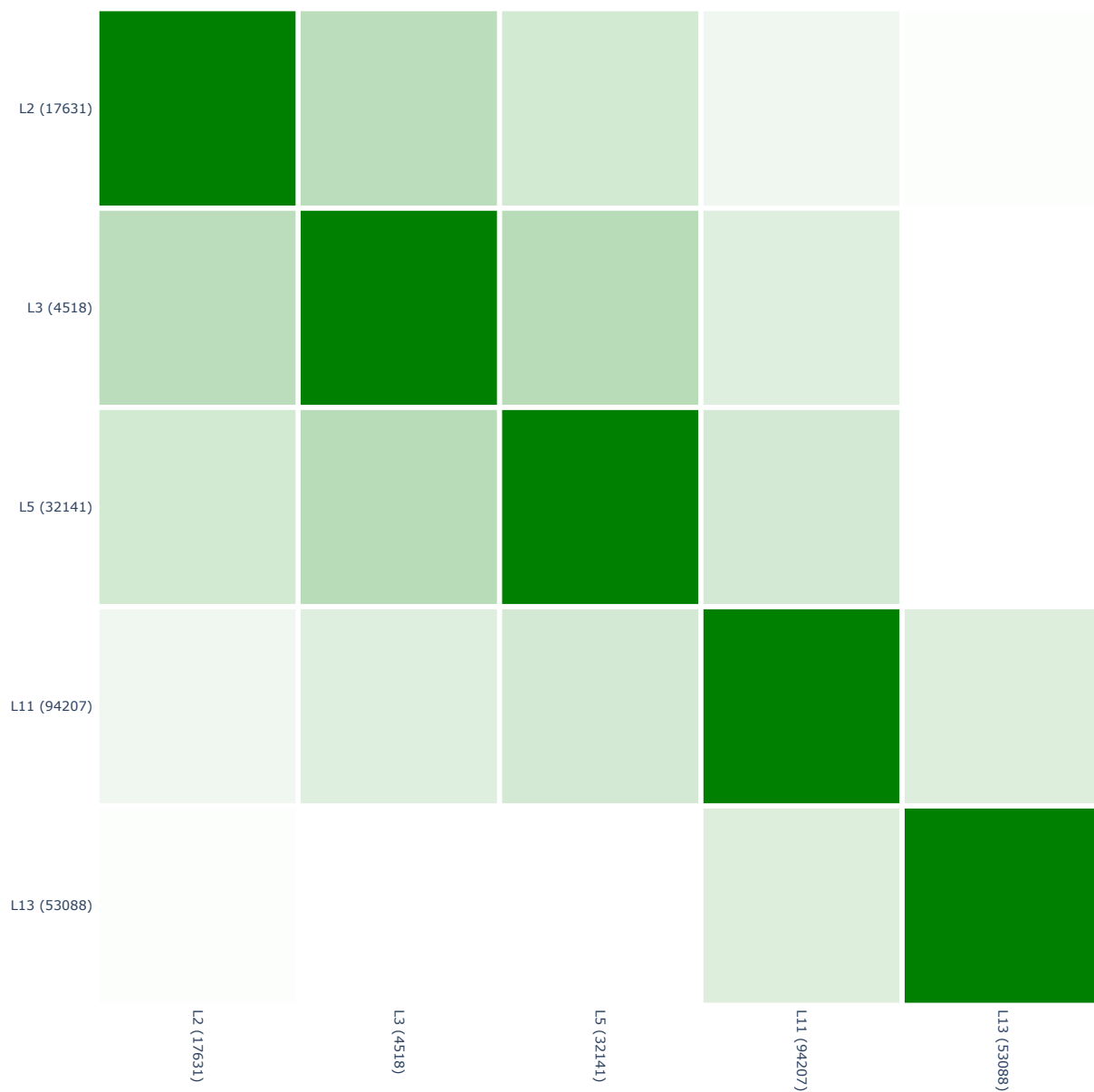


Figure 43: Intersection over union of activating tokens in Llama 3.2 1B for Spanish-specific features across layers.

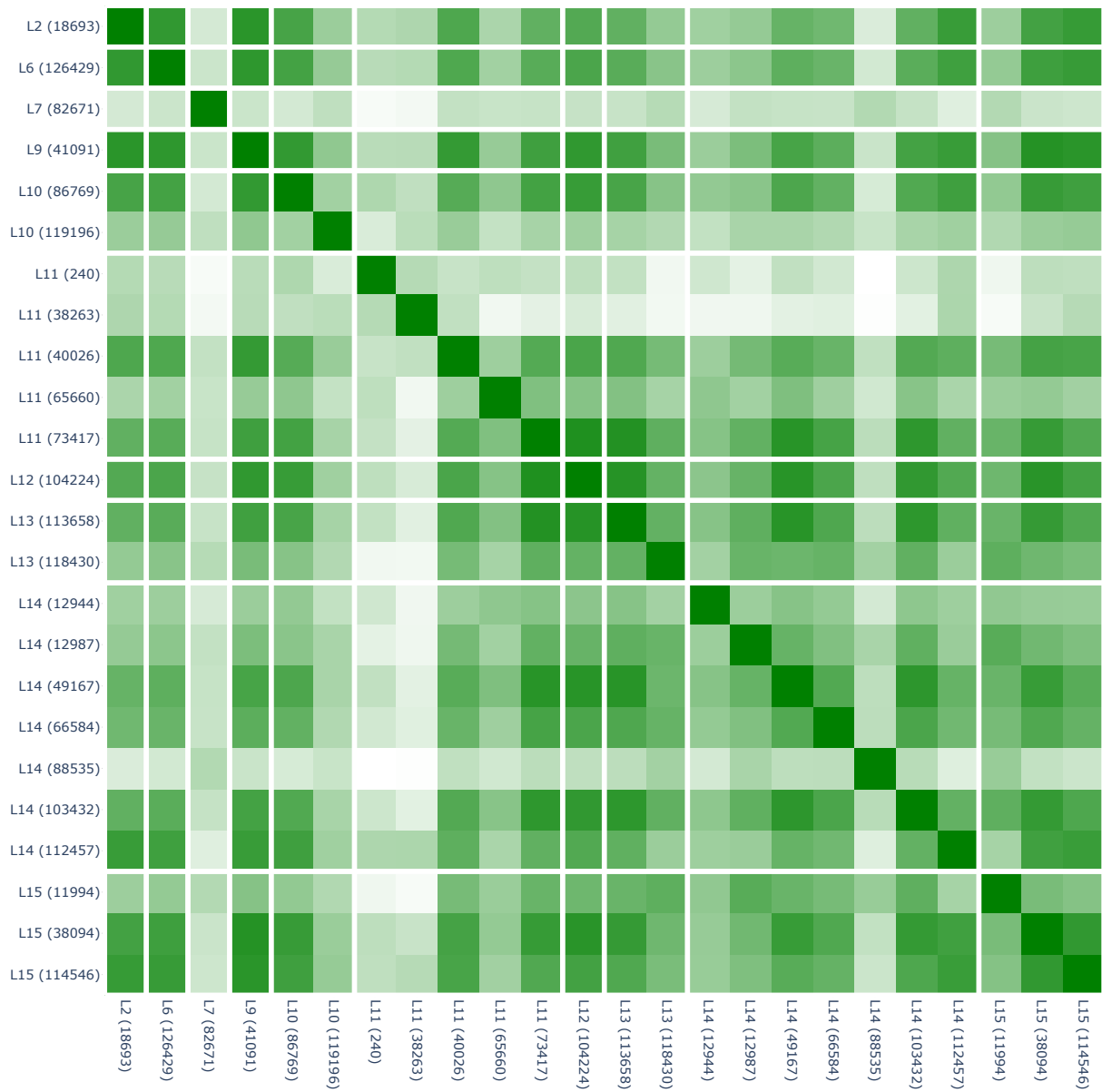


Figure 44: Intersection over union of activating tokens in Llama 3.2 1B for Chinese-specific features across layers.

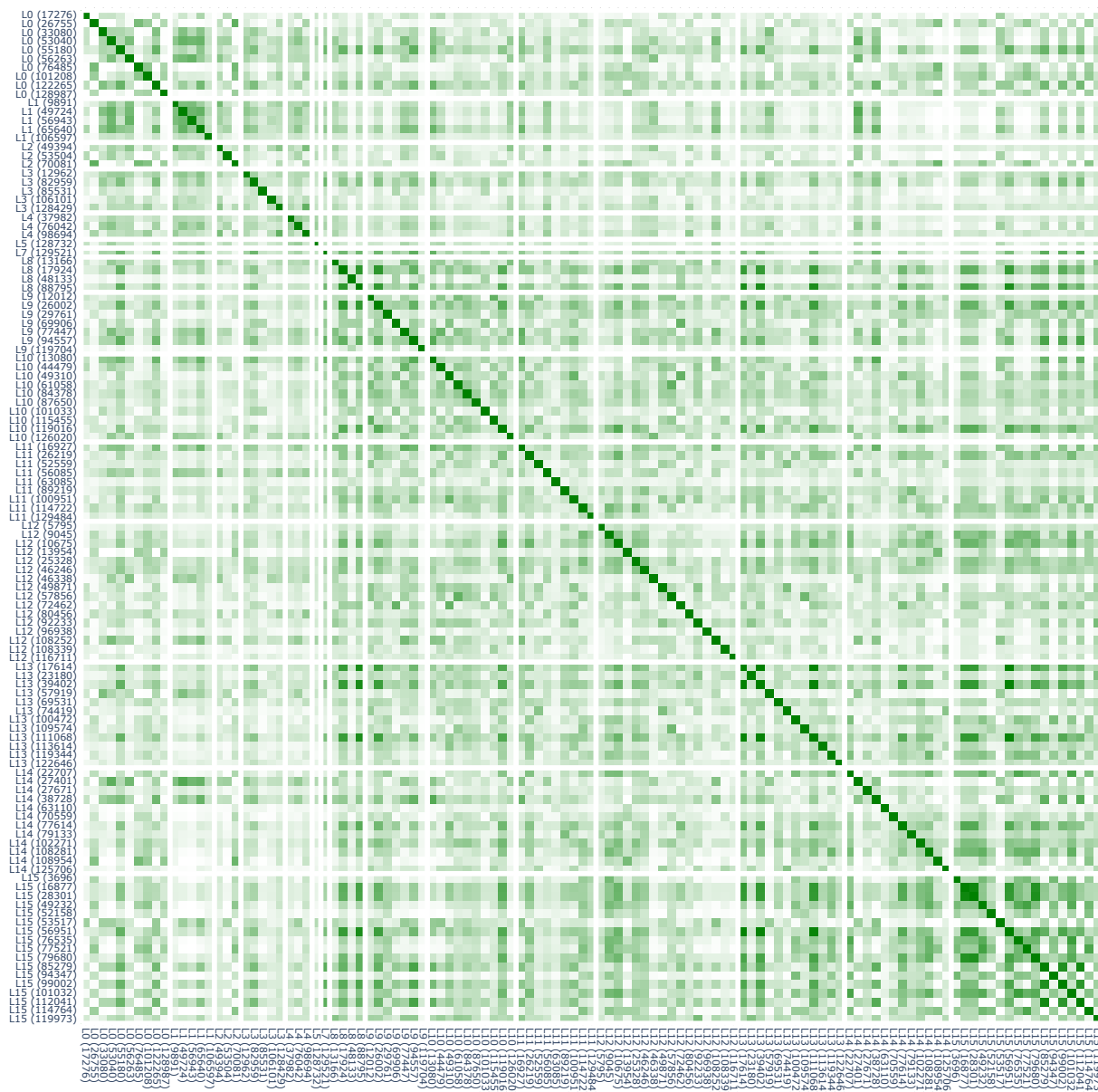


Figure 45: Intersection over union of activating tokens in Llama 3.2 1B for Hindi-specific features across layers.

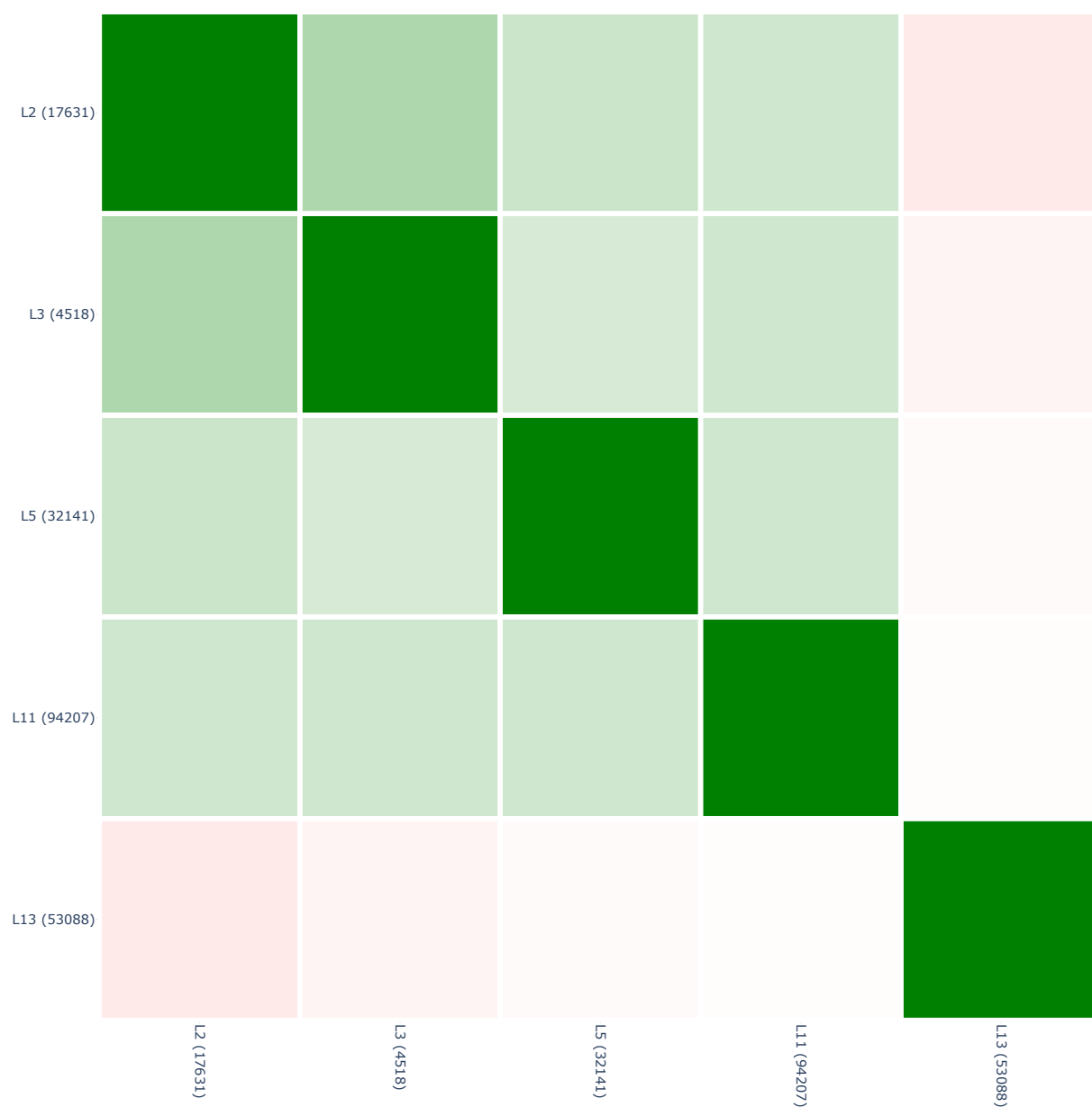


Figure 46: Pearson correlation of activating tokens in Llama 3.2 1B for Spanish-specific features across layers.



Figure 47: Pearson correlation of activating tokens in Llama 3.2 1B for Chinese-specific features across layers.

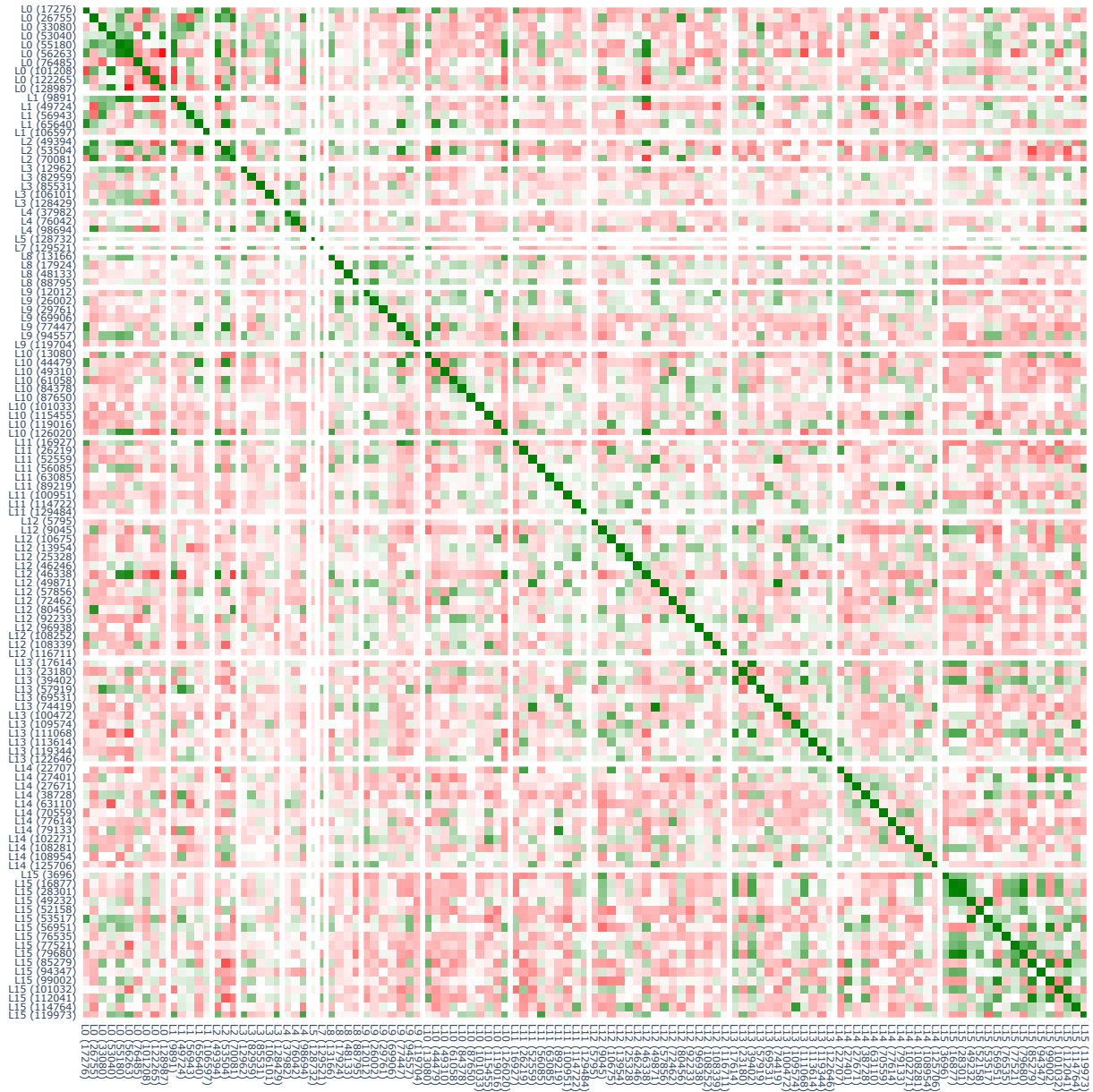


Figure 48: Pearson correlation of activating tokens in Llama 3.2 1B for Hindi-specific features across layers.

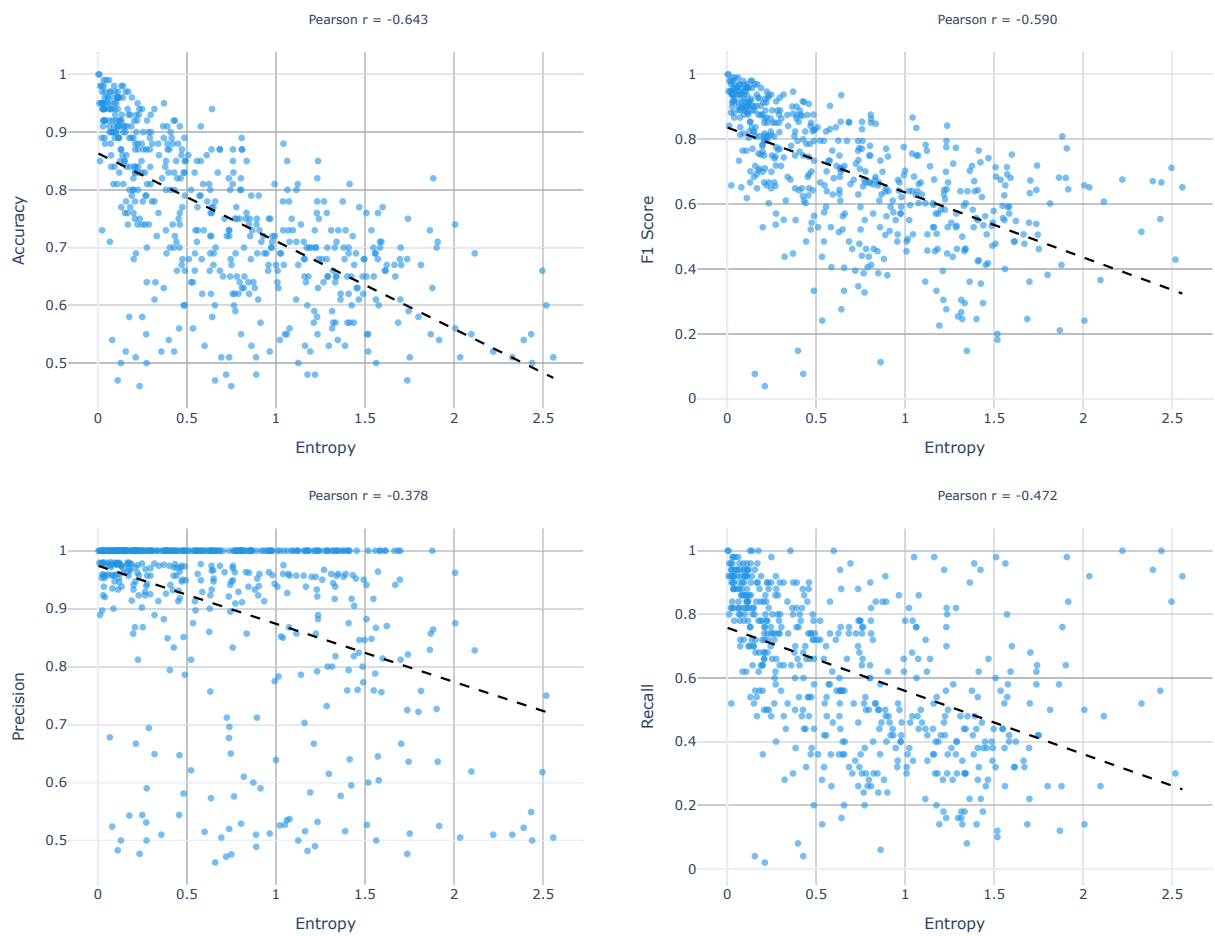


Figure 49: Pearson correlations between interpretation scores and entropy for detection scoring in Llama 3.2 1B.

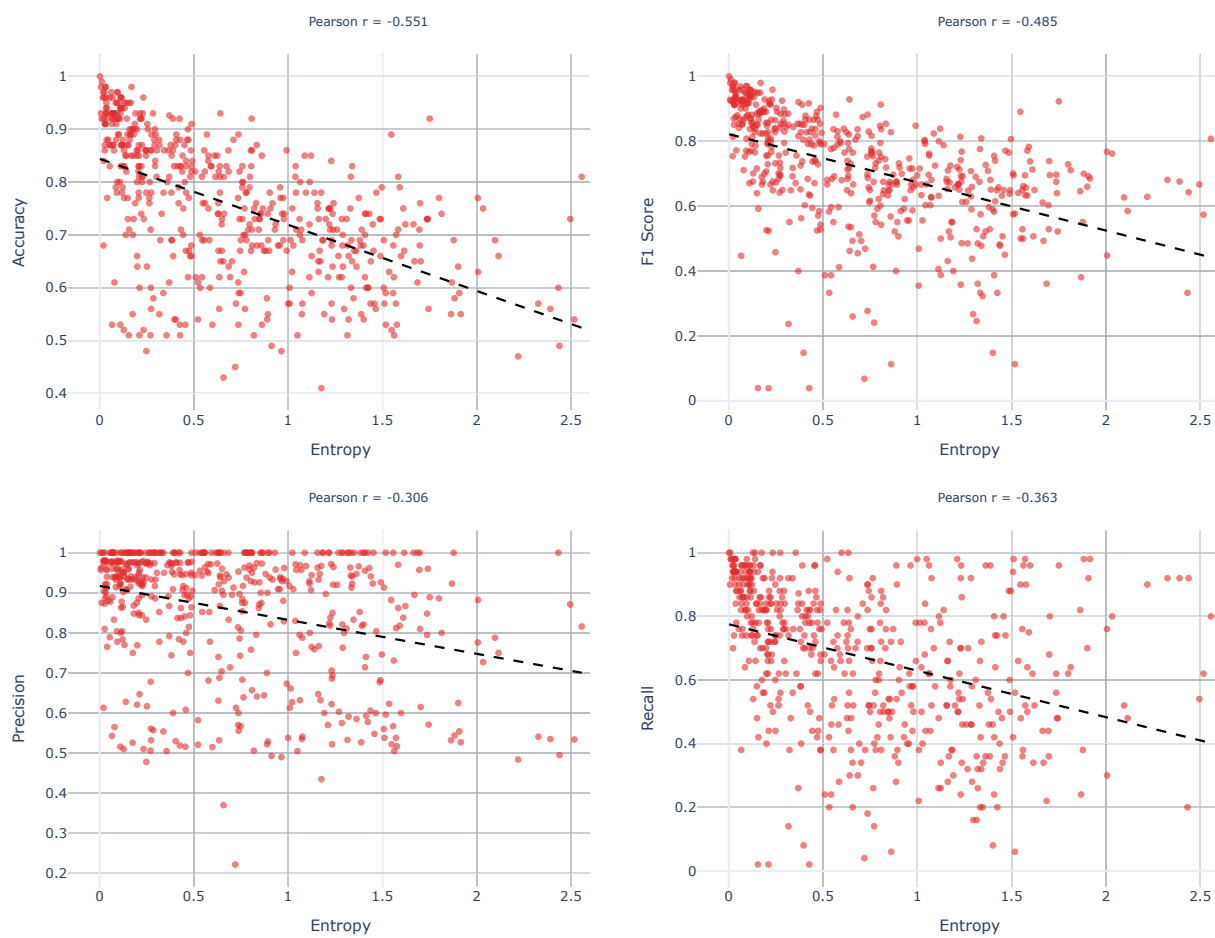


Figure 50: Pearson correlations between interpretation scores and entropy for fuzzing scoring in Llama 3.2 1B.

Feature 67343

Activation Range

Language: English

Model: meta-llama/llama-3.2-1B

Layer: model.layers.15.mlp

SAE Model: EleutherAI/sae-llama-3.2-1B-131k

Selected Token Probability: 0.308

Entropy: 0.723

0.332-1.973	1.973-3.614	3.614-5.255	5.255-6.896	6.896-8.537	8.537-10.179	10.179-11.82
11.82-13.461	13.461-15.102	15.102-16.743				

Interpretation

"The text contains a wide variety of conversational and narrative fragments, including idiomatic expressions, filler words, discourse markers, and references to people, places, and actions. There is a frequent use of pronouns, conjunctions, and auxiliary verbs, as well as sequences that reflect spoken language patterns such as repetitions, hesitations, and clarifications. Many segments highlight the structure of dialogue, question-answer formats, and descriptive or explanatory statements, often focusing on actions, states, or relationships between entities."

Score Type	Accuracy	Precision	Recall	F1 score	TPR	TNR	FPR	FNR
detection	0.72	0.712	0.74	0.725	0.74	0.7	0.3	0.26
fuzz	0.57	0.613	0.38	0.469	0.38	0.76	0.24	0.62

English

#examples: [['xnli', 1000], ('paws-x', 986), ('flores', 990)]

xnli-188. <|begin_of_text|> yeah yeah glad to see y'all taken care of well the i think what changed everything and uh is uh y'all were y'all the only ones that make any money for TI here in the last two years In the past few years , no one has made any money for TI .

xnli-34. <|begin_of_text|> Felicia 's Journey takes place behind the eyes of its central a young Irish girl , Felicia , who crosses the sea to England in a hopeful quest to find the father of her unborn child ; and the fat , middle-aged catering manager , Hiditch , who takes a paternal interest in the lass when it becomes clear that her young man has caddishly given her the slip . The woman did not care where the man was as long as it was far .

xnli-936. <|begin_of_text|> was this uh done not by TI people was it It wasn't done by TI people .

xnli-230. <|begin_of_text|> i um it 's it 's amazingly less expensive than it is It 's much more expensive that way .

xnli-85. <|begin_of_text|> I 'm not into this Multivista stuff , because what is some dumpy system when compared to a beautiful hyperextension of the sun somewhere over Kuchara , when compared to the golden hue of onion fried with the kse-fi waves , when compared to the number of dividers for credit membranes in a wallet of a rich man , when compared to the magnificent smell of a briessante roll dunked in wholesome milk synthetically enriched with substances boosting the secretion of happiness hormones , that one from two years ago , not three , ' Gonzo said in a tone characteristic for a man who just discovered a solution to his life problem . He 's not into this multivista stuff because what is some dumpy system compared to the magnificent smell of a cinnamon roll dunked in wholesome milk .

xnli-743. <|begin_of_text|> yes it really is and that way we 're not really missing anything out you know of those children uh-huh We 're not really missing out on childcare benefits .

xnli-317. <|begin_of_text|> um-hum yeah i end up well yeah i mean i do a lot of like even weekender kind of things i go out for just one or two nights uh it 's not my favorite thing but i can do it a lot more often so like over at the uh particularly in the uh the fall and spring when the insects haven 't come out yet uh where they 're already gone uh you know i spend almost every weekend up in the mountains and i guess i live about five hours away from some place to go hiking where i am now I don 't mind camping for a couple nights in the fall and spring .

xnli-169. <|begin_of_text|> and if you yeah if you want to so at least you you know you wouldn 't have any surprises you can order any type of you know service you want but you stand a good chance you know having to pick up at least part of the bill yourself You may have to foot the bill if you order any service you want .

xnli-754. <|begin_of_text|> two yeah yeah and then of course Indianapolis which is only a hundred miles away from us has the Colts yeah We ill go to Indianapolis to see the Colts .

xnli-811. <|begin_of_text|> and so um yeah uh yeah i guess i usually do i like to cook um heavy sauces and um i generally like to prepare heavy sauces .

xnli-778. <|begin_of_text|> it is comforting to me to see uh more concern about some of these things that that cost us money especially when we have dwindling uh resources such as oil that 's burning out of control in in the Persian Gulf and and so forth just just every little bit does it makes me feel better it it makes me feel like well there may be something left for my children my nieces and nephews and so forth It makes me feel good to see people be concerned about our nation 's expenses .

Figure 51: Activated examples for English-specific feature 67343 in layer 15 of Llama 3.2 1B. This feature is activated in 1000 XNLI examples, 986 PAWS-X examples, and 990 FLORES+ examples from the English-specific corpus.

Feature 5889

Activation Range

Language: English

Model: meta-llama/Llama-3.2-1B

Layer: model.layers.15.mlp

SAE Model: EleutherAI/sae-Llama-3.2-1B-131k

Selected Token Probability: 0.183

Entropy: 1.901

0.213-1.443	1.443-2.673	2.673-3.904	3.904-5.134	5.134-6.364	6.364-7.594	7.594-8.824
8.824-10.055	10.055-11.285	11.285-12.515				

Interpretation

"The highlighted tokens frequently correspond to structural elements in formal or factual writing, such as numbers, dates, conjunctions, prepositions, punctuation, and proper nouns, which are often used to organize, enumerate, or specify information in lists, references, or descriptive passages."

Score Type	Accuracy	Precision	Recall	F1 score	TPR	TNR	FPR	FNR
detection	0.7	0.727	0.64	0.681	0.64	0.76	0.24	0.36
fuzz	0.64	0.625	0.7	0.66	0.7	0.58	0.42	0.3

English

#examples: [('xnli', 1000), ('paws-x', 1000), ('flores', 997)]

xnli-720. <|begin_of_text|>Comments on classified or restricted reports should be transmitted in the manner agreed to by GAO and the agency. There are specific transmission methodologies for classified reports.

xnli-575. <|begin_of_text|>Analysis of this information revealed that no two of these organizations defined the design review process and its various elements in exactly the same manner. Many organizations have a radically different approach to their design review process.

xnli-36. <|begin_of_text|>there 's a uh a couple called um oh i 'm going to forgot his name now uh Dirkson I can 't remember their name

xnli-589. <|begin_of_text|>i 've had a chance to i had a chance to terminate and uh L O A and back to L O A I never had any opportunity to end LOA.

xnli-732. <|begin_of_text|>Advertisement, replied Tuppence promptly. Tuppence replied right away.

xnli-345. <|begin_of_text|>2 House Judiciary Committee Chairman Henry Hyde said he has begun to examine impeachment procedures in the event they are justified. Henry Hyde is a new house judiciary.

xnli-650. <|begin_of_text|>(5) (A) The term baseline heat input means, except under subpart 1 of part B and section 407, the average annual heat input used by a unit during the three years in which the unit had the highest heat input for the period 1997 through 2001. The baseline heat input is the lowest annual heat input over 3 years.

xnli-177. <|begin_of_text|>okay pro football i like two teams one the New York Giants and the second is the Raiders The New York Giants and Raiders are my favorite teams in football.

xnli-550. <|begin_of_text|>so i i know the people at TI who are doing this and i heard about it so i called them and ask if i could could participate and uh I don 't know anyone at TI and that 's okay because I don 't want to participate anyway.

xnli-618. <|begin_of_text|>1.7, formerly methodology transfer paper 7. It used to be paper 7.

xnli-943. <|begin_of_text|>Also the Treasury requires departments to disclose instances of irregular expenditures arising from erroneous benefit awards and fraud by claimants. Departments are typically reluctant to disclose mistakes in the awarding of benefits.

xnli-478. <|begin_of_text|>Breyiana Breyiana look at me please just a second be quite please i can not hear i will talk to you in just a second go ahead so she went to school out here Please be quiet for a second; I cannot hear and will talk to you in a second.

xnli-119. <|begin_of_text|>In any case--it 's over. It is over.

xnli-301. <|begin_of_text|>Inside, a huge mountain of a man, even bigger than Thorn, fought with a huge two-handed axe. The man was extremely large.

xnli-54. <|begin_of_text|>Analyzing Postal Service accounts for depreciation, fuel, and maintenance for city delivery carriers, we have estimated the average city delivery vehicle cost per route. Driving cost estimates can be averaged will sufficient data.

xnli-13. <|begin_of_text|>Thebes held onto power until the 12th Dynasty, when its first king, Amenemhet I who reigned between 1980 1951 b.c. established a capital near Memphis. The capital near Memphis lasted only half a century before its inhabitants abandoned it for the next capital.

Figure 52: Activated examples for English-specific feature 5889 in layer 15 of Llama 3.2 1B. This feature is activated in 1000 XNLI examples, 1000 PAWS-X examples, and 997 FLORES+ examples from the English-specific corpus.

Feature 125019

Language: Korean

Model: meta-llama/llama-3.2-1B

Layer: model.layers.10.mlp

SAE Model: EleutherAI/sae-llama-3.2-1B-131k

Selected Token Probability: 0.266

Entropy: 0.155

Activation Range

0.082-0.209	0.209-0.335	0.335-0.462	0.462-0.589	0.589-0.715	0.715-0.842	0.842-0.969
0.969-1.096	1.096-1.222	1.222-1.349				

Interpretation

"The highlighted tokens consistently mark proper nouns, numerals, and key terms related to King Sejong, the Joseon Dynasty, and the invention of the Hangeul alphabet, as well as their equivalents in multiple languages. Dates, names, and specific terminology are emphasized, often in the context of historical or factual statements."

Score Type	Accuracy	Precision	Recall	F1 score	TPR	TNR	FPR	FNR
detection	0.52	1.0	0.04	0.077	0.04	1.0	0.0	0.96
fuzz	0.51	1.0	0.02	0.039	0.02	1.0	0.0	0.98

Korean

#examples: [(['paws-x', 997), ('fiores', 997)]

paws-x-162. <|begin_of_text|>연락 원은 58 이었지만이 중 20 은 메시지를 위해서만 사용되었습니다.

paws-x-979. <|begin_of_text|>쇼는 MBC 2에서 큰 상금 \$ 100 000으로 방송되었지만 취소로 인해 이겼습니다.

paws-x-828. <|begin_of_text|>Holy Trinity (가 리그) (1443 W. Division St) 또한 카 리그 팀이 있습니다. 마지막 호랑이 축구 시즌은 1965 년이 있습니다.

paws-x-208. <|begin_of_text|>Togdheer (소 말리 'Wabi Togdheer')는 소말리랜드 동부의 Togdheer River 지역의 계절 강입니다.

Text Examples for Each Interval

interval 1

Range: 1.222-1.349

#examples: 4

paws-x-162. <|begin_of_text|>연락 원은 58 이었지만이 중 20 은 메시지를 위해서만 사용되었습니다.

xnli-463. <|begin_of_text|>Một khu phố châu á thứ ba , Hàn Quốc , nằm phía tây của trung tâm dọc theo đại lộ olympic giữa vermont và Phương Tây . Đại lộ Olympic là phía đông của khu vực Hàn Quốc , và là vermont và Phương Tây .

paws-x-979. <|begin_of_text|>Die Sendung wurde auf MBC 2 mit einem großen Preis von 100.000 \$ ausgestrahlt, die aufgrund der Absage nicht gewonnen wurde.

fiores-385. <|begin_of_text|>Hangeul is the only purposely invented alphabet in popular daily use. The alphabet was invented in 1444 during the reign of King Sejong (1418 – 1450).

interval 2

Range: 1.096-1.222

#examples: 8

paws-x-979. <|begin_of_text|>The show was broadcast on MBC 2 with a big prize \$ 100 000 , not won because of the cancellation .

paws-x-979. <|begin_of_text|>쇼는 MBC 2에서 큰 상금 \$ 100 000으로 방송되었지만 취소로 인해 이겼습니다.

paws-x-828. <|begin_of_text|>Holy Trinity (가 리그) (1443 W. Division St) 또한 카 리그 팀이 있습니다. 마지막 호랑이 축구 시즌은 1965 년이 있습니다.

fiores-385. <|begin_of_text|>Hangeul là bảng chữ cái được phát minh chỉ nhằm mục đích sử dụng thông dụng hàng ngày. Bảng chữ cái được phát minh vào năm 1444 trong triều đại Vua Sejong (1418 - 1450)

Figure 53: Activated examples for Korean-specific feature 125019 in layer 10 of Llama 3.2 1B. This feature is activated in 997 PAWS-X examples and 997 FLORES+ examples from the Korean-specific corpus. The replacement character (the one that looks like a question mark inside a diamond) indicates that Llama 3.2 1B does not cover that token, due to its limited vocabulary size.

Language: Korean
Model: meta-llama/Llama-3.2-1B
Layer: model.layers.12.mlp
SAE Model: EleutherAI/sae-Llama-3.2-1B-131k
Selected Token Probability: 0.136
Entropy: 0.861

0.071-0.804 0.804-1.537 1.537-2.271 2.271-3.004 3.004-3.737 3.737-4.47 4.47-5.203
5.203-5.937 5.937-6.67 6.67-7.403

"The text consistently centers on King Sejong and the Joseon Dynasty, the invention of the Hangeul (Hangul) alphabet, and the name \"Hunmin Jeongeum,\" with key tokens marking royal titles, dynastic references, invention dates (especially 1444 and 1418–1450), and the names of the alphabet and its inventor, across multiple languages."

Score Type	Accuracy	Precision	Recall	F1 score	TPR	TNR	FPR	FNR
detection	0.53	1.0	0.06	0.113	0.06	1.0	0.0	0.94
fuzz	0.53	1.0	0.06	0.113	0.06	1.0	0.0	0.94

```
#examples: [('paws-x', 994), ('flores', 995)]
```

paws-x-462. <|begin_of_text|> 그녀는 그에게 크게 매료되어 그를 성적 관계로 유혹하려고 했지만 Hanuwant Singh는 종교적인 생각으로 근친 상간에 가지 않았습니다.

paws-x-155. <[begin_of_text]>Neyab (또한 Neyab로 로마자 표기)은 Esfarayen 카운티, 북부 Khorasan 주, 이란, Bam 및 Safiabad District, Safiabad Rural District에 있는 마을입니다.

Range: 6.67-7.403

flores-386. <|begin_of_text|>Il re Sejong fu il quarto re della dinastia Joseon e uno dei sovrani più stimati.

Range: 5.937-6.67

#examples: 7

flores-385. <[begin_of_text]>Lo hangeul è l'unico alfabeto inventato intenzionalmente per l'uso quotidiano da parte del popolo. Fu ideato durante il regno del Re Sejong (1418-1450) nel 1444.

flores-385. <|begin_of_text|>ハングルは、日常的に使われている唯一の字母です。字母は世宗時代(1418~1450)の1444年に発明されました。

flores-386. <|begin_of_text|>King Sejong was the fourth king of the Joseon Dynasty and is one of the most highly regarded.

flores-385. <[begin_of_text]> ฮันกิลงเป็นอักษรที่ประดิษฐ์ขึ้นโดยเจดนาเพียงชุดเดียวที่นิยมใช้ในชีวิตประจำวัน ชุดอักษรนี้ประดิษฐ์ขึ้นในปีค.ศ. 1444 ในรัชสมัยของกษัตริย์เซจง (ค.ศ. 1418 - 1450)

Figure 54: Activated examples for Korean-specific feature 67845 in layer 12 of Llama 3.2 1B. This feature is activated in 994 PAWS-X examples and 995 FLORES+ examples from the Korean-specific corpus. The replacement character (the one that looks like a question mark inside a diamond) indicates that Llama 3.2 1B does not cover that token, due to its limited vocabulary size.

Feature 118169

Language: Thai

Model: meta-llama/llama-3.2-1B

Layer: model.layers.5.mlp

SAE Model: EleutherAI/sae-llama-

3.2-1B-131k

Selected Token Probability: 0.819

Entropy: 0.397

Activation Range

0.085-0.179	0.179-0.273	0.273-0.366	0.366-0.46	0.46-0.554	0.554-0.648	0.648-0.742
0.742-0.835	0.835-0.929	0.929-1.023				

Interpretation

"The highlighted tokens correspond to the word \"Thailand\" and its demonyms or derivatives, as well as related country names, written in various languages and scripts. The pattern is the consistent identification of the country \"Thailand\" or its linguistic equivalents across multilingual contexts."

Score Type	Accuracy	Precision	Recall	F1 score	TPR	TNR	FPR	FNR
detection	0.54	1.0	0.08	0.148	0.08	1.0	0.0	0.92
fuzz	0.54	1.0	0.08	0.148	0.08	1.0	0.0	0.92

Thai

#examples: [('xnli', 1000), ('flores', 997)]

xnli-183. <|begin_of_text|>เรา ได้ ไป ใน ทริป ที่ เรา ได้ อามน้ำ ใน ลำธาร นอกจาก การอามน้ำ ใน ลำธาร แล้ว เรา ก็ ไป สป่า และ ชาว นา ด้วย

xnli-277. <|begin_of_text|>ยัง มี พิพิธภัณฑ์ โบราณคดี ที่ แสดง ศิลปวัตถุ ที่ เก่า กว่า รวมถึง ตัวอย่าง ของ เครื่องปั้นดินเผา เครื่องปั้นดินเผา พิพิธภัณฑ์ วัง มาก และ ไม่มี อะไร อยู่ ใน นั้น เลย

xnli-127. <|begin_of_text|>ใน หมู่ พวก นี้ คือ รัง อีร แดง ซึ่ง ตอนนี บ้าน พิพิธภัณฑ์ patan (เนปาล ' s ที่ ดี ที่สุด และ พิพิธภัณฑ์ หันสมัย ที่สุด) และ หินหน้า เข ่า รัง ตรงข้าม อีร แดง พลาซา แปด วัด ของ สไตส์ และ ขนาด ต่างๆ พิพิธภัณฑ์ patan อยู่ ริมถนน จาก รัง อีร แดง ราชบุรี

xnli-824. <|begin_of_text|>แค ล ผลัก จะ ' daan นอกจาก มือ ที่มีพลัง เดียว Ca ' daan ถูก ผลัก ออก ไป และ เคาะ พื้น โดย มือ ที่ มีพลัง ของ คา ล

Text Examples for Each Interval

interval 1

Range: 0.929-1.023

#examples: 1

flores-441. <|begin_of_text|>Nel XVIII secolo la Cambogia fu invasa più volte dai thailandesi, che nel 1772 distrussero la capitale Phnom Phen.

interval 2

Range: 0.835-0.929

#examples: 5

flores-663. <|begin_of_text|>São Francisco é também um dos melhores lugares do país no que diz respeito a outras opções de culinária asiática: comida coreana, tailandesa, indiana e japonesa.

flores-663. <|begin_of_text|>San Francisco ist auch einer der landesweit besten Orte für andere asiatische Küche: Koreanisch, Thailändisch, Indisch und Japanisch.

flores-440. <|begin_of_text|>През 18 -ти век се оказва, че Камбоджа е притисната между двама могъщи съседни, Тайланд и Виетнам.

flores-440. <|begin_of_text|>EL Reino de Camboya quedó acorralado entre dos poderosos vecinos durante el siglo XVIII. Ellos eran Tailandia y Vietnam.

flores-440. <|begin_of_text|>Nel corso del XVIII secolo la Cambogia si trovava stretta tra due vicini molto potenti, la Thailandia e il Vietnam.

interval 3

Range: 0.742-0.835

#examples: 7

flores-440. <|begin_of_text|>18. asırda Kamboçya kendisini iki güçlü komşu olan Tayland ve Vietnam arasında buldu.

Figure 55: Activated examples for Thai-specific feature 118169 in layer 5 of Llama 3.2 1B. This feature is activated in 1000 XNLI examples and 997 FLORES+ examples from the Thai-specific corpus.

Feature 118682

Language: Portuguese
Model: meta-llama/Llama-3.2-1B
Layer: model.layers.7.mlp
SAE Model: EleutherAI/sae-Llama-3.2-1B-131k
Selected Token Probability: 0.523
Entropy: 0.533

Activation Range

0.095-0.167 0.167-0.239 0.239-0.311 0.311-0.383 0.383-0.455 0.455-0.527 0.527-0.599
0.599-0.671 0.671-0.743 0.743-0.815

Interpretation

"The highlighted tokens consistently correspond to the morphemes or substrings forming the country name \"Brazil\" and its derivatives across multiple languages and scripts, often marking the root or core segment of the word regardless of linguistic context."

Score Type	Accuracy	Precision	Recall	F1 score	TPR	TNR	FPR	FNR
detection	0.56	0.875	0.14	0.241	0.14	0.98	0.02	0.86
fuzz	0.6	1.0	0.2	0.333	0.2	1.0	0.0	0.8

Portuguese

#examples: [('flores', 977)]

flores-216. <|begin_of_text|>Esportes como squash, caratê e patins tentaram entrar para a programação das Olimpíadas, da mesma forma que o beisebol e o softbol, que foram excluídos dos Jogos Olímpicos de 2005.

flores-269. <|begin_of_text|>Gabriel Jesus, de 21 anos, foi contratado pelo Manchester City em janeiro de 2017. Ele veio do Palmeiras, um clube brasileiro, no valor de 27 milhões de euros.

flores-194. <|begin_of_text|>O brasileiro sofreu um ferimento grave na cabeça depois de uma colisão no Grande Prêmio da Hungria de 2009.

flores-270. <|begin_of_text|>O brasileiro, desde então, marcou 24 gols e jogou 53 partidas pelo clube em todas as competições.

Text Examples for Each Interval

interval 1

Range: 0.743-0.815

#examples: 6

flores-269. <|begin_of_text|>A gennaio 2017, il ventunenne Jesus ha lasciato il club brasiliano Palmeiras per il Manchester City per un compenso dichiarato di 27 milioni di sterline.

paws-x-202. <|begin_of_text|>Le Tetra côtier se trouve autour du sud-est du Brésil et du bassin du Paraná dans les rivières jaunes.

flores-269. <|begin_of_text|>Jesus, 21 ans, a rejoint Manchester City l'année dernière en janvier 2017, en provenance du club brésilien Palmeiras, pour un montant annoncé de 27 millions de livres sterling.

flores-63. <|begin_of_text|>Le Congrès national du Brésil débat de la légalisation depuis 10 ans, et ces mariages civils ne sont actuellement légaux que dans le Rio Grande do Sul.

paws-x-202. <|begin_of_text|>El Tetra costero se encuentra alrededor del sureste de Brasil y la cuenca del río Paraná en ríos amarillos.

flores-62. <|begin_of_text|>Roman Katolik ülkeler arasından en büyükü Brezilya ve Roma Katolik Kilisesi, ülkede aynı cinsiyetle evliliğin yasal hale gelmesine sürekli karşı çıkmıştır.

interval 2

Range: 0.671-0.743

#examples: 20

flores-270. <|begin_of_text|>Depuis lors, le Brésilien a disputé 53 matches pour le club toutes compétitions confondues et a marqué 24 buts.

flores-63. <|begin_of_text|>От 10 години насам националният конгрес на Бразилия обсъжда легализацията и за момента такъв тип граждански брак ове са законни единствено в Рио Гранде до Сул.

Figure 56: Activated examples for Portuguese-specific feature 118682 in layer 7 of Llama 3.2 1B. This feature is activated in 997 FLORES+ examples from the Portuguese-specific corpus.

Feature 84378

Activation Range

Language: Hindi

Model: meta-llama/llama-3.2-1B

Layer: model.layers.10.mlp

SAE Model: EleutherAI/sae-llama-

3.2-1B-131k

Selected Token Probability: 0.296

Entropy: 0.427

0.074-0.487	0.487-0.9	0.9-1.313	1.313-1.726	1.726-2.139	2.139-2.551	2.551-2.964	2.964-3.377
3.377-3.79	3.79-4.203						

Interpretation

"The text contains references to the year 1947, the country Pakistan, and related political or administrative terms, often in the context of independence or governance, and these patterns appear across multiple languages and scripts."

Score Type	Accuracy	Precision	Recall	F1 score	TPR	TNR	FPR	FNR
detection	0.52	1.0	0.04	0.077	0.04	1.0	0.0	0.96
fuzz	0.51	1.0	0.02	0.039	0.02	1.0	0.0	0.98

Hindi

#examples: [('xnli', 1000), ('flores', 997)]

xnli-886. <|begin_of_text|>मुझे लगता है कि यह काम कर रहा है, ' 'उसने एक महीने से पहले एक महीने के बारे में कहा था, और उसके पति का एक अस् प ष ट प रभाव था कि उसकी बेटा के चेहरे पर कम या कम तीन zits कम थे. पिता ने सोचा कि उसकी बेटा आमतौर पर बहुत zits थी.

xnli-981. <|begin_of_text|>हर जगह मोन, और धीरे विडोज ♦♦. एक तूफान आ रहा था तो खिड ♦♦ की धीरे थी।

xnli-352. <|begin_of_text|>वह एक चेतवनी लिख रहा था, कोई नुस् खा नहीं. पुलिस अधिकारी एक चेतवनी टिकट लिख रहा था, डॉक ट र से कोई नुस् खा नहीं.

xnli-844. <|begin_of_text|>जैसा कि मैं लिखता हूँ, घर फूलों और फलों से जीवित है, और यह फिर से बर् फ शुरु कर रहा है। यह गर् म और गर् म है.

Text Examples for Each Interval

interval 1

Range: 3.79-4.203

#examples: 1

paws-x-915. <|begin_of_text|>Ayeza Khan (née le 15 janvier 1991 sous le nom d'Aiza Khan), également connue sous le nom de Kinza Khan, est une actrice et mannequin télévisée pakistanaise.

interval 2

Range: 3.377-3.79

#examples: 0

interval 3

Range: 2.964-3.377

#examples: 8

paws-x-915. <|begin_of_text|>Ayeza Khan (* 15. Januar 1991 als Aiza Khan), auch als Kinza Khan bekannt, ist eine pakistanische Fernsehschauspielerin und -modell.

flores-246. <|begin_of_text|>Since Pakistani independence from British rule in 1947, the Pakistani President has appointed "Political Agents" to govern FATA, who exercise near-complete autonomous control over the areas.

flores-246. <|begin_of_text|>1947年にイギリスの支配からパキスタンが独立して以来、パキスタン大統領はFATAを統治するために「政治エージェント」を任命しており、FATAはほぼ完全な自治権を行使しています。

paws-x-915. <|begin_of_text|>Ayeza Khan (nacida el 15 de enero de 1991 como Aiza Khan), también conocida como Kinza Khan, es una actriz y modelo de televisión pakistaní.

Figure 57: Activated examples for Hindi-specific feature 84378 in layer 10 of Llama 3.2 1B. This feature is activated in 1000 XNLI examples and 997 FLORES+ examples from the Hindi-specific corpus.

Feature 97688

Language: Turkish
Model: meta-llama/llama-3.2-1B
Layer: model.layers.10.mlp
SAE Model: EleutherAI/sae-llama-3.2-1B-131k
Selected Token Probability: 0.639
Entropy: 2.096

Activation Range

0.065-0.586 0.586-1.107 1.107-1.627 1.627-2.148 2.148-2.669 2.669-3.19 3.19-3.711
3.711-4.231 4.231-4.752 4.752-5.273

Interpretation

"The highlighted tokens are verb endings or suffixes in Romance languages, often marking tense, person, or number, and are typically found at the end of verbs in various conjugated forms."

Score Type	Accuracy	Precision	Recall	F1 score	TPR	TNR	FPR	FNR
detection	0.55	0.619	0.26	0.366	0.26	0.84	0.16	0.74
fuzz	0.69	0.788	0.52	0.627	0.52	0.86	0.14	0.48

Turkish

#examples: 978, 997

xnli-517. <|begin_of_text|>Yatak odasındaki masada kilitli , bayan ingletthorp ' un bir vasiyet bulmuşlar , evlenmeden önce , servetini Alfred Ingletthorp ' a bırakmış . Bayan Ingletthorp ' un will ' i buldular , bu da tüm servetini Alfred Ingletthorp ' a bıraktı .

xnli-1. <|begin_of_text|>Sezon boyunca bilirsın ve sanırım senin seviyesinde onları bir sonraki seviyeye kaybedersin . Eğer ana takımı geri almaya karar verir sen yiğitlere , üçlü bir adamı hatırlamak için aramaya karar verir . yerini değiştir ve tek bir adam onun yerine geçecek Eğer insanlar hani , aşağıdaki seviyeye .

xnli-41. <|begin_of_text|>Güven Fonu 2016 ' de nakit açıkları yayınlamaya başladığında , hükümet , Sosyal Güvenlik ' in nakit açığını , sosyal güvenlik fazlaları , halktan ödünç alarak , diğer vergileri yükselterek , Sosyal Güvenlik ' in nakit açığını finanse etmek için para olmalı . veya diğer hükümet harcamalarını azaltmak . Halk Genel olarak hükümetin sosyal güvenliği finanse etmek için diğer alanlarda harcamalarını azaltmayı tercih eder .

xnli-691. <|begin_of_text|>Sadece onları kurtarabiliriz diye umuyorum jon . Jon Onları öfkeli mafyadan kurtarmak istedi .

Text Examples for Each Interval

interval 1

Range: 4.752-5.273

#examples: 5

flores-411. <|begin_of_text|>Le loro balestre mortali scagliavano frecce capaci di penetrare nell'armatura dei soldati nemici. Attorno al 1000 a.C. gli Assiri introdussero la prima cavalleria.

flores-118. <|begin_of_text|>L'editore di videogiochi Konami ha detto oggi a un giornale giapponese che non pubblicherà il gioco Sei Giorni a Fallujah.

flores-615. <|begin_of_text|>Nuove evidenze, tuttavia, fanno pensare che i Moriori fossero un gruppo di Maori della terraferma che dalla Nuova Zelanda migrò sulle isole Chatham, sviluppando una propria peculiare cultura pacifica.

flores-441. <|begin_of_text|>Nel XVIII secolo la Cambogia fu invasa più volte dai thailandesi, che nel 1772 distrussero la capitale Phnom Penh.

flores-800. <|begin_of_text|>Se le persone sottovalutano la potenziale pericolosità degli alci, potrebbero avvicinarsi troppo a loro e correre dei rischi.

interval 2

Range: 4.231-4.752

#examples: 11

flores-26. <|begin_of_text|>il secondo gol della serata è stato il suo 60° gol della stagione, e questo fa di lui il primo giocatore a realizzare 60 gol (o più) in una stagione dal 1995-96, stagione in cui Jaromir Jagr e Mario Lemieux raggiunsero entrambi questo traguardo.

flores-107. <|begin_of_text|>Quando la capsula raggiungerà la Terra ed entrerà nell'atmosfera, verso le 05:00 (orario della costa est), si prevede che gli abitanti della California settentrionale, dell'Oregon, del Nevada e dello Utah assisteranno ad un notevole spettacolo di luci.

Figure 58: Activated examples for Turkish-specific feature 84378 in layer 10 of Llama 3.2 1B. This feature is activated in 978 XNLI examples and 997 FLORES+ examples from the Turkish-specific corpus.

Feature 32154

Language: Japanese
Model: meta-llama/Llama-3.2-1B
Layer: model.layers.8.mlp
SAE Model: EleutherAI/sae-Llama-3.2-1B-131k
Selected Token Probability: 0.396
Entropy: 1.069

Activation Range

0.092-0.202

0.202-0.312

0.312-0.422

0.422-0.532

0.532-0.642

0.642-0.752

0.752-0.862

0.862-0.972

0.972-1.082

1.082-1.192

Interpretation

"The highlighted tokens are primarily Bulgarian morphemes or word segments, especially those forming relative pronouns (such as \"/>

Score Type	Accuracy	Precision	Recall	F1 score	TPR	TNR	FPR	FNR
detection	0.73	0.96	0.48	0.64	0.48	0.98	0.02	0.52
fuzz	0.71	0.957	0.44	0.603	0.44	0.98	0.02	0.56

Japanese

#examples: [(['paws-x', 991), ('flores', 997)]

paws-x-138. <|begin_of_text|>最大濃度に達した場合、毒性毒性が発生するまでに1〜2日の◆◆延があるかもしれません。

paws-x-957. <|begin_of_text|>ESAによってeNLP（European NLP Solver）とも呼ばれるWORHPは、連続的な数学的最適化問題を数値的に解くための非線形ソフトウェアライブラリです。

paws-x-883. <|begin_of_text|>Pidouxは、Pablo Casalsの` Jackie "でチェロ奏者PabloLarraiñとして登場しました。

paws-x-519. <|begin_of_text|>もう1つのマーストン社の製品ラインは1931年に始まり、後に英国のシー◆◆ルとして知られるマーストンシー◆◆ルとして最初に販売されたマリン船外機エンジンで始まりました。

Text Examples for Each Interval

interval 1

Range: 1.082-1.192

#examples: 5

flores-412. <|begin_of_text|>Кавалерията е армия, която се сражава на кон. Асирийската кавалерия се сражавала на голите гърбове на конете си, тъй като седлото още не било измислено.

flores-276. <|begin_of_text|>В ролята му на председател на подкомисията за надзор и разследване, която е под егидата на Комисията по енергетика и търговия на Парламента Стърнс разследва дали данъците се използват за финансиране на аборти чрез Планирано Родителство.

flores-719. <|begin_of_text|>Карибска държава, която заема източната половина на остров Испаньола, който дели с Хаити, е Доминиканската република (на испански: Република Доминикана).

flores-952. <|begin_of_text|>Въпреки че според вас е логично да изберете авиокомпанията, с която летите най-много, трябва да знаете, че предлаганите привилегии често са различни и друга авиокомпания в същия съюз може да бъде много по-щедра по отношение на точките за редовни клиенти.

xnli-274. <|begin_of_text|>знам , че не познавам никого с ума си , който казва , че го правя , защото искам да го направя . знам , че има хора , които мислят , че го правя , защото го желая .

interval 2

Range: 0.972-1.082

#examples: 24

xnli-601. <|begin_of_text|>когато американският дипломат , който наблюдава международните мирни наблюдатели в косово , обвини сръбската полиция в клането , югославската (т.е. американският дипломат е изключително доволен от работата и постиженията си .

Figure 59: Activated examples for Japanese-specific feature 32154 in layer 10 of Llama 3.2 1B. This feature is activated in 991 PAWS-X examples and 997 FLORES+ examples from the Japanese-specific corpus.

Feature 56594

Activation Range

Language: Russian

Model: meta-llama/Llama-3.2-1B

Layer: model.layers.10.mlp

SAE Model: EleutherAI/sae-Llama-3.2-1B-131k

Selected Token Probability: 0.29

Entropy: 0.887

0.072-0.326	0.326-0.581	0.581-0.835	0.835-1.089	1.089-1.344	1.344-1.598	1.598-1.852
1.852-2.106	2.106-2.361	2.361-2.615				

Interpretation

"The highlighted tokens are predominantly proper nouns, technical terms, and named entities—such as personal names, place names, and institutional or scientific terms—often in multiple languages. These tokens frequently appear in contexts involving formal identification, attribution, or description of people, locations, organizations, or specialized concepts."

Score Type	Accuracy	Precision	Recall	F1 score	TPR	TNR	FPR	FNR
detection	0.51	0.51	0.5	0.505	0.5	0.52	0.48	0.5
fuzz	0.58	0.7	0.28	0.4	0.28	0.88	0.12	0.72

Russian

#examples: [('xnli', 1000), ('flores', 997)]

xnli-335. <|begin_of_text|> - хорошо . Наталья Плевал . Наталья улыбнулась .

xnli-703. <|begin_of_text|> Федеральный закон об упорядочении закупок 1994 года (общественности) (публичный закон 103-355) этот закон требует от учреждений определять стоимость , график и показатели работы для федеральных программ приобретения (включая проекты ИТ) и осуществлять контроль за этими программами , с тем чтобы обеспечить , чтобы они были оставаться в пределах установленных допусков . Этот закон требует , чтобы учреждения определили различные цели для программ приобретения .

xnli-688. <|begin_of_text|> Он это сделал ? ' Наталья спросила , указывая на окно . Наталья молча смотрела в окно .

xnli-622. <|begin_of_text|> Армейский корпус инженеров США (usace) , институт строительной промышленности (мосф) , 3 год и другие ffc усилия , а также другие , в целях повышения эффективности исследования . В ведет к увеличению исследования путем финансирования первоначальных исследователей еще на три года .

Text Examples for Each Interval

interval 1

Range: 2.361-2.615

#examples: 1

paws-x-28. <|begin_of_text|> Aksuat Dendropark (en russe: Dendroparks d'Akshout) est un monument naturel (zones protégées de l'oblast d'Oulianovsk)

interval 2

Range: 2.106-2.361

#examples: 6

paws-x-28. <|begin_of_text|> Akshuat dendropark (Russian : Акшутский дендропарк) is a natural monument (Ulyanovsk Oblast protected areas)

xnli-335. <|begin_of_text|> - хорошо . Наталья Плевал . Наталья улыбнулась .

xnli-703. <|begin_of_text|> Федеральный закон об упорядочении закупок 1994 года (общественности) (публичный закон 103-355) этот закон требует от учреждений определять стоимость , график и показатели работы для федеральных программ приобретения (включая проекты ИТ) и осуществлять контроль за этими программами , с тем чтобы обеспечить , чтобы они были оставаться в пределах установленных допусков . Этот закон требует , чтобы учреждения определили различные цели для программ приобретения .

paws-x-28. <|begin_of_text|> Aksuat Dendropark (러시아어 : Akshout Dendroparks) 자연 기념물 (Ulyanovsk Oblast 보호 지역)

paws-x-226. <|begin_of_text|> Le premier laser au monde a été développé en 1960 par les scientifiques américains Nikolay Basov et Alexander Prokhorov et par le scientifique russe Charles H. Townes.

Figure 60: Activated examples for Russian-specific feature 56494 in layer 10 of Llama 3.2 1B. This feature is activated in 1000 XNLI examples and 997 FLORES+ examples from the Russian-specific corpus.

Feature 49501

Language: Vietnamese

Model: meta-llama/Llama-3.2-1B

Layer: model.layers.5.mlp

SAE Model: EleutherAI/sae-Llama-3.2-1B-131k

Selected Token Probability: 0.589

Entropy: 1.191

Activation Range

0.043-0.258	0.258-0.474	0.474-0.689	0.689-0.904	0.904-1.119	1.119-1.335	1.335-1.55
1.55-1.765	1.765-1.981	1.981-2.196				

Interpretation

"The highlighted tokens are characteristic morphemes, suffixes, or letter clusters commonly found in Romanian and related Eastern European place names, river names, and surnames, often marking grammatical or regional features."

Score Type	Accuracy	Precision	Recall	F1 score	TPR	TNR	FPR	FNR
detection	0.52	0.583	0.14	0.226	0.14	0.9	0.1	0.86
fuzz	0.55	0.6	0.3	0.4	0.3	0.8	0.2	0.7

Vietnamese

#examples: [('xnli', 995), ('flores', 989)]

xnli-229. <|begin_of_text|>Chiến dịch hồi giáo cho phân vùng được dẫn dắt bởi luật sư bombay được đào tạo Luân Đôn , muhammad ali jinnah . Muhammad Ali Jinnah đã đến Luân Đôn để chiến dịch cho phân vùng .

xnli-386. <|begin_of_text|>Chiến tranh không phải là người apache , chắc chắn là vậy . Chắc chắn đó không phải là người apache , chắc chắn rồi .

xnli-172. <|begin_of_text|>Căn nhà này rất nhỏ bé và đơn giản , với một phòng ngủ , một nhà bếp nhỏ , và một vài phòng xã hội . Ngôi nhà rất lớn và tự hào hơn phòng ngủ , một nhà bếp khổng lồ , và một hồ bơi olympic đầy đủ .

xnli-707. <|begin_of_text|>Đây là tất cả những gì trong sạch và khó chịu , rung động với những người khó chịu , nhưng cũng có những câu hỏi khó khăn và những câu trả lời không được đáp ứng . Điều này thật khó chịu và khó chịu .

Text Examples for Each Interval

interval 1

Range: 1.981-2.196

#examples: 4

paws-x-921. <|begin_of_text|>Cirja # 7 überholte 2 TD Scramble und erreichte außerdem 2 TD mit Florin Oltean # 85 und mit Dan Crasnic # 93.

paws-x-933. <|begin_of_text|>Der Fluss Crisul Mic ist ein Nebenfluss des Flusses Doba in Rumänien.

paws-x-921. <|begin_of_text|>Cirja 7は2 TD Scrambleを通過し、さらに2 TD 1をFlorin Oltean 85に、1 TDをDan Crasnic 93に達成しました。

paws-x-921. <|begin_of_text|>Cirja 7通过了2次TD Scramble, 并且还获得了2个TD, 其中一个是Florin Oltean 85, 另一个是Dan Crasnic 93。

interval 2

Range: 1.765-1.981

#examples: 7

paws-x-933. <|begin_of_text|>La rivière Crisul Mic est un affluent de la rivière Doba en Roumanie.

paws-x-478. <|begin_of_text|>Der Fluss Fadimac ist ein Nebenfluss des Flusses Bega in Rumänien.

paws-x-921. <|begin_of_text|>Cirja # 7 aprobó 2 TD Scramble y también logró 2 TD uno para Florin Oltean # 85 y otro para Dan Crasnic # 93.

paws-x-933. <|begin_of_text|>El río Crisul Mic es un afluente del río Doba en Rumania.

paws-x-766. <|begin_of_text|>La rivière Valea Turcului est un affluent de la rivière Azuga en Roumanie.

paws-x-780. <|begin_of_text|>La rivière Vidăcut ou rivière Hidecut est un affluent de la rivière Eliseni, en Roumanie.

paws-x-868. <|begin_of_text|>Olt River 또는 Pârâul Sec는 루마니아의 Seaca 강 지류입니다.

Figure 61: Activated examples for Vietnamese-specific feature 49501 in layer 5 of Llama 3.2 1B. This feature is activated in 995 XNLI examples and 989 FLORES+ examples from the Vietnamese-specific corpus.

Feature 61383

Activation Range

Language: Vietnamese

Model: meta-llama/llama-3.2-1B

Layer: model.layers.9.mlp

SAE Model: EleutherAI/sae-llama-

3.2-1B-131k

Selected Token Probability: 0.881

Entropy: 0.211

0.097-0.368	0.368-0.639	0.639-0.911	0.911-1.182	1.182-1.453	1.453-1.724	1.724-1.995
1.995-2.267	2.267-2.538	2.538-2.809				

Interpretation

"The highlighted tokens correspond to names of Cambodian places and landmarks, especially those related to Tonle Sap, Phnom Krom, Angkor, and Siem Reap, often appearing in multiple languages and scripts, with activations on both full names and their constituent parts."

Score Type	Accuracy	Precision	Recall	F1 score	TPR	TNR	FPR	FNR
detection	0.51	1.0	0.02	0.039	0.02	1.0	0.0	0.98
fuzz	0.51	1.0	0.02	0.039	0.02	1.0	0.0	0.98

Vietnamese

#examples: [('xnli', 1000), ('fiores', 997)]

xnli-298. <|begin_of_text|> Bình luận về các vấn đề về thực thi và câu hỏi được nuôi dưỡng bởi công nghiệp bị kiểm soát được gửi bởi epa trong bốn câu hỏi quy tắc và trả lời các tài liệu. Tất cả các vấn đề và câu hỏi được nuôi dưỡng bởi các công nghiệp bị kiểm soát được đưa ra bởi epa.

xnli-931. <|begin_of_text|> Hôm nay nó nhà một bảo tàng nhỏ ? . Nó chứa một viện bảo tàng nhỏ .

xnli-568. <|begin_of_text|> Họ thậm chí còn có thể ngồi cạnh tài xế. Chỗ ngồi bên cạnh tài xế cung cấp những quan điểm tốt nhất.

xnli-906. <|begin_of_text|> Nghe có vẻ như đó là một ý tưởng rất hay đấy. Nghe có vẻ là một ý tưởng khủng khiếp.

Text Examples for Each Interval

interval 1

Range: 2.538-2.809

#examples: 9

fiores-690. <|begin_of_text|> A atmosfera sombria do templo e a vista do lago Tonle Sap fazem a subida a colina valer a pena.

fiores-441. <|begin_of_text|> Durante el siglo XVIII Camboya sufrió varias invasiones de los tailandeses, quienes, en 1772, destruyeron Phnom Phen.

fiores-690. <|begin_of_text|> Tapınağın kasvetli atmosferi ve Tonle Sap gölü manzarası, tepeye tırmanmayı zahmete değer bir hale getiriyor.

fiores-690. <|begin_of_text|> 寺的阴围和洞里萨湖 (Tonle Sap) 的景色, 让人感到到山顶是值得的。

fiores-690. <|begin_of_text|> Vale la pena scalare la collina per godersi l'atmosfera tenebrosa di questo tempio e il panorama sul lago Tonle Sap.

fiores-692. <|begin_of_text|> Bạn cần vé vào Angkor để vào đền, vậy nên đừng quên mang theo hộ chiếu của bạn khi đi đến Tonle Sap.

fiores-692. <|begin_of_text|> 需要购买吴哥门票才能进入圣殿, 所以前往洞里萨湖 (Tonle Sap) 时别忘了带上护照。

fiores-690. <|begin_of_text|> The gloomy atmosphere of the temple and the view over the Tonle Sap lake make the climb to the hill worthwhile.

fiores-692. <|begin_of_text|> The Angkor Pass is needed to enter the temple so do not forget to bring your passport along when heading to Tonle Sap.

interval 2

Range: 2.267-2.538

#examples: 18

fiores-690. <|begin_of_text|> La atmósfera sombría del templo y la vista del lago Tonle Sap hacen que el ascenso por la colina tenga su recompensa.

fiores-441. <|begin_of_text|> 18. yüzyılda birçok kez Kamboçya'yı işgal eden Taylandlılar 1772'de Phnom Phen'i yerle bir etmiştir.

Figure 62: Activated examples for Vietnamese-specific feature 61383 in layer 9 of Llama 3.2 1B. This feature is activated in 1000 XNLI examples and 997 FLORES+ examples from the Vietnamese-specific corpus.

Feature 8775

Language: Thai
Model: meta-llama/Llama-3.2-1B
Layer: model.layers.12.mlp
SAE Model: EleutherAI/sae-Llama-3.2-1B-131k
Selected Token Probability: 0.253
Entropy: 1.799

Activation Range

0.076-0.347	0.347-0.618	0.618-0.889	0.889-1.16	1.16-1.431	1.431-1.702	1.702-1.973
1.973-2.244	2.244-2.515	2.515-2.786				

Interpretation

"The highlighted tokens are often found in polite, formal, or explanatory constructions in Japanese and Korean, including suggestions, indirect statements, and expressions of possibility or uncertainty. These tokens frequently appear in verb endings, auxiliary forms, and set phrases that convey nuance, deference, or hypothetical meaning."

Score Type	Accuracy	Precision	Recall	F1 score	TPR	TNR	FPR	FNR
detection	0.58	0.722	0.26	0.382	0.26	0.9	0.1	0.74
fuzz	0.77	0.886	0.62	0.729	0.62	0.92	0.08	0.38

Thai

#examples: [('xnli', 1000), ('flores', 996)]

xnli-282. <|begin_of_text|> ไข่ เอ๋ เรา ไม่ได้ เป็น ทหารเรือ จริงๆ แต่ ฉัน ต้องการ ♦♦♦ ร้อนใน เร็ว ๆ นี้ เพื่อ อิ่ม จ้าง เรือใบ กับ กัปตัน เนื่องจาก เรา ไม่ได้ รู้ ว่า มีความรู้ ใน สิ่ง นั้น และ เอ๋ ไป ที่ แคริบเบียน เขา ได้ ล่องเรือ ไป นิดนึง

xnli-326. <|begin_of_text|> เอ๋ ไม่ ฉัน ไม่ได้ ฉัน ตัดสินใจ ที่จะ ทำ มัน และ ฉัน ก็ ทำ

xnli-300. <|begin_of_text|> คุณ โกรธ แล้ว นะ คุณ ยัง สด ดี อยู่

xnli-743. <|begin_of_text|> ไข่ มัน เป็น แบบ นั้น จริงๆ และ ทาง นั้น เรา ก็ ไม่ได้ พลาด อะไร ไป เลย ที่ คุณ รู้ ว่า เด็ก พวก นั้น เอ๋ เรา ไม่ได้ หาย ไป จาก ประโยชน์ ของ การ ดูแล เด็ก จริงๆ

Text Examples for Each Interval

interval 1

Range: 2.515-2.786

#examples: 1

flores-873. <|begin_of_text|> もっと冒険したいと思ったら、この機会にスムージーを♦♦♦るかブレンドしてみたいか♦♦♦でしょうか。

interval 2

Range: 2.244-2.515

#examples: 1

flores-868. <|begin_of_text|> 特に、最初の数日間は、♦♦♦れるために、欧風の高級なホテルや食事、サービスにお金をかけることを検討してみたいか♦♦♦でしょうか。

interval 3

Range: 1.973-2.244

#examples: 15

flores-591. <|begin_of_text|> もしかしたら、あなたのひ♦♦♦たちが異星人の惑星に降り立って、古代の祖先に思いを♦♦♦せる日が来るかもしれません。

flores-235. <|begin_of_text|> "아버지가 무슨 말을 했냐고 ♦♦♦자 그녀는 ""아무 말도 할 수가 없어서 그저 눈만 ♦♦♦♦이며 서 계♦♦♦어""라고 말했습니다."

flores-245. <|begin_of_text|> 応援してくださる方々がいて幸せです。

flores-244. <|begin_of_text|> 異を唱える人もいるかもしれないが、私は気にしません。

flores-406. <|begin_of_text|> そうです! ツタンカー♦♦♦ン王は、「少年王」と呼ばれることもあり、現代では最も有名な古代エジプトの王の1人です。

paws-x-769. <|begin_of_text|> 林氏は、マッキーは「テンチのオリジナルモデルだ」と述べた。

Figure 63: Activated examples for Thai-specific feature 8775 in layer 12 of Llama 3.2 1B. This feature is activated in 1000 XNLI examples and 996 FLORES+ examples from the Thai-specific corpus.

Question: Write a Python code snippet to Display Low Nail care: Trimming Nails for Decision Making for Experts. Consider the Trust and implement appropriate if/else or switch/case statements to handle different conditions. Dry-run, then provide comments explaining your decision-making process and how the control flow is structured.

Solution: Here is some sample python code which implements low nail care trimming based off of expert input:

```
```python
def trim_nails(input):
 # dry run input
 print("Dry Run Input:", input)

 if input == "Yes":
 return "Trim nails"
 elif input == "No":
 return "No Action Required"
 else:
 raise ValueError("Invalid Input")
...
```
```

In this function we first perform a dry run of our input by printing it out. Then we use an `if`/`elif` statement to check whether the user has answered 'yes' or 'no'. If they have answered yes, we will return the message "trim nails". If they have answered no, we will return the message "no action required". Finally, if the user provides any other input, we will raise a value error indicating that the input was invalid. This allows us to easily add more options in the future without having

Figure 64: Example of text generated in English by Llama 3.2 1B, with top-p sampling (seed = 0, max_new_tokens = 256). The model was prompted with the [BOS] token.

Question: Write a Python code snippet to Display a List of Python Libraries. 使用Python的库和包列表显示列表。请使用Python3.5。显示列表时，使用Python3.5的方法。使用Python3.5的方法。

```
...
#Python代码
print('Libraries')
print('1.Python')
print('2.PyQt')
print('3.PyQt5')
print('4.Python3')
print('5.Python3.5')
print('6.Python3.6')
print('7.Python3.7')
print('8.Python3.8')
print('9.Python3.9')
print('10.Python3.10')
print('11.Python3.11')
print('12.Python3.12')
print('13.Python3.13')
print('14.Python3.14')
print('15.Python3.15')
print('16.Python3.16')
print('17.Python3.17')
print('18.Python3.18')
print('19.Python3.19')
print('20.Python3.20')
print('21.Python3.21')
print('22.Python3.22')
print('
```

Figure 65: Example of text generated when steered using Chinese-specific features with $\alpha = 0.3$ in Llama 3.2 1B, with top-p sampling (seed = 0, max_new_tokens = 256). The model was prompted with the [BOS] token.

Question: Create a Python script snippet that Determines Extreme Fasting: Diaper Rash на Англии на русизации. Сделайте ее как можно более простой и понятной. Обойдите всех своих знания и навыства, включая использования командной лодки, проверки данных, использования библиотек и использования других модальных программ. Используйте свои знания и навыства для создания такого кода, который будет иметь самый простой и понятный пример.

Solution: Here is a python programme that determines extreme fasting: Diaper Rash на английской версии. С помощью командной строки, проверенных данных, использования библиотек и других модальных программ. Используя свои знания и навыства для создания такого кода, который будет иметь самый простой и понятный пример.

```
```python
import sys
from datetime import datetime

Функции для работы с командной строкой
def time_1():
 print("Введите даты в формате " \
 "dd. MM. ГГГГ. года. года. месяца. дня. месяца. дня. месяца. дня. года. года. года. года. года. года.")
```

Figure 69: Example of text generated when steered using Russian-specific features with  $\alpha = 0.5$  in Llama 3.2 1B, with top-p sampling (seed = 0, max\_new\_tokens = 256). The model was prompted with the [BOS] token.

Question: Write a Python code snippet that Determines Extreme Weather: Precipitación by City (data.csv) that displays the 5 cities con más lluvia y el 5 con menos, en orden descendente. Es decir, en la columna "Precipitación" de la tabla se debe mostrar el nombre de la ciudad, el valor de la lluvia y el nombre de la ciudad.

Solution:

```
```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
df = pd.read_csv("data.csv")
df.head()
df.describe()
df["Precipitación"].value_counts()
df.groupby("Pais").agg([sum, max, min]).plot(kind="bar")
df.groupby("Pais").agg([sum, max, min]).plot(kind="bar", title="Precipitación por País")
df.groupby("Pais").agg([sum, max, min]).plot(kind="bar", title="Precipitación por País")
df.groupby("Pais").agg([sum, max, min]).plot(kind="bar", title="Precipitación por País")
df.groupby("Pais").agg([sum, max, min]).
```

Figure 70: Example of text generated when steered using Spanish-specific features with $\alpha = 0.5$ in Llama 3.2 1B, with top-p sampling (seed = 0, max_new_tokens = 256). The model was prompted with the [BOS] token.

Question: Build a Python module snippet that 计算出指定范围内的最长公共部分
 (Translation: Question: Build a Python module snippet that calculates the longest common prefix within a specified range.)
 #解答
 (Translation: # Answer)
 def getLCP(str1,str2):
 #求最长公共部分
 (Translation: # Find the longest common prefix)
 lcp =0
 #判断str1的长度
 (Translation: # Determine the length of str1)
 if len(str1) > len(str2):
 lcp = len(str1)
 else:
 lcp = len(str2)
 #求最长公共部分
 (Translation: # Find the longest common prefix)
 for i in range(0,lcp):
 if str1[i] == str2[i]:
 lcp = i
 #返回最长公共部分
 (Translation: # Return the length of the longest common prefix)
 return lcp
 <end of text>

Figure 71: Example of text generated when steered using Chinese-specific features with $\alpha = 0.3$ in Llama 3.2 1B, with top-p sampling (seed = 48, max_new_tokens = 256). The model was prompted with the [BOS] token.

```
def nested_ternary_dosun(a, b, c):
    """
    verilen üç sayıları birer birliği olarak yazınız.
    Bu sayıların toplamını bulalım
    :param a: bir sayı
    :param b: bir sayı
    :param c: bir sayı
    :return: topluyor
    """
    (Translation: """Write the given three numbers as a unit.
    Let's find the sum of these numbers
    :param a: a number
    :param b: a number
    :param c: a number
    :return: adds""")
    return a * b * c
<end of text>
```

Figure 72: Example of text generated when steered using Turkish-specific features with $\alpha = 0.25$ in Llama 3.2 1B, with top-p sampling (seed = 85, max_new_tokens = 256). The model was prompted with the [BOS] token.

```

def create_mixed_sau_oi(mixed_sau_oi):
    """Viết chương trình để tạo danh sách của 3 món ăn: cá, tôm và mực, với lượng mỗi món ăn là 100g.
    Sắp xếp các món ăn theo thứ tự: cá, tôm, mực.
    Viết chương trình để tính tổng lượng của 3 món ăn và xuất hóa đơn với giá trị 2000.
    """
    (Translation: """Write a program to create a list of 3 dishes: fish, shrimp and squid, with each dish weighing 100g.
    Arrange the dishes in order: fish, shrimp, squid.
    Write a program to calculate the total quantity of the 3 dishes and issue an invoice with the value of 2000.
    """)
    # Viết chương trình
    (Translation: # Write a program)
    # Tạo danh sách với 3 món ăn: cá, tôm và mực
    (Translation: # Create a list with 3 dishes: fish, shrimp and squid)
    mixed_sau_oi = ['Cá', 'Tôm', 'Mực']
    # Tạo danh sách 3 món ăn với lượng 100g
    (Translation: # Create a list of 3 dishes with 100g quantity)
    mixed_sau_oi = [100, 100, 100]
    # Sắp xếp các món ăn theo thứ tự: cá, tôm, mực
    (Translation: # Arrange the dishes in order: fish, shrimp, squid)
    mixed_sau_oi.sort()
    # Viết chương trình để tính tổng lượng của 3 món ăn và xuất hóa đơn với giá trị 2000
    (Translation: # Write a program to calculate the total amount of 3 dishes and issue a bill with the value of 2000.)
    # Lệ toán hóa đơn
    (Translation: # Invoice payment)
    total = 0
    for i in range(3):
        total += mixed_sau_oi[i] * 100
    # Xử lý hóa đơn
    (Translation: # Bill processing)
    print("Sản phẩm: " + mixed_sau_oi[0] + " - " + mixed_sau_oi[1] + " - " + mixed_sau_oi[2])
    (Translation: "Product: " + mixed_sau_oi[0] + " - " + mixed_sau_oi[1] + " - " + mixed_sau_oi[2])
    print("Tổng tiền: " + str(total))
    (Translation: "Total amount: " + str(total))
<end of text>

```

Figure 73: Example of text generated when steered using Vietnamese-specific features with $\alpha = 0.4$ in Llama 3.2 1B, with top-p sampling (seed = 52, max_new_tokens = 512). The model was prompted with the [BOS] token.

Intervened Language	α	Change				Unchange			
		Count	Incoherent	Partially Coherent	Coherent	Count	Incoherent	Partially Coherent	Coherent
-	-	-	-	-	-	100	1	0	99
de	0.4	19	0	7	12	81	3	5	73
fr	0.3	29	0	6	23	71	0	0	71
it	0.4	11	1	6	4	89	3	5	81
pt	0.2	17	0	13	4	83	6	12	65
hi	0.175	34	14	14	6	66	1	2	63
es	0.5	41	0	7	34	59	0	0	59
th	0.375	20	1	19	0	80	4	2	74
bg	0.4	12	1	9	2	88	14	8	66
ru	0.5	30	2	16	12	70	3	2	65
tr	0.25	15	0	3	12	85	6	2	77
vi	0.3	11	0	5	6	89	5	1	83
ja	0.3	10	1	4	5	90	12	1	77
ko	0.4	16	2	9	5	84	9	7	68
zh	0.3	29	1	8	20	71	3	1	67

Table 9: Text generation results Llama 3.2 1B for each language with various scaling factors α . Note that without intervention, the generated text from seed 0 to 100 is in English.

Intervened Language	α	Change				Unchange			
		Count	Incoherent	Partially Coherent	Coherent	Count	Incoherent	Partially Coherent	Coherent
en	-1.2	8	3	5	0	92	16	6	70
de	0.5	28	2	16	10	72	7	6	59
fr	0.4	39	0	23	16	61	1	0	60
it	0.5	15	3	12	0	85	13	10	62
pt	0.25	17	1	13	3	83	30	10	43
hi	0.2	49	25	18	6	51	2	2	47
es	0.8	44	0	17	27	56	1	0	55
th	0.4	49	20	28	1	51	0	1	50
bg	0.5	22	8	13	1	78	30	16	32
ru	0.6	51	9	16	26	49	10	2	37
tr	0.3	26	1	12	13	74	7	3	64
vi	0.4	16	0	5	11	84	16	1	67
ja	0.4	20	3	13	4	80	12	6	62
ko	0.5	46	13	28	5	54	9	4	41
zh	0.4	66	2	25	39	34	4	1	29

Table 10: Text generation results Llama 3.2 1B for each language with either a negative scaling factor or a higher positive scaling factor of α than the one used in Table 9. Note that when English was steered with $\alpha = -1.2$, the generated text changed to other languages, primarily Chinese. See Figure 74 in Appendix A for examples.

Language	SAEs Classifier			FastText			Neurons Classifier		
	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1
bg	0.990	0.966	0.978	1.000	0.964	0.982	0.264	0.974	0.415
de	0.979	0.938	0.958	0.965	0.984	0.974	0.966	0.978	0.972
en	0.836	0.980	0.902	0.849	1.000	0.918	0.000	0.000	0.000
fr	0.969	0.994	0.981	0.984	0.994	0.989	0.965	0.990	0.977
hi	0.990	0.984	0.987	1.000	0.978	0.989	0.493	0.978	0.656
it	0.990	0.956	0.973	0.996	0.966	0.981	0.866	0.956	0.909
ja	0.998	0.994	0.996	1.000	0.994	0.997	0.543	0.970	0.696
ko	1.000	0.990	0.995	1.000	0.990	0.995	0.000	0.000	0.000
pt	0.929	0.938	0.933	0.998	0.936	0.966	0.493	0.930	0.644
ru	0.986	0.998	0.992	0.994	0.998	0.996	0.000	0.000	0.000
es	0.998	0.878	0.934	0.996	0.974	0.985	0.000	0.000	0.000
th	0.998	0.988	0.993	1.000	0.988	0.994	1.000	0.972	0.986
tr	0.959	0.994	0.976	0.996	0.992	0.994	0.976	0.990	0.983
vi	1.000	0.988	0.994	1.000	0.986	0.993	1.000	0.520	0.684
zh	0.990	0.996	0.993	0.990	0.996	0.993	1.000	0.002	0.004
Macro Avg	0.974		0.972	0.984	0.983	0.983	0.571	0.617	0.528
Accuracy	0.972			0.983			0.617		

Table 11: Results of SAE-based and neuron-based LID derived from Llama 3.2 1B, and fastText LID, applied to WiLI-2018 across 15 languages. **Bold** text indicates the performance is equal or greater than the counterpart.

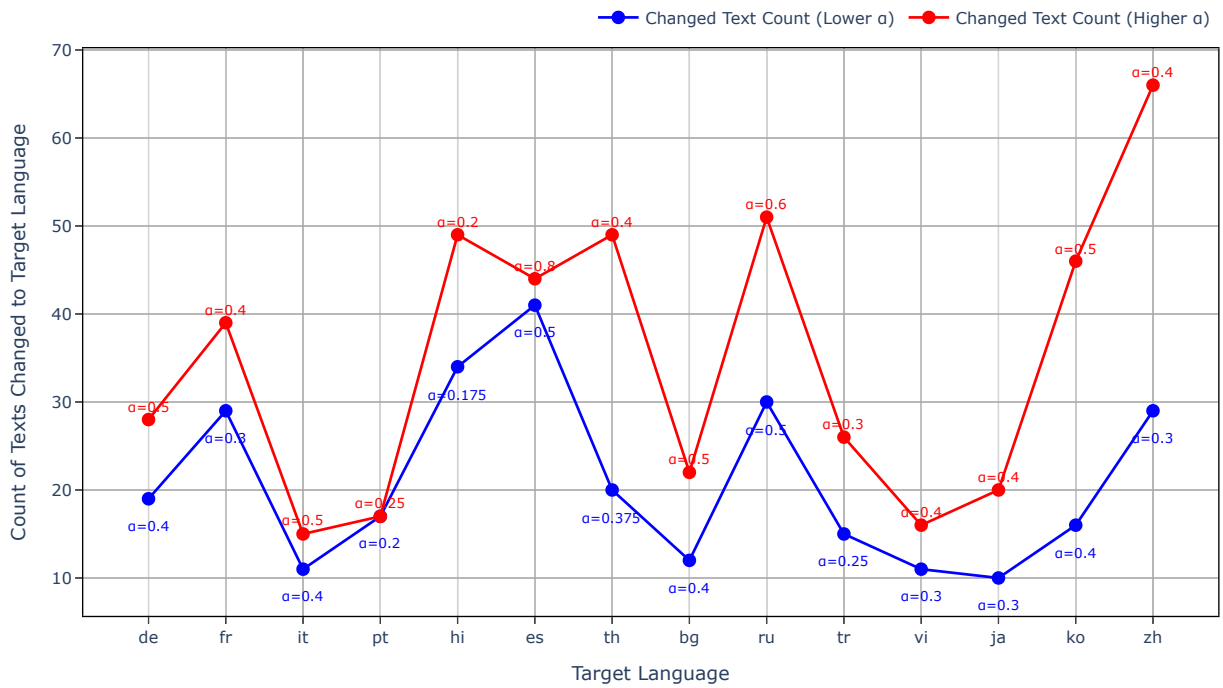


Figure 75: The impact of increasing the scaling factor α on the frequency of changes across different languages in unconditional text generation with Llama 3.2 1B.

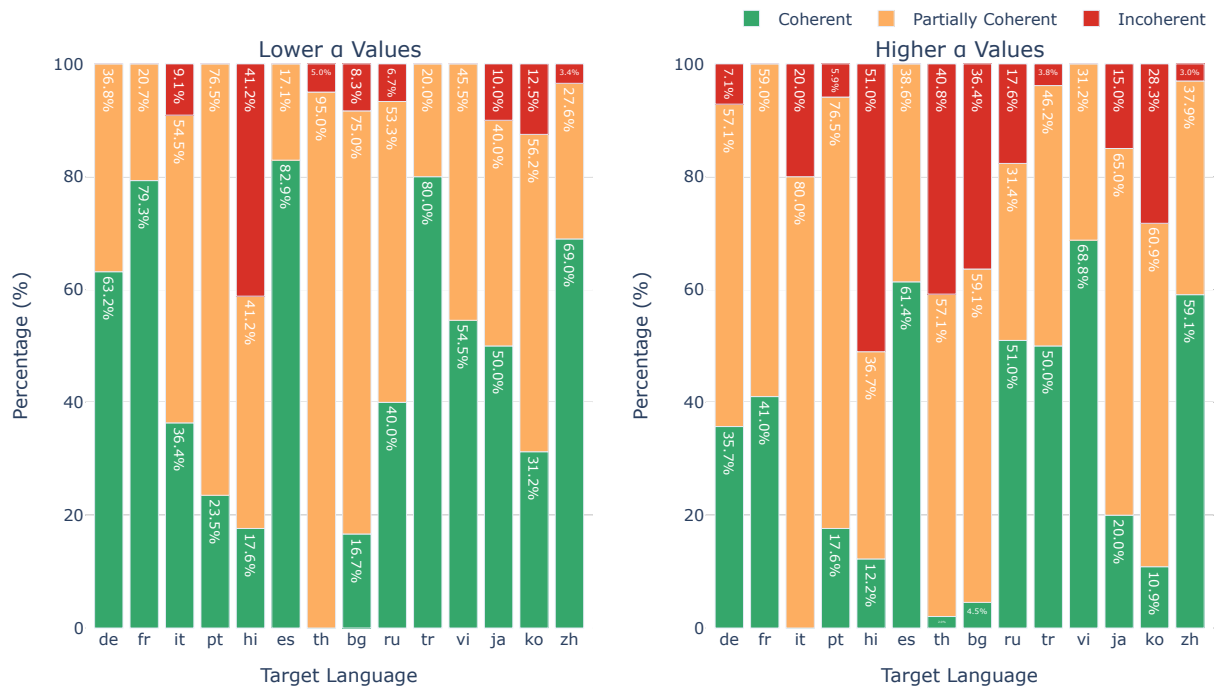


Figure 76: The impact of increasing the scaling factor α on language coherence in changed portions across different languages during unconditional text generation with Llama 3.2 1B.

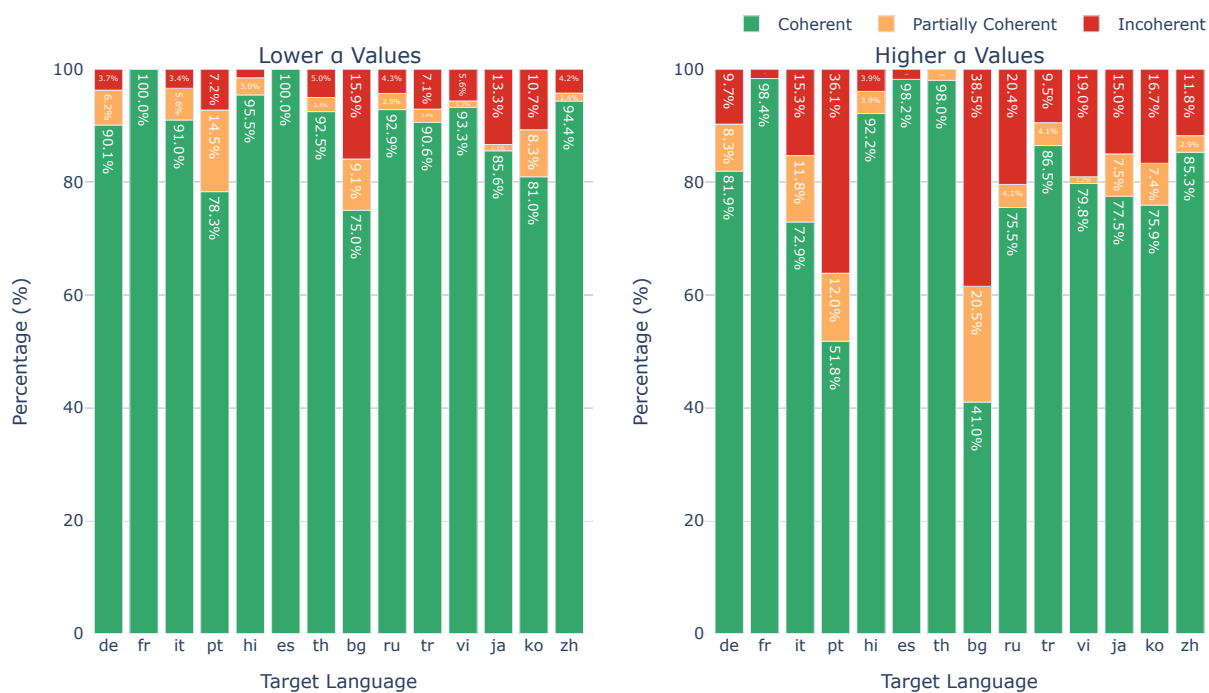


Figure 77: The impact of increasing the scaling factor α on language coherence in unchanged portions across different languages during unconditional text generation with Llama 3.2 1B.

Layer	Lang	Feature ID	Interpretation	Tokens
model.layers.0.mlp	English	93066	"Highly frequent activations on punctuation marks, especially commas and spaces, often at the start of sentences or clauses, and across multiple languages, indicating a focus on structural or delimiting tokens in multilingual text."	has, been, the, ,, ,, and, for, ,, as, well, as, co, for, all, of
model.layers.0.mlp	English	128132	"The highlighted tokens are primarily function words (such as articles, conjunctions, prepositions, and auxiliary verbs) and common noun phrases that serve as structural anchors in sentences. These tokens often mark the boundaries of clauses, introduce or connect ideas, and are essential for grammatical cohesion and meaning flow in both narrative and informational contexts."	has, been, the, ,, ,, and, for, ,, as, well, as, co, for, all, of
model.layers.2.mlp	English	14207	"The highlighted tokens are often function words, conjunctions, prepositions, or common short phrases that serve as syntactic connectors or modifiers, as well as inflectional or derivational morphemes in various languages. These tokens are crucial for sentence structure, meaning transitions, and grammatical relationships across diverse linguistic contexts."	<[begin_of_text]>, the, primary, as, as, co, all
model.layers.4.mlp	English	34138	"The highlighted text segments are typically multi-word phrases or clauses that convey specific actions, states, or relationships, often providing key contextual or descriptive information within a sentence. These segments frequently include verbs, objects, and modifiers, forming meaningful units that contribute to the overall narrative or informational structure of the text."	been, the, primary, songwriter, ,, guitarist, ,, and, for, as, as, co, -produ, cer, for, all, of, their, recorded
model.layers.5.mlp	Bulgarian	12333	"The highlighted tokens are predominantly suffixes, prefixes, or inflectional endings in Bulgarian (and occasionally other languages), marking grammatical features such as tense, number, gender, case, or aspect, as well as forming participles, adjectives, and nouns. These morphemes are essential for word formation and grammatical agreement in the language."	primary
model.layers.8.mlp	English	123973	"The highlighted spans often represent multi-word phrases or clauses that function as cohesive semantic units, such as prepositional phrases, idiomatic expressions, or descriptive noun phrases. These segments frequently provide contextual detail, specify relationships, or add nuance to the main clause, and are typically composed of function words combined with content words to form meaningful, context-dependent chunks."	has, been, the, primary, ,
model.layers.11.mlp	Portuguese	19293	"The highlighted tokens are predominantly roots, stems, or affixes within Portuguese words, often marking the beginning or internal structure of nouns, verbs, and adjectives. These segments are morphologically significant, frequently corresponding to common derivational or inflectional morphemes, and are central to word formation and meaning in the language."	Brothers
model.layers.11.mlp	Turkish	37895	"The highlighted tokens are primarily suffixes, inflections, or short function words in Turkish and German, often marking grammatical relationships such as possession, plurality, tense, or case, as well as prepositions and names in German. These tokens are crucial for the syntactic and semantic structure of sentences in agglutinative and inflectional languages."	co, recorded
model.layers.15.mlp	English	5889	"The highlighted tokens frequently correspond to structural elements in formal or factual writing, such as numbers, dates, conjunctions, prepositions, punctuation, and proper nouns, which are often used to organize, enumerate, or specify information in lists, references, or descriptive passages."	Hamilton, has, been, the, primary, songwriter, ,, ,, and, vocalist, for, Brothers, ,, as, well, as, cer, for, all, of, their, releases, .
model.layers.15.mlp	English	67343	"The text contains a wide variety of conversational and narrative fragments, including idiomatic expressions, filler words, discourse markers, and references to people, places, and actions. There is a frequent use of pronouns, conjunctions, and auxiliary verbs, as well as sequences that reflect spoken language patterns such as repetitions, hesitations, and clarifications. Many segments highlight the structure of dialogue, question-answer formats, and descriptive or explanatory statements, often focusing on actions, states, or relationships between entities."	Hamilton, has, been, the, primary, songwriter, ,, ,, and, vocalist, for, Brothers, ,, as, well, as, cer, for, all, of, their, releases, .

Figure 78: Full feature activation details in Llama 3.2 1B for the sentence "Hamilton has been the primary songwriter, guitarist, and vocalist for Brothers Past, as well as co-producer for all of their recorded releases."

Layer	Lang	Feature ID	Interpretation	Tokens
model.layers.0.mlp	Bulgarian	123218	"The text highlights the frequent use of single Cyrillic letters, especially at the beginning of sentences or phrases, functioning as prepositions, conjunctions, or initials in Russian and Bulgarian, often with high activation when capitalized and sentence-initial."	О, га, на, Г
model.layers.0.mlp	English	93066	"Highly frequent activations on punctuation marks, especially commas and spaces, often at the start of sentences or clauses, and across multiple languages, indicating a focus on structural or delimiting tokens in multilingual text."	(, :, *, ., , in,), und, , (
model.layers.0.mlp	Turkish	64723	"Suffixes such as '\i', '\lar', '\ler', '\lik', '\lk', '\im', '\in', '\isî', '\ğî', '\larî', '\lerî', '\mî', '\udur', '\ildî', '\mî', '\isî', '\arî', '\ah', '\isî', '\lk'"	up
model.layers.1.mlp	Bulgarian	32578	"The highlighted tokens are common Bulgarian suffixes used to form adjectives, nouns, and participles, indicating grammatical features such as gender, number, definiteness, and case. These suffixes are essential for word formation and inflection in Bulgarian morphology."	на, на
model.layers.1.mlp	Turkish	84692	"The highlighted tokens are predominantly prefixes, roots, or short morphemes within words, often marking the beginning or core of nouns, verbs, or adjectives across multiple languages. These segments frequently correspond to meaningful subword units, such as roots, affixes, or syllables, and are often found in proper names, technical terms, or compound words. The pattern reflects a focus on linguistically significant subword structures that contribute to word formation and meaning."	Gir
model.layers.2.mlp	English	14207	"The highlighted tokens are often function words, conjunctions, prepositions, or common short phrases that serve as syntactic connectors or modifiers, as well as inflectional or derivational morphemes in various languages. These tokens are crucial for sentence structure, meaning transitions, and grammatical relationships across diverse linguistic contexts."	< begin_of_text >,
model.layers.2.mlp	German	56064	"The highlighted tokens are primarily German morphemes, word stems, or inflections, often marking noun or verb endings, compound word parts, or grammatical function words. Many are common in German-language text, including noun endings (-en, -heit, -keit, -ung), verb forms (-iert, -te, -en), and connecting words (und, von, in, der, die, das, eine). There is also frequent activation on recurring roots in compound nouns and on inflectional suffixes, reflecting the morphological structure of German."	isch, Juni, Lang, ist, ß
model.layers.2.mlp	Thai	100192	"The highlighted tokens are often function words, affixes, or short morphemes in various languages, including prepositions, conjunctions, pronouns, and grammatical particles, as well as some high-activation non-standard or corrupted characters. These tokens are typically important for sentence structure, meaning, or language identification."	OI, O
model.layers.2.mlp	Thai	105874	"The highlighted tokens are often morphemes, syllables, or short word fragments that serve as meaningful units in various languages, including Thai, Hindi, and others. These units frequently appear at the beginning, middle, or end of words and are important for word formation, inflection, or conveying grammatical and semantic information."	OI, O
model.layers.3.mlp	Bulgarian	98476	"The highlighted tokens are predominantly Bulgarian suffixes, pronoun and conjunction forms, and inflectional endings that mark grammatical relationships such as gender, number, case, and verb tense, as well as function words that connect clauses or indicate possession and agency. These elements are essential for the syntactic structure and meaning in Bulgarian sentences."	Alexand
model.layers.3.mlp	Italian	120200	"The highlighted tokens are often morphemes, suffixes, or name fragments within words, especially in Italian and other Romance languages, marking comparative forms, diminutives, or parts of proper nouns and loanwords. These tokens frequently appear at word endings or within multi-syllabic names and terms."	O
model.layers.3.mlp	Spanish	4518	"The highlighted tokens are often morphemes, word stems, or affixes from various languages, especially Spanish, marking grammatical or semantic units such as verb endings, noun forms, or prepositions, and sometimes appear in named entities or set phrases."	ga
model.layers.4.mlp	Russian	48343	"The highlighted tokens are primarily short morphemes, prefixes, or function words in Slavic and related languages, often marking grammatical relationships, word formation, or serving as connectors within or between words and phrases."	Gir

Figure 79: Full feature activation details in Llama 3.2 1B for the sentence "Olga Alexandrowna Girja (russisch Ольга Александровна Гиря; * 4. Juni 1991 in Langeepas) ist eine russische Schachspielerin und seit 2009 Großmeister der Frauen (WGM)." from layers 0–4.

model.layers.5.mlp	Bulgarian	12333	"The highlighted tokens are predominantly suffixes, prefixes, or inflectional endings in Bulgarian (and occasionally other languages), marking grammatical features such as tense, number, gender, case, or aspect, as well as forming participles, adjectives, and nouns. These morphemes are essential for word formation and grammatical agreement in the language."	ja, *
model.layers.5.mlp	Bulgarian	30265	"The highlighted tokens are predominantly suffixes, inflections, or short morphemes in Bulgarian (and some other languages), such as grammatical endings for tense, case, gender, number, or diminutives, as well as common function words and particles. These elements are crucial for the grammatical structure and meaning of words and sentences."	ja, *
model.layers.5.mlp	German	31366	"The highlighted tokens are primarily German (and some Russian) morphemes, word stems, and inflectional endings, often marking grammatical features such as case, number, tense, or forming compound nouns and verbs. Many are parts of longer words or compounds, reflecting the agglutinative and inflectional nature of German, and are often found at the beginning, middle, or end of words."	*, ist
model.layers.5.mlp	Vietnamese	49501	"The highlighted tokens are characteristic morphemes, suffixes, or letter clusters commonly found in Romanian and related Eastern European place names, river names, and surnames, often marking grammatical or regional features."	Alexand
model.layers.6.mlp	Russian	121507	"The highlighted tokens are predominantly morphemes, roots, prefixes, suffixes, or short function words within Russian, Bulgarian, and related Slavic texts. These segments often mark grammatical features, word formation, or serve as connectors, and are frequently found at the beginning, middle, or end of words, reflecting the agglutinative and inflectional nature of these languages."	ль, га, Александ, Г
model.layers.6.mlp	Thai	68498	"The highlighted tokens are morphemes or syllables at word boundaries across multiple languages, often marking inflections, derivations, or meaningful subword units. These include suffixes, prefixes, or roots that contribute to the grammatical or semantic structure of words."	spl, Gro
model.layers.7.mlp	Thai	70835	"The highlighted tokens are primarily morphemes, syllables, or short word fragments from multiple languages, especially Thai, Russian, and Bulgarian, often marking the start or core of content words such as nouns, verbs, or adjectives. These fragments frequently appear at the beginning or within important words, indicating their role in word formation and semantic content across diverse scripts and languages."	О, ль, Г, ир
model.layers.8.mlp	English	123973	"The highlighted spans often represent multi-word phrases or clauses that function as cohesive semantic units, such as prepositional phrases, idiomatic expressions, or descriptive noun phrases. These segments frequently provide contextual detail, specify relationships, or add nuance to the main clause, and are typically composed of function words combined with content words to form meaningful, context-dependent chunks."	ga
model.layers.8.mlp	German	79729	"The highlighted tokens are primarily German function words, verb forms, noun and adjective endings, and common prefixes or suffixes, often marking grammatical relationships, tense, plurality, or case, as well as frequent collocations and set phrases. These tokens are essential for sentence structure and meaning in German text."	Juni, ist
model.layers.8.mlp	Russian	109317	"The highlighted tokens are primarily morphemes, word stems, suffixes, and grammatical endings in Russian (and some in English), often marking case, number, tense, or forming nouns, adjectives, and verbs. There is a focus on functional elements that contribute to word formation, inflection, and syntactic structure, as well as on common connective words and phrase boundaries."	ja
model.layers.9.mlp	German	9083	"The highlighted tokens are primarily function words, verb forms, noun and adjective endings, and common collocations in German sentences, often marking grammatical structure, tense, or case, as well as frequent phrase boundaries and connectors. These tokens are crucial for sentence construction and meaning in German text."	*, Juni,), ist, eine,).
model.layers.9.mlp	Portuguese	77250	"The highlighted tokens are predominantly prefixes, suffixes, or stems within Portuguese words, often marking verb conjugations, noun/adjective forms, or common morphemes. These segments frequently appear at the start or end of words, reflecting morphological structure and inflectional patterns in the language."	eine
model.layers.9.mlp	Russian	59767	"The highlighted tokens are predominantly Russian word stems, prefixes, and suffixes that form the core of verbs, nouns, and adjectives, often marking grammatical or semantic roles such as actions, objects, or qualities. These morphemes are crucial for word formation and meaning in Russian, frequently appearing at the beginning or within words to indicate tense, aspect, subject, or object, and are central to the structure and interpretation of Russian sentences."	Г, О, ль, га, Александ, ров, на, Г, ир, я

Figure 80: Full feature activation details in Llama 3.2 1B for the sentence "Olga Alexandrowna Girja (russisch Ольга Александровна Гиря; * 4. Juni 1991 in Langeepas) ist eine russische Schachspielerin und seit 2009 Großmeister der Frauen (WGM)." from layers 5–9.

model.layers.10.mlp	Bulgarian	41383	"The highlighted tokens are predominantly proper nouns, especially names of people, places, and media titles, often appearing in multiple languages and scripts. These tokens frequently occur in contexts involving legal, journalistic, or entertainment references, and are often accompanied by titles, roles, or institutional affiliations."	Alexand, r
model.layers.10.mlp	Bulgarian	43549	"The highlighted tokens are primarily function words, pronouns, conjunctions, prepositions, and common verb forms in Bulgarian (and some in other languages), as well as frequent morphemes and endings. These elements are essential for sentence structure, grammatical relationships, and meaning, often marking tense, negation, possession, or subordination. The activations focus on the connective tissue of language that enables coherent and contextually appropriate communication."	Alexand, r
model.layers.10.mlp	German	44371	"The highlighted spans are predominantly German multi-word expressions, noun or verb phrases, and compound words, often marking grammatical constructs, inflections, or idiomatic units. Many are functionally important for sentence structure, such as verb-final clauses, participial forms, or noun compounds, and frequently include suffixes or prefixes that signal tense, plurality, or case. These patterns reflect the morphological richness and syntactic dependencies characteristic of German text."	ist
model.layers.10.mlp	Russian	56594	"The highlighted tokens are predominantly proper nouns, technical terms, and named entities—such as personal names, place names, and institutional or scientific terms—often in multiple languages. These tokens frequently appear in contexts involving formal identification, attribution, or description of people, locations, organizations, or specialized concepts."	ga, Alexand, row, na, Gir, ja, (, r, uss, isch, О, ль, га, Александ, ров, на, Г, ир, я, , , in, Lang, as, eine, russ, ach, Gro, der, (, W
model.layers.11.mlp	Bulgarian	52975	"The highlighted tokens are predominantly short function words, prefixes, or single letters from various languages, often marking grammatical relationships, verb forms, or serving as conjunctions and prepositions. These tokens are crucial for sentence structure and meaning, especially in morphologically rich or agglutinative languages."	r
model.layers.11.mlp	German	27401	"The highlighted tokens are predominantly German morphemes, word stems, and grammatical endings, often marking inflections, compounds, or function words. These tokens frequently appear at the boundaries of words or phrases, indicating their importance for understanding German syntax, morphology, and sentence structure."	;, *, ., Juni,), ist, eine, ische, elerin, und, seit, 9, ister, Frauen,).
model.layers.11.mlp	German	52681	"The highlighted tokens are primarily German function words, verb forms, pronouns, and common noun or adjective suffixes, especially those marking person, tense, or plurality. There is a strong focus on grammatical structures, such as verb conjugations, modal verbs, and case endings, as well as on frequently used collocations and phrasal patterns in German sentences."	;, *, ., Juni,), ist, eine, ische, elerin, und, seit, 9, ister, Frauen,).
model.layers.11.mlp	Hindi	63085	"The text contains frequent use of Hindi conjuncts and half-letters, especially the 'ँ' (virama) and combinations with 'ँ', 'ँ', 'ँ', and other consonants, which are characteristic of Hindi morphology and word formation, particularly in formal or technical contexts."	ep
model.layers.11.mlp	Korean	102511	"The highlighted tokens are predominantly Korean noun and suffix forms, often ending with '함', '것', '음', '법', '형', '을', '량', '군', '객', '면', '성', '전', '편', '심', '득', '상', '시간', and similar morphemes. These tokens typically denote abstract concepts, objects, people, or states, and are frequently used as nominalizers or to form compound nouns, reflecting a pattern of marking key semantic units or grammatical roles in Korean text."	rov
model.layers.11.mlp	Russian	8452	"The highlighted tokens are predominantly Russian morphemes, words, or short phrases that serve as key semantic or syntactic units within sentences. They include verb roots, noun endings, pronouns, conjunctions, and common collocations, often marking the core meaning, grammatical structure, or transitions in the text. The activations focus on elements that define actions, states, relationships, or important contextual information, reflecting the building blocks of Russian sentence construction and meaning."	О, ль, га, Александ, ров, на, Г, ир
model.layers.11.mlp	Russian	74675	"The highlighted tokens are primarily Russian morphemes, roots, and affixes that form the core semantic or grammatical structure of words, often marking verbs, nouns, or adjectives, and are crucial for conveying meaning, tense, aspect, or function within the sentence."	О, ль, га, Александ, ров, на, Г, ир
model.layers.11.mlp	Turkish	37895	"The highlighted tokens are primarily suffixes, inflections, or short function words in Turkish and German, often marking grammatical relationships such as possession, plurality, tense, or case, as well as prepositions and names in German. These tokens are crucial for the syntactic and semantic structure of sentences in agglutinative and inflectional languages."	in, eine, Gro, ß, me, W

Figure 81: Full feature activation details in Llama 3.2 1B for the sentence "Olga Alexandrowna Girja (russisch Ольга Александровна Гиря; * 4. Juni 1991 in Langeepas) ist eine russische Schachspielerin und seit 2009 Großmeister der Frauen (WGM)." from layers 10–11.

